

This chapter discusses several extensions of the classical linear model. We first describe in Sect. 4.1 the general linear model and its applications. This model allows for correlated errors and heteroscedastic variances of the errors. Section 4.2 discusses several techniques to regularize the least squares estimator. Such a regularization may be useful in cases where the design matrix is highly collinear or even rank deficient. Moreover, regularization techniques allow for built-in variable selection. Section 4.4 describes Bayesian linear models as an alternative to the frequentist linear model framework. In modern statistics, Bayesian approaches have become increasingly more important and widely used. Hence, Sect. 4.4 (in combination with Appendix B.5) serves as a basis for the Bayesian approaches to be described in chapters to follow. Moreover, we present powerful Bayesian techniques for model choice, in particular variable selection.

4.1 The General Linear Model

Thus far, we have addressed the classical linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with uncorrelated and homoscedastic errors, i.e., $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. Least squares estimation based on these assumptions is referred to as *ordinary least squares*. The discussion of the model has shown that the assumption of uncorrelated and homoscedastic errors is not always satisfied. In this section, we want to extend the class of linear models such that correlated and heteroscedastic errors are possible. The resulting model is called the *general linear model*. The classical linear model considered so far is an important special case. In fact, many inference problems in the general linear model can be traced back to the classical linear model.

4.1.1 Model Definition

In the general linear model, we replace

$$\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$$

by the more general assumption

$$\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1},$$

where $\mathbf{W}$ is a positive definite matrix. In the special case of heteroscedastic and (still) uncorrelated errors, we have

$$\mathbf{W} = \text{diag}(w_1, \dots, w_n).$$

The heteroscedastic error variances are then given by $\text{Var}(\varepsilon_i) = \sigma_i^2 = \sigma^2/w_i$.

When introducing a more general model class, more complicated inference techniques are usually needed than in the simpler special case. Thus, it is worth investigating whether or not the use of the more general model is really necessary. We therefore study the consequences of using the comparably simple inference techniques of the classical linear model if the true model is the general linear model, i.e., $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1}$ and not $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. We first examine the consequences of using the ordinary least squares estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}$ within the general linear model. Analogous to the derivations on p. 115 for the classical linear model, we obtain

$$\mathbf{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{W}^{-1} \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}. \quad (4.1)$$

Hence, ordinary least squares developed for the classical linear model is still unbiased for $\boldsymbol{\beta}$ within the general linear model setting. However, the covariance matrix in the general linear model does not correspond with the one found for the classical model, $\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$. Thus, all derivations that are based on the covariance matrix of $\hat{\boldsymbol{\beta}}$ are wrong. In particular, we obtain incorrect variances and standard errors for the estimated regression coefficients, and thus incorrect tests and confidence intervals.

In the following section, we discuss several alternatives to obtain improved estimators within the general linear model. Section 4.1.2 develops a *weighted least squares estimator* (WLS) as a generalization of the ordinary least squares estimator. The WLS estimator shares many of the desirable properties as found with the ordinary least squares estimator in the classical setting. When using the WLS estimator, the matrix $\mathbf{W}$ must be known, which in practice is usually not the case. Therefore Sects. 4.1.3 and 4.1.4 address inference in the case of unknown $\mathbf{W}$. We restrict ourselves to the two important special cases of heteroscedastic and autocorrelated errors.

4.1.2 Weighted Least Squares

We now develop an estimator that bypasses the above mentioned problems when using the ordinary least squares estimator in the general linear model. The idea is to transform the response variable, design matrix, and errors such that the

4.1 The General Linear Model

The model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

is called general linear model, if the following assumptions hold:

1. $E(\boldsymbol{\varepsilon}) = \mathbf{0}$.
2. $\text{Cov}(\boldsymbol{\varepsilon}) = E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') = \sigma^2 \mathbf{W}^{-1}$, where $\mathbf{W}$ is a known positive definite matrix.
3. The design matrix $\mathbf{X}$ has full column rank, i.e., $\text{rk}(\mathbf{X}) = k + 1 = p$.

We refer to general normal regression when the additional assumption

4. $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{W}^{-1})$

applies.

transformed variables follow a classical linear model. For illustration, we first look at a model with uncorrelated and heteroscedastic errors, i.e., $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1} = \sigma^2 \text{diag}(1/w_1, \dots, 1/w_n)$. Multiplication of the errors ε_i with $\sqrt{w_i}$ produces the transformed errors $\varepsilon_i^* = \sqrt{w_i} \varepsilon_i$ with constant variances $\text{Var}(\varepsilon_i^*) = \text{Var}(\sqrt{w_i} \varepsilon_i) = \sigma^2$. To make sure that the model is unchanged, we must transform the response variable and all covariates (including the constant) accordingly. We obtain $y_i^* = \sqrt{w_i} y_i$, $x_{i0}^* = \sqrt{w_i}$, $x_{i1}^* = \sqrt{w_i} x_{i1}$, $\dots$, $x_{ik}^* = \sqrt{w_i} x_{ik}$, and, thus, the classical linear model

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \dots + \beta_k x_{ik}^* + \varepsilon_i^*$$

with homoscedastic errors ε_i^* . This transformation formally corresponds to multiplication of the model equation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, from the left, with the matrix $\mathbf{W}^{1/2} = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})$, yielding

$$\mathbf{W}^{1/2} \mathbf{y} = \mathbf{W}^{1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{W}^{1/2} \boldsymbol{\varepsilon}.$$

Using the transformed values $\mathbf{y}^* = \mathbf{W}^{1/2} \mathbf{y}$, $\mathbf{X}^* = \mathbf{W}^{1/2} \mathbf{X}$, and $\boldsymbol{\varepsilon}^* = \mathbf{W}^{1/2} \boldsymbol{\varepsilon}$, we finally obtain

$$\mathbf{y}^* = \mathbf{X}^* \boldsymbol{\beta} + \boldsymbol{\varepsilon}^*. \quad (4.2)$$

The transformation allows us to work in the familiar framework of the classical linear model. For the least squares estimator we get

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}^{*'} \mathbf{X}^*)^{-1} \mathbf{X}^{*'} \mathbf{y}^* \\ &= (\mathbf{X}' \mathbf{W}^{1/2} \mathbf{W}^{1/2} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W}^{1/2} \mathbf{W}^{1/2} \mathbf{y} \\ &= (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{y}, \end{aligned}$$

which is referred to as the *Aitken* or *WLS estimator*. It can be shown that the Aitken estimator minimizes the “weighted” residual sum of squares

$$\text{GLS}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \mathbf{W} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \sum_{i=1}^n w_i (y_i - \mathbf{x}_i' \boldsymbol{\beta})^2.$$

Observations with a higher variance σ^2/w_i have less weight w_i than observations with a smaller variance.

Assuming normal errors, we can also show that the WLS estimator coincides with the ML estimator for $\boldsymbol{\beta}$, i.e., $\hat{\boldsymbol{\beta}}_{ML} = \hat{\boldsymbol{\beta}}$. For the ML estimator for σ^2 , we obtain

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})' \mathbf{W} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \frac{1}{n} \hat{\boldsymbol{\varepsilon}}' \mathbf{W} \hat{\boldsymbol{\varepsilon}}.$$

As in the classical linear model, this estimator is biased. An unbiased estimator is given by

$$\hat{\sigma}^2 = \frac{1}{n-p} \hat{\boldsymbol{\varepsilon}}' \mathbf{W} \hat{\boldsymbol{\varepsilon}} = \frac{1}{n-p} \sum_{i=1}^n w_i (y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}})^2.$$

This estimator can be seen as a restricted maximum likelihood estimator; see Sect. 3.2.2 (p. 108). All derivations and proofs are completely analogous to the classical linear model.

The approach exemplified for heteroscedastic errors can be extended to the case of a general covariance matrix $\sigma^2 \mathbf{W}^{-1}$. When doing so, we use a “square root” $\mathbf{W}^{1/2}$ of $\mathbf{W}$ with $\mathbf{W}^{1/2} (\mathbf{W}^{1/2})' = \mathbf{W}$. The matrix $\mathbf{W}^{1/2}$ is not uniquely determined. It can be obtained using, e.g., the spectral decomposition

$$\mathbf{W} = \mathbf{P} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{P}'$$

of $\mathbf{W}$ (see result A.25 (p. 633) in Appendix A), resulting in

$$\mathbf{W}^{1/2} = \mathbf{P} \text{diag}(\lambda_1^{1/2}, \dots, \lambda_n^{1/2}) \mathbf{P}'.$$

We can then transform the response vector, the design matrix, and the errors with $\mathbf{W}^{1/2}$. This results in a model of the form (4.2), which is a classical linear model since we have

$$\mathbf{E}(\boldsymbol{\varepsilon}^*) = \mathbf{E}(\mathbf{W}^{1/2} \boldsymbol{\varepsilon}) = \mathbf{W}^{1/2} \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\text{Cov}(\boldsymbol{\varepsilon}^*) = \mathbf{E}(\mathbf{W}^{1/2} \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' \mathbf{W}^{1/2}) = \sigma^2 \mathbf{W}^{1/2} \mathbf{W} \mathbf{W}^{1/2} = \sigma^2 \mathbf{I}.$$

Finally, the WLS estimator shares the same stochastic properties as ordinary least squares; see the following Box 4.2. The proof for the properties of the weighted estimator is completely analogous to the unweighted estimator. Moreover, the tests

4.2 Estimators in the General Linear Model

Weighted Least Squares or ML Estimator for β

$$\hat{\beta} = (X' W X)^{-1} X' W y.$$

Properties of the Weighted Least Squares Estimator

1. *Expectation*: $E(\hat{\beta}) = \beta$, i.e., the WLS estimator is unbiased.
2. *Covariance matrix*: $\text{Cov}(\hat{\beta}) = \sigma^2 (X' W X)^{-1}$.
3. *Gauß-Markov Theorem*: Among all linear and unbiased estimators $\hat{\beta}^L = A y$, the WLS estimator has minimal variance, i.e.,

$$\text{Var}(\hat{\beta}_j) \leq \text{Var}(\hat{\beta}_j^L), \quad j = 0, \dots, k.$$

REML Estimator for σ^2

$$\hat{\sigma}^2 = \frac{1}{n - p} \hat{\epsilon}' W \hat{\epsilon}.$$

The REML estimator is unbiased.

Note that all properties hold only for known W and correctly specified model.

and confidence intervals of the classical linear model can be easily adapted. We simply have to replace the standard errors of the classical linear model by those of the general linear model. They are computed as usual by inserting $\hat{\sigma}^2$ into the covariance $\text{Cov}(\hat{\beta}) = \sigma^2 (X' W X)^{-1}$ (see Box 4.2) to obtain $\widehat{\text{Cov}}(\hat{\beta})$. The standard errors are then the square roots of the main diagonal.

A first application of the WLS estimator with known weight matrix W is given for grouped data as follows.

Grouped Data

Thus far we have considered *individual* or *ungrouped data*, which implies that we have an observation $(y_i, \mathbf{x}_i)$ for every individual or subject i in a sample of size n . Every response y_i and every covariate vector $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})'$ belongs to exactly one subject i :

$$\begin{array}{c} \text{Unit 1} \\ \vdots \\ \text{Unit } i \\ \vdots \\ \text{Unit } n \end{array} \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix} \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{i1} & & x_{ik} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}$$

If several covariate vectors or rows of the design matrix are identical, we can *group* the data. After sorting and combining the data, the grouped design matrix now only contains rows with *different* covariate vectors $\mathbf{x}_i$. We denote by n_i the number of replicates of $\mathbf{x}_i$ within the original sample of individual data. We further denote by $\bar{y}_i$ the arithmetic mean of the respective individual response values that were observed for $\mathbf{x}_i$:

$$\begin{array}{c} \text{Group 1} \\ \vdots \\ \text{Group } i \\ \vdots \\ \text{Group } G \end{array} \begin{bmatrix} n_1 \\ \vdots \\ n_i \\ \vdots \\ n_G \end{bmatrix} \begin{bmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_i \\ \vdots \\ \bar{y}_G \end{bmatrix} \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{i1} & \cdots & x_{ik} \\ \vdots & \vdots & & \vdots \\ 1 & x_{G1} & \cdots & x_{Gk} \end{bmatrix}$$

Thereby G represents the number of unique covariate vectors in the sample. In many cases G is much smaller than the sample size n , especially with binary or categorical covariates.

Grouped data can be easily handled within the scope of the general linear model by defining $\mathbf{y} = (\bar{y}_1, \dots, \bar{y}_G)'$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \text{diag}(1/n_1, \dots, 1/n_G)$, i.e., $\mathbf{W} = \text{diag}(n_1, \dots, n_G)$.

Other applications of the WLS estimator with known weight matrix $\mathbf{W}$ will be discussed in Chaps. 5 and 8. In Chap. 5, the WLS estimator will be the basis for parameter estimates in generalized linear models. In Chap. 8, WLS estimation will be of importance in relation to *local smoothers*; see Sect. 8.1.7.

In the following two sections, we consider situations in which the weight matrix $\mathbf{W}$ is, to some extent, unknown. More specifically, we will discuss heteroscedastic and autocorrelated errors.

4.1.3 Heteroscedastic Errors

In the linear model with heteroscedastic errors, the covariance matrix of the errors is given by $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \text{diag}(1/w_1, \dots, 1/w_n)$. In the literature, especially in the econometrics literature, we find a large variety of estimators and modeling approaches, which are beyond the scope of this book. A good overview can be

found in [Greene \(2000\)](#). Here, we discuss a two-stage least squares estimator and we briefly sketch the ML estimator. The drawback of both estimators is that they require knowledge of the specific form of heteroscedasticity. We therefore describe an alternative approach developed by [White \(1980\)](#), which avoids assumptions about the form of heteroscedasticity. Before we present the different estimators, we show how to detect heteroscedastic errors.

Detecting Heteroscedastic Errors

As a starting point for detecting heteroscedastic errors, a classical linear model is typically estimated, followed by a careful residual analysis. In the literature, we generally find two different strategies. The statistics and biometry literature proposes graphical tools, residual plots in particular. The econometrics literature rather develops a variety of statistical tests to discover heteroscedasticity.

Residual Plots

In order to detect heteroscedastic errors, it is useful to plot the residuals against the predicted values $\hat{y}_i$ and the covariates x_{ij} . Covariates *not included in the model* should also be considered. Note also that a plot of the residuals against y (rather than the predicted values $\hat{y}$) is not recommended because the residuals $\hat{\varepsilon}$ depend (by definition) on the response y and a plot would reveal this dependency. The standardized or studentized residuals (see [Box 3.12](#) on p. 126) are preferred over the raw residuals, since the latter are heteroscedastic with $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$. Consequently, raw residuals are less appropriate to examine heteroscedasticity (see p. 122). In the case of homoscedastic error variances, the standardized or studentized residuals exhibit random fluctuation around zero with a *constant variance*. If this is *not* the case, there is evidence for heteroscedastic variances.

Example 4.1 Munich Rent Index—Diagnostics for Heteroscedastic Errors

We illustrate the detection of heteroscedastic errors with the rent index data. For simplicity we restrict ourselves to a model with the net rent as response variable and the covariates living area and year of construction. In [Fig. 1.5](#) (p. 15), the scatter plots between net rent and living area and year of construction show a linear effect of the living area and a slightly nonlinear effect of the year of construction. We model the effect of the year of construction using an orthogonal polynomial of degree three. Thus, we assume the classical linear model

$$\text{rent}_i = \beta_0 + \beta_1 \text{area}_i + \beta_2 \text{yearco}_i + \beta_3 \text{yearco}2_i + \beta_4 \text{yearco}3_i + \varepsilon_i. \quad (4.3)$$

Estimation results can be found in [Table 4.1](#). [Figure 4.1a, b](#) shows the estimated effects of living area and year of construction, including the corresponding partial residuals. Panels (c), (d), and (e) show the studentized residuals as a function of the estimated net rent, the living area, and the year of construction. There is clear evidence of heteroscedastic variances depending on both the living area and the year of construction. $\triangle$



Testing Heteroscedasticity

Tests of heteroscedasticity are treated extensively in the econometrics literature, as, for example, in [Greene \(2000\)](#) and [Judge, Griffith, Hill, Lütkepohl, and Lee \(1980\)](#).

Table 4.1 Munich rent index: estimation results for the unweighted regression

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	459.437	2.631	174.600	<0.001	454.278	464.597
<i>area</i>	121.817	2.742	44.430	<0.001	116.441	127.193
<i>yearco</i>	54.336	2.706	20.080	<0.001	49.030	59.642
<i>yearco2</i>	31.484	2.668	11.800	<0.001	26.252	36.715
<i>yearco3</i>	−0.198	2.631	−0.080	0.940	−5.358	4.961

We illustrate heteroscedasticity tests with a test according to [Breusch and Pagan \(1979\)](#). The basis for the test is a multiplicative model for the error variances:

$$\sigma_i^2 = \sigma^2 \cdot h(\alpha_0 + \alpha_1 z_{i1} + \cdots + \alpha_q z_{iq}),$$

where h is a function not depending on the unit index i and $z_1, \dots, z_q$ are covariates that may influence the variance. The hypothesis of homoscedastic variances is equivalent to $\alpha_1 = \cdots = \alpha_q = 0$. The Breusch–Pagan test is based on the hypothesis

$$H_0 : \alpha_1 = \dots = \alpha_q = 0 \quad \text{against} \quad H_1 : \alpha_j \neq 0 \text{ for at least one } j.$$

In order to conduct the test, an auxiliary regression between the response variable

$$g_i = \frac{\hat{\varepsilon}_i^2}{\hat{\sigma}_{ML}^2}$$

and the explanatory variables $z_1, \dots, z_q$ is performed. Here, $\hat{\varepsilon}_i$ are the residuals and $\hat{\sigma}_{ML}^2$ is the ML estimator for σ^2 for the linear model $y = \mathbf{x}'\boldsymbol{\beta} + \varepsilon$ with homoscedastic errors. The test statistic is given by

$$T = \frac{1}{2} \sum_{i=1}^n (\hat{g}_i - \bar{g})^2.$$

In the case that none of the covariates influence the variance, i.e., heteroscedasticity does not exist, then we have $\hat{g}_i \approx \bar{g}$. The larger T is, i.e., the larger the sum of squared deviations of the estimated values $\hat{g}_i$ from the mean $\bar{g}$, the more evidence in favor of the alternative hypothesis of heteroscedastic variances. Under H_0 , the distribution of T is independent of the function h and is asymptotically χ^2 -distributed with q degrees of freedom, i.e., $T \stackrel{a}{\sim} \chi_q^2$. With a significance level of α , we reject the null hypothesis if the observed test statistic is greater than the $(1 - \alpha)$ -quantile of the χ_q^2 -distribution.

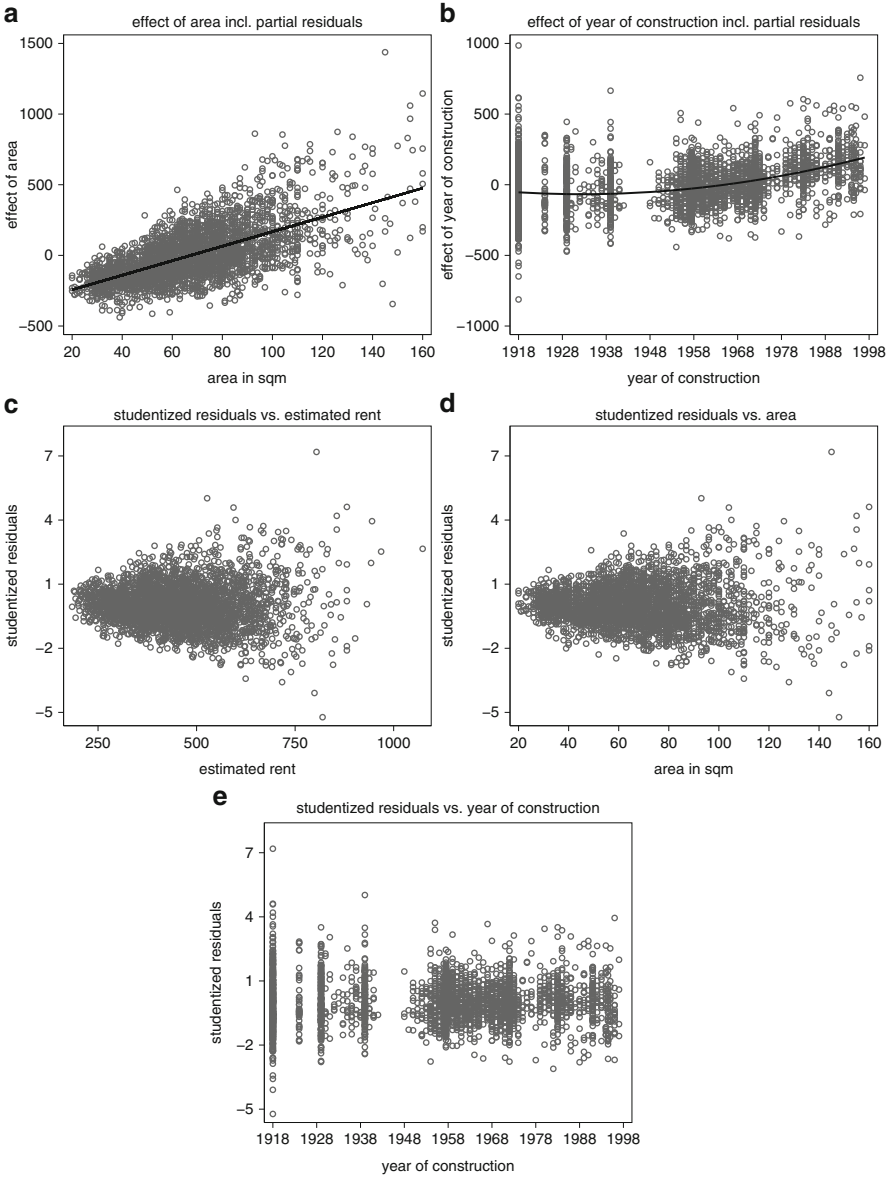


Fig. 4.1 Munich rent index: Panels (a) and (b) show the effects of living area and year of construction including partial residuals. Panels (c–e) display studentized residuals against the estimated net rent, living area, and year of construction

Example 4.2 Munich Rent Index—Breusch–Pagan Test



It is reasonable to assume that the error variances both depend (possibly nonlinearly) on living area and year of construction; see Example 4.1. We therefore assume the variance model

$$\sigma_i^2 = \sigma^2 h(\alpha_0 + \alpha_1 \text{areao}_i + \alpha_2 \text{areao2}_i + \alpha_3 \text{areao3}_i \\ + \alpha_4 \text{yearco}_i + \alpha_5 \text{yearco2}_i + \alpha_6 \text{yearco3}_i),$$

where *areao*, *areao2*, and *areao3* are cubic orthogonal polynomials (see Example 3.5 on p. 90) for living area. Based on this model, we obtain $T = 997.164$ as the Breusch–Pagan test statistic. The corresponding p-value is essentially zero so that the Breusch–Pagan test (in addition to the studentized residuals) provides further evidence for heteroscedastic variances. The Breusch–Pagan test has been carried out using function `hettest` of STATA.

△

Some concluding (critical) remarks regarding tests on heteroscedasticity are in order. For some of the readers, applying a formal test might seem more exact than the inspection of residual plots. However, in most cases, heteroscedasticity is diagnosed with exploratory techniques, while substantial scientific theory regarding the type and magnitude of heteroscedasticity almost never exists. This is the reason why we face even more uncertainty when modeling error variances in a linear model than when modeling expectations. The validity of statistical tests is extremely dependent on the correctness of models. The Breusch–Pagan test assumes, for example, multiplicative variances with exactly defined covariates. The number of covariates in the variance expression determines the distribution of the test statistic. Hence, tests on heteroscedasticity should be seen as a (heuristic) exploratory tool, similar to residual plots. These tests should by no means be the only device to diagnose heteroscedasticity. Yet, this is suggested by most econometrics textbooks, in which a battery of heteroscedasticity tests are described, while rarely is there a mention of residual plots. Residual plots should always be part of heteroscedasticity analysis, since they are in many cases the only tool to detect the *specific type* of heteroscedasticity or to determine which of the covariates influence the error variances.

Treating Heteroscedastic Variances

When diagnosing heteroscedastic errors, we should react appropriately to prevent incorrect conclusions that may result from ignoring heteroscedasticity. The most widely used approaches are described below.

Variable Transformation

When discussing the assumptions of the linear model, we have encountered models with multiplicative errors (see p. 83). A popular model with multiplicative errors is the exponential model

$$y_i = \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i) \\ = \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdot \dots \cdot \exp(\beta_k x_{ik}) \exp(\varepsilon_i).$$

If the errors are normally distributed $\varepsilon_i \sim N(0, \sigma^2)$, then $\exp(\varepsilon_i)$ and y_i are log-normally distributed. Using the variance of the log-normal distribution (see p. 641 in Appendix B), we obtain

$$\text{Var}(\exp(\varepsilon_i)) = \exp(\sigma^2) \cdot (\exp(\sigma^2) - 1)$$

and, thus

$$\text{Var}(y_i) = (\exp(\mathbf{x}'_i \boldsymbol{\beta}))^2 \exp(\sigma^2) \cdot (\exp(\sigma^2) - 1).$$

This implies that the variances of y_i are heteroscedastic in models with multiplicative errors, even though the variances of the errors are homoscedastic. As mentioned before, it is possible to log-transform the exponential model and obtain the general linear model $\log(y_i) = \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i$, with homoscedastic variances; see p. 83ff. This result provides us with a simple tool if heteroscedastic variances are diagnosed. In case of a multiplicative model, we can take a simple logarithmic transformation of the response and estimate a classical linear model using the transformed response.

Two-Stage Least Squares

In addition to the regression coefficients $\boldsymbol{\beta}$ and the variance parameter σ^2 , the weights w_i are also unknown with heteroscedastic variances. Thus, a reasonable approach is *joint* estimation of all unknown parameters. Since $E(\varepsilon_i) = 0$, we have $E(\varepsilon_i^2) = \text{Var}(\varepsilon_i) = \sigma_i^2$, and we can represent ε_i^2 as

$$\varepsilon_i^2 = \sigma_i^2 + v_i,$$

where v_i are the deviations of the squared errors from their expectations. In most cases σ_i^2 depends on one or more covariates. It seems natural to assume

$$\sigma_i^2 = \alpha_0 + \alpha_1 z_{i1} + \dots + \alpha_q z_{iq} = \mathbf{z}'_i \boldsymbol{\alpha},$$

where the vector $\mathbf{z}$ consists of all covariates that influence the variance. In many cases, the vector $\mathbf{z}$ is identical with the covariate vector $\mathbf{x}$. In order to estimate the unknown parameters $\boldsymbol{\alpha}$, we could fit a linear model using the squared errors ε_i^2 as dependent variable and $\mathbf{z}_i$ as independent variables. Since the errors are unobserved, they have to be replaced by the residuals $\hat{\varepsilon}_i = y_i - \mathbf{x}'_i \hat{\boldsymbol{\beta}}$ resulting from an unweighted regression between y and $\mathbf{x}$. As a result, we obtain the following two-stage method:

1. Obtain preliminary estimates $\hat{\boldsymbol{\beta}}$ from an unweighted regression between y and $\mathbf{x}$. Compute the residuals $\hat{\varepsilon}_i$.
2. Obtain estimates $\hat{\boldsymbol{\alpha}}$ from an unweighted regression between the squared residuals $\hat{\varepsilon}_i^2$ and the variance explanatory variables $\mathbf{z}_i$. Fit a general linear model using the weights

$$\hat{w}_i = \frac{1}{\mathbf{z}'_i \hat{\boldsymbol{\alpha}}}.$$

However, this method is not always practical, since we cannot guarantee that $\mathbf{z}_i' \hat{\boldsymbol{\alpha}}$ is greater than zero. If $\mathbf{z}_i' \hat{\boldsymbol{\alpha}} < 0$, then negative weights $\hat{w}_i$ and negatively estimated variances $\hat{\sigma}_i^2$ would result. Alternatively, we can assume the model

$$\sigma_i^2 = \exp(\mathbf{z}_i' \boldsymbol{\alpha})$$

similar to the Breusch–Pagan test. The exponential function ensures that the estimated expected variances are positive. In this model, $\boldsymbol{\alpha}$ can be estimated with the help of the regression

$$\log(\hat{\epsilon}_i^2) = \mathbf{z}_i' \boldsymbol{\alpha} + v_i.$$

The weights for the regression between y and $\mathbf{x}$ are then given by

$$\hat{w}_i = \frac{1}{\exp(\mathbf{z}_i' \hat{\boldsymbol{\alpha}})}.$$

Example 4.3 Munich Rent Index—Two-Stage Estimation



We illustrate the two-stage least squares approach with the rent index data. Recall that we diagnosed heteroscedastic variances for model (4.3) in Examples 4.1 and 4.2. We obtain the two-stage estimator for this model with the following three steps:

1. *First step: Classical linear model*

In the first step, we estimate a classical linear model; see the results in Example 4.1. Panels (a) and (b) of Fig. 4.2 show scatter plots between $\log(\hat{\epsilon}_i^2)$ and living area or year of construction. Both panels indicate that the variances σ_i^2 may depend on both covariates.

2. *Second step: Auxiliary regression*

To determine the weights $\hat{w}_i$ for the weighted regression, we estimate the regression model

$$\begin{aligned} \log(\hat{\epsilon}_i^2) = & \alpha_0 + \alpha_1 \text{area}_i + \alpha_2 \text{area}_i^2 + \alpha_3 \text{area}_i^3 \\ & + \alpha_4 \text{yearco}_i + \alpha_5 \text{yearco}_i^2 + \alpha_6 \text{yearco}_i^3 + v_i \end{aligned}$$

using ordinary least squares. Figure 4.2c,d show the estimated effects of the living area and the year of construction. The variability of the net rent appears to increase monotonically with increased living area. The effect of the year of construction is slightly S-shaped, with a considerably smaller effect than that associated with living area. The interpretation of this effect is that the variability of the net rents for older apartments is higher than for more modern apartments.

3. *Third step: Weighted linear model*

In the last step we reestimate the regression model (4.3) with the weights

$$\hat{w}_i = \frac{1}{\exp(\hat{\eta}_i)}$$

using $\hat{\eta}_i = \hat{\alpha}_0 + \hat{\alpha}_1 \text{area}_i + \dots + \hat{\alpha}_6 \text{yearco}_i^3$. We obtain the results listed in Table 4.2. The estimated standard errors are based on the estimated covariance matrix

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\mathbf{X}' \text{diag}(\hat{w}_1, \dots, \hat{w}_n) \mathbf{X})^{-1},$$

which are generally “smaller” than those of the unweighted regression. Using these corrected standard errors, we can perform tests and construct confidence intervals,

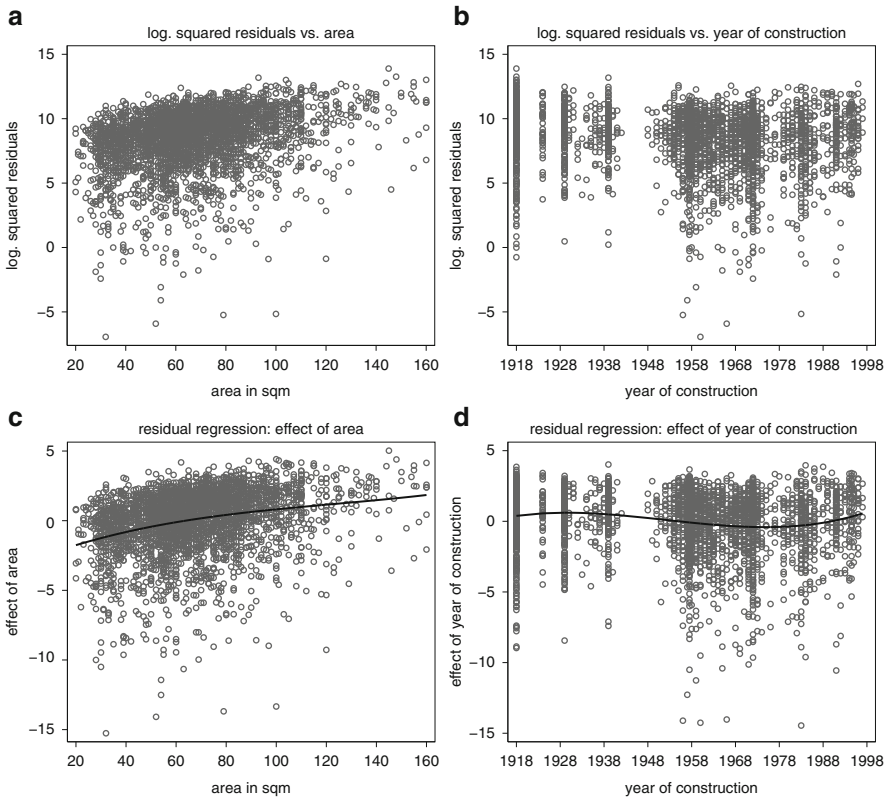


Fig. 4.2 Munich rent index: Panels (a) and (b) show scatter plots between $\log(\hat{\epsilon}_i^2)$ and area respectively year of construction. Panels (c) and (d) display effects of area and year of construction including partial residuals for the variance regression

Table 4.2 Munich rent index: estimation results for the weighted regression

Variable	Coefficient	Standard error	t-value	p-Wert	95 % Confidence interval	
<i>intercept</i>	458.840	2.608	175.900	<0.001	453.726	463.955
<i>areao</i>	117.736	2.381	49.430	<0.001	113.066	122.406
<i>yearco</i>	48.697	2.656	18.330	<0.001	43.488	53.906
<i>yearco2</i>	25.282	2.334	10.830	<0.001	20.706	29.858
<i>yearco3</i>	−1.323	2.381	−0.560	0.578	−5.991	3.344

like the ones that were developed for the classical linear model. In the present case, confidence intervals are somewhat narrower than those based on the unweighted least squares estimator. △

Simultaneous Estimation

In comparison to a two-stage method, *simultaneous estimation* of β , α , and σ^2 is more favorable, such as full ML estimation or a combination of ML estimation for

β and REML estimation for the variance parameters α and σ^2 . At this point, we omit a detailed description of these estimation approaches, as they are technically involved and software is limited. Chapter 12 of [Greene \(2000\)](#) describes in detail a maximum likelihood method for a model with multiplicative variances $\sigma_i^2 = \sigma^2 \exp(z_i' \alpha)$. An implementation exists, for example, in STATA (function `regh`). The R package `gamlss` allows full maximum likelihood estimation of Gaussian regression models with heteroscedastic variances; see also Sect. 2.9. Both the mean and the standard deviation or variance may depend on covariates. The approach even allows for semiparametric predictors as described in Chaps. 8 and 9. Details can be found in [Rigby and Stasinopoulos \(2005\)](#) and in the documentation to `gamlss`, see the web page gamlss.org.

White-Estimators

One of the prerequisites for using the methods discussed thus far is knowledge about the *type of heteroscedasticity*. An alternative approach by [White \(1980\)](#) proposes to *correct* the usual standard errors, confidence intervals, and tests associated with ordinary least squares. The correction is based on a consistent estimate of the covariance matrix (4.1) on p. 178. Under general conditions [White \(1980\)](#) shows that

$$\widehat{\text{Cov}}(\hat{\beta}) = (X'X)^{-1} X' \text{diag}(\hat{\varepsilon}_1^2, \dots, \hat{\varepsilon}_n^2) X (X'X)^{-1} \quad (4.4)$$

is a consistent estimator for the covariance matrix (4.1) of $\hat{\beta}$. The variances $\sigma_i^2 = \sigma^2/w_i$ in Eq. (4.1) are thus replaced or estimated by the squared residuals $\hat{\sigma}_i^2 = \hat{\varepsilon}_i^2$. Due to its form the matrix (4.4) is referred to as “sandwich matrix.” In Sects. 7.3.3 (p. 378) and 8.1.2 (p. 439) we will encounter similar forms for covariance matrices. The estimated robust covariance matrix for $\hat{\beta}$ can be used for computing the F-test statistic when testing general linear hypotheses. In doing so, we obtain asymptotically correct tests and confidence intervals.

The approach has the advantage that it is *not* necessary to know the weight matrix W as is the case for the other approaches. However, there is no free lunch. The price we pay for the generality is less accuracy of estimators through increased variances of the estimators.

Example 4.4 Munich Rent Index—White-Estimator

We again estimate model (4.3) on p. 183. However, this time we use the White covariance matrix (4.4) for the construction of tests and confidence intervals. Using the `regress` function of STATA with additional option `robust` we obtain the results listed in Table 4.3. The estimated standard errors tend to be much higher in comparison to the results based on an unadjusted covariance matrix, as well as in comparison to the two-stage estimation method. This should not be surprising: we have less information because we make no specific assumptions regarding the type of heteroscedasticity.

We can conclude as follows: In comparison to the classical linear model, the different variants of heteroscedastic errors provide only little qualitative differences in interpretation. The effect of the living area and the year of construction is significant in all estimated models. Moreover there is evidence that a second degree polynomial for modeling the effect of the year of construction is sufficient. Such relative robustness of results can be observed in

Table 4.3 Munich rent index: estimation results for the unweighted regression with corrected standard errors

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	459.437	2.631	174.600	<0.001	454.277	464.596
<i>areao</i>	121.817	3.681	33.090	<0.001	114.599	129.035
<i>yearco</i>	54.336	3.098	17.530	<0.001	48.260	60.412
<i>yearco2</i>	31.484	2.810	11.200	<0.001	25.974	36.994
<i>yearco3</i>	−0.198	2.919	−0.070	0.946	−5.923	5.526

many applications. Although heteroscedastic variances have been considered for estimation, we often do not find significant changes in the interpretation compared to the classical linear model.

△

Example 4.5 Prices of Used Cars—Weighted Regression

In Chap. 3 we illustrated the practical application of the classical linear model with a detailed case study on prices of used cars in the presence of covariates; see the Examples 3.19 (p. 152), 3.20 (p. 156), 3.21 (p. 158), 3.22 (p. 162), 3.23 (p. 166), and 3.24 (p. 166). Example 3.20 provided some evidence for heteroscedastic errors. We therefore applied the two-stage method outlined on p. 187 for the AIC optimal model with second-degree orthogonal polynomials of *age* and *kilometer*. For the variance equation, we used the covariates *age*, *kilometer*, and *TIA*. We modeled the effects of the age and the kilometer reading using third-degree polynomials. Overall, we found no significant differences relative to unweighted regression.

△

4.1.4 Autocorrelated Errors

In addition to heteroscedastic errors, correlated errors are one of the main reasons why the assumptions of the classical linear model are violated in applications. Section 3.1.2 (p. 80) pointed out that autocorrelated errors may arise if the model is not correctly specified. We mentioned the following reasons for misspecified models:

- *Misspecified covariate effect:* The effect of an explanatory variable is not modeled correctly. The examples in Sect. 3.1.3 could prove helpful in this respect as they discuss various approaches for nonlinear modeling of continuous covariate effects.
- *Omitted variables:* Relevant explanatory variables cannot be observed and consequently cannot be considered in the model. In relation to autocorrelated errors, this problem occurs especially with time series, panel, or longitudinal data, especially if the missing explanatory variables show a temporal trend. In this case, we can use the estimation methods provided in this section in order to achieve more precise estimates and better prediction.

First-Order Autocorrelation

We restrict ourselves to one of the simplest and most frequently used error process, first-order autocorrelation. More specifically, we assume that the errors follow an autoregressive process of first-order (AR(1)) with

$$\varepsilon_i = \rho \varepsilon_{i-1} + u_i,$$

where $-1 < \rho < 1$. Regarding the u_i , we assume:

1. $E(u_i) = 0$
2. $\text{Var}(u_i) = E(u_i^2) = \sigma_u^2, \quad i = 1, \dots, n$
3. $\text{Cov}(u_i, u_j) = E(u_i u_j) = 0, \quad i \neq j$

Related to sample size, we additionally assume that the data are sufficiently historical in nature. Based on these assumptions, a specific linear model with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and with covariance matrix

$$\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1} = \frac{\sigma_u^2}{1 - \rho^2} \begin{pmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{pmatrix} \quad (4.5)$$

results; see Sect. 4.5.2 on p. 258 for a derivation.

With the help of the covariance matrix, we are able to calculate the correlation coefficients between the errors ε_i and the errors ε_{i-j} lagged by j periods. We obtain the *autocorrelation function*

$$\text{ACF}(j) = \frac{\text{Cov}(\varepsilon_i, \varepsilon_{i-j})}{\text{Var}(\varepsilon_j)} = \rho^j \quad j = 0, 1, 2, \dots$$

An obvious characteristic of first-order autocorrelation is a slow decay of the correlation between ε_i and ε_{i-j} as the lag j increases. If ρ is positive, the correlation decreases geometrically. If ρ is negative, the correlation decreases with alternating signs. This is illustrated in the left column of Fig. 4.3, which shows the autocorrelation function for different values of ρ . These graphical representations are usually referred to as *correlograms*.

In addition to the autocorrelation function, the *partial autocorrelation function* is another tool to characterize correlated errors or, more generally, stochastic processes. The partial autocorrelation PACF(j) between ε_i and ε_{i-j} is defined as the regression coefficient α_j in the model

$$\varepsilon_i = \alpha_1 \varepsilon_{i-1} + \dots + \alpha_j \varepsilon_{i-j} + v_i. \quad (4.6)$$

We have $\text{PACF}(1) = \text{ACF}(1)$, independent of the order of the autocorrelation. For first-order autocorrelated errors, we have $\text{PACF}(1) = \rho$ and $\text{PACF}(j) = 0$ for $j > 1$ by definition of the AR(1) process. The right column of Fig. 4.3 shows the

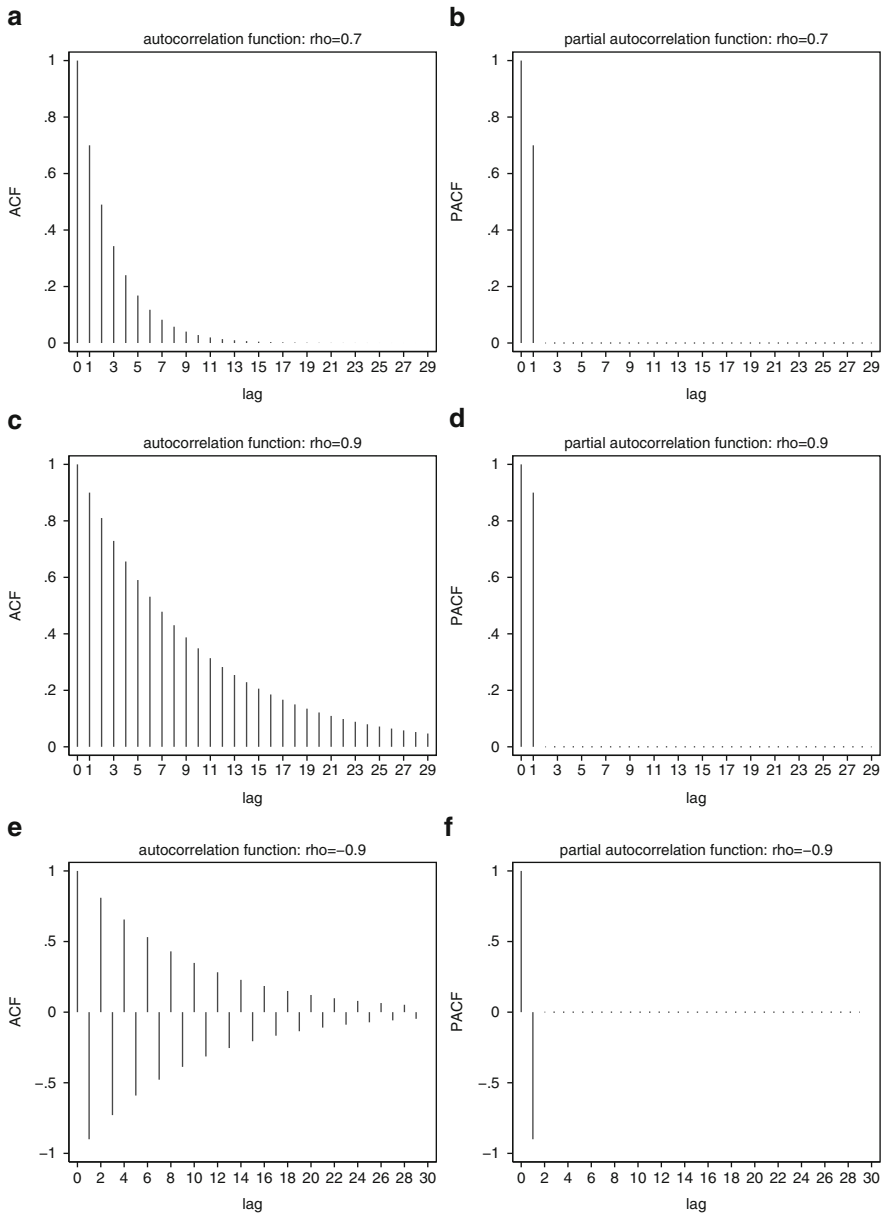


Fig. 4.3 Autocorrelation function and partial autocorrelation function for autocorrelated errors of first order. The functions for $\rho = 0.7, 0.9, -0.9$ are shown

partial autocorrelation function for some AR(1) processes. A characteristic property is the abrupt decline of the partial autocorrelation for $j > 1$. We can interpret the coefficient α_j in Eq. (4.6) as the correlation coefficient between $\varepsilon_i - \alpha_1 \varepsilon_{i-1} - \dots - \varepsilon_{i-j+1} \alpha_{j-1}$ and ε_{i-j} . The terminology partial correlation coefficient is due to the fact that we do not simply calculate the correlation between ε_i and ε_{i-j} , but rather the correlation between ε_i and ε_{i-j} while controlling the effect of intermediate errors.

Diagnosing Autocorrelated Errors

Similar to the diagnosis of heteroscedastic errors, we first rely on the residuals from a classical linear model. The following diagnostic tools work well in practice:

Graphical Illustration of the Residuals Over Time

A straightforward tool for diagnosing correlated errors are residual plots, more specifically scatter plots of residuals or studentized residuals over time. If a positive (negative) residual is followed in tendency by a positive (negative) residual the errors are positively autocorrelated. Likewise, if in tendency residuals with alternating signs are observed, we have negative autocorrelation.

Empirical Autocorrelation Function

Other useful instruments for diagnosing autocorrelation are the empirical autocorrelation and the partial autocorrelation functions along with their display in correlograms. The empirical autocorrelation function is an estimate for $\text{ACF}(j)$ and is given by

$$\widehat{\text{ACF}}(j) = \frac{\widehat{\text{Cov}}(\varepsilon_i, \varepsilon_{i-j})}{\widehat{\text{Var}}(\varepsilon_i)} \quad \text{with} \quad \widehat{\text{Cov}}(\varepsilon_i, \varepsilon_{i-j}) = \frac{1}{n} \sum_{i=j+1}^n \hat{\varepsilon}_i \hat{\varepsilon}_{i-j}.$$

Basically, the estimates are empirical correlation coefficients between the residuals and the lagged residuals by j periods.

The partial autocorrelations are obtained by repeatedly estimating the regression model (4.6) for $j = 1, 2, 3, \dots$. This yields $\widehat{\text{PACF}}(j) = \hat{\alpha}_j$, thereby substituting the errors ε_i in Eq. (4.6) with $\hat{\varepsilon}_i$. By inspecting both the empirical correlogram and the partial correlogram, we obtain information about existing autocorrelation. If all empirical autocorrelations and partial autocorrelations are near zero, we can reasonably assume uncorrelated errors. If the empirical autocorrelations are similar to the theoretical (partial) autocorrelation function of AR(1)-errors, then we have evidence for the existence of first-order autocorrelation. If the correlograms do not follow the typical form of an AR(1)-process, then there is evidence of a more complex correlation structure. Such error structures go beyond the scope of this book and are examined in the literature on statistical time series analysis; see, e.g., [Hamilton \(1994\)](#).

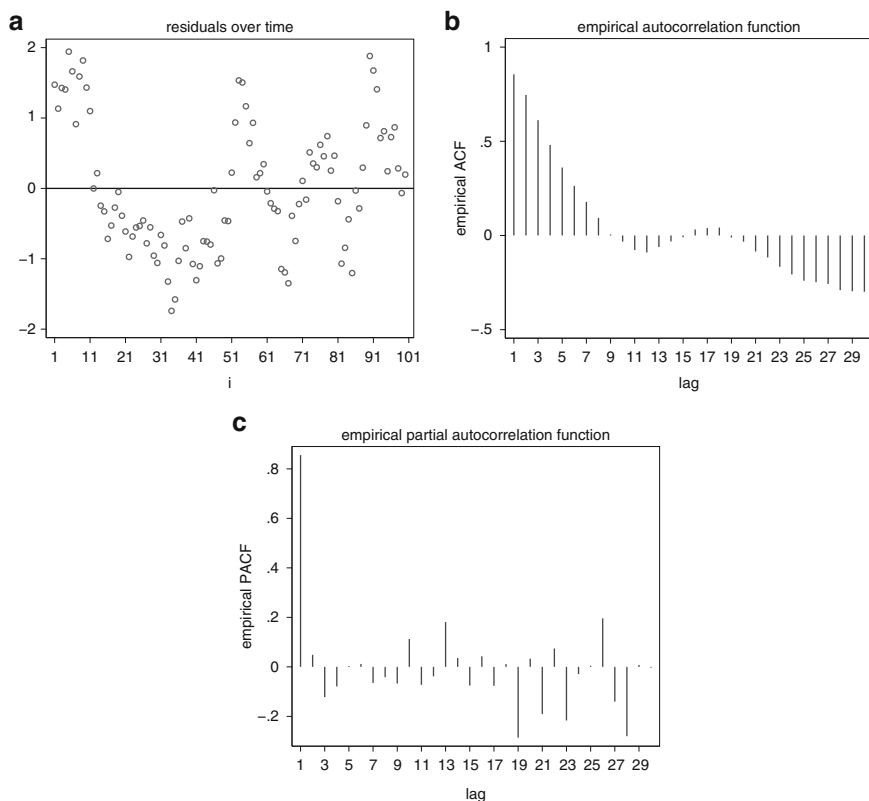


Fig. 4.4 Residuals over time (a), empirical autocorrelation function (b), and empirical partial autocorrelation function (c) based on simulated data with first-order autocorrelation

Example 4.6 Simulated Data—Graphical Diagnosis of Autocorrelated Errors

For illustration purposes, we examine the simulated regression model with positive autocorrelation, as shown in Fig. 3.3 (p. 81). We simulated the model $y_i = -1 + 2x_i + \varepsilon_i$ with $\varepsilon_i = 0.9\varepsilon_{i-1} + u_i$. To show how autocorrelated errors are diagnosed using residual plots and correlograms, we assume a classical linear model for the data. We obtain the estimated model $\hat{y}_i = -1.08 + 2.13x_i$. Figure 4.4 shows the residuals over time and the empirical (partial) autocorrelation function. Note that the residual plot in panel (a) differs slightly from the errors over time as presented in Fig. 3.3b, since the residuals are only estimates of the true errors. The plots display almost perfect first-order autocorrelation. The residuals are highly correlated, the empirical autocorrelation decreases geometrically, and the partial autocorrelations are almost zero for $j > 1$.

△

Tests for Autocorrelation—Durbin–Watson Test

In addition to graphical techniques, we can also use statistical tests to uncover autocorrelation. In this context, a test for serial correlation developed by Durbin

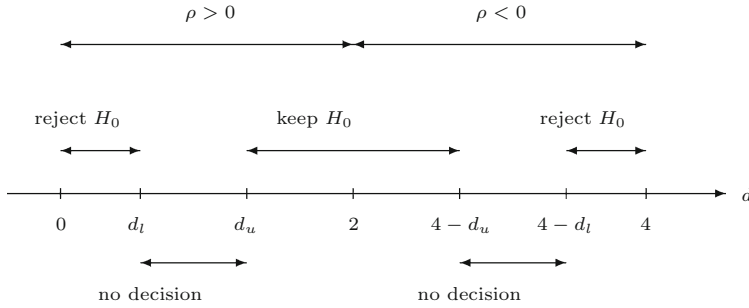


Fig. 4.5 Acceptance and rejection ranges for the Durbin–Watson test

and Watson (1950, 1951, 1971) is widely used. The Durbin–Watson test considers the hypothesis

$$H_0 : \rho = 0 \quad \text{versus} \quad H_1 : \rho \neq 0$$

and is based on the test statistic

$$d = \frac{\sum_{i=2}^n (\hat{\varepsilon}_i - \hat{\varepsilon}_{i-1})^2}{\sum_{i=1}^n \hat{\varepsilon}_i^2}.$$

The rationale behind the test statistic is as follows: For large sample size n we obtain

$$d = \frac{\sum_{i=2}^n \hat{\varepsilon}_i^2 + \sum_{i=2}^n \hat{\varepsilon}_{i-1}^2 - 2 \sum_{i=2}^n \hat{\varepsilon}_i \hat{\varepsilon}_{i-1}}{\sum_{i=1}^n \hat{\varepsilon}_i^2} \approx 1 + 1 - 2\hat{\rho} = 2(1 - \hat{\rho}).$$

Since $-1 < \hat{\rho} < 1$, we have $0 < d < 4$. If d is close to 2, then $\hat{\rho} \approx 0$, and we fail to reject the null hypothesis. The closer the test statistic d is to either 0 or 4, the closer is $\hat{\rho}$ to 1 or -1 , and we then generally reject the null hypothesis.

The distribution of the test statistic under H_0 is relatively difficult to obtain, since it depends on the design matrix. Hence, a test decision is sometimes difficult. Durbin and Watson partially solved the problem, and were able to provide a decision for certain interval ranges of d . The intervals depend on some lower and upper limits d_l and d_u , which vary for different sample size n and numbers of regressors k . Figure 4.5 provides a graphical presentation of the acceptance and rejection ranges for the Durbin–Watson test.

Nowadays, it has become more routine to obtain the p-values for the Durbin–Watson test numerically using statistical software. An implementation can be found, for instance, in the function `dwtest` of `matlab`. The R function `dwtest` of the package *lmtest* turned out to be erroneous in the following example.

Example 4.7 Simulated Data—Durbin–Watson Test

For the data used in Example 4.6, we obtain the test statistic $d = 0.2619$ and a very small p-value with the `matlab` function `dwtest`. Thus the Durbin–Watson test, in addition to the previously presented graphical diagnostics, provides clear evidence of first-order autocorrelation. △

Treating First-Order Autocorrelated Errors

As already pointed out, autocorrelated errors are an indicator for model misspecification. Hence, we should first examine whether or not it is possible to correct the specification problem. Possible improvements can result from the inclusion of other covariates into the model or nonlinear modeling of some continuous covariates.

If these attempts fail to correct for the correlation, we can use estimation procedures for models with autocorrelated errors. In a model with first-order autocorrelated errors, we need estimates for the regression coefficients β , the variance parameter σ^2 , and the correlation coefficient ρ . The statistical literature provides a wealth of methods to estimate these parameters. We first present a two-stage procedure. In the first step, the correlation coefficient is estimated using the ordinary least squares estimates for the regression coefficients. The regression coefficients are then re-estimated with the help of WLS. As an alternative to the two-stage procedure, we briefly describe maximum likelihood estimation. An overview of alternative approaches is provided by [Judge et al. \(1980\)](#).

Two-Stage Estimation

We estimate a model with autocorrelated errors as follows:

1. Obtain a preliminary estimate $\hat{\beta}$ for β using the unweighted regression between y and x . Denote the resulting residuals with $\hat{\varepsilon}_i$.
2. Obtain an estimate $\hat{\rho}$ of the correlation coefficient through

$$\hat{\rho} = \frac{\sum_{i=2}^n \hat{\varepsilon}_i \hat{\varepsilon}_{i-1}}{\sqrt{\sum_{i=2}^n \hat{\varepsilon}_i^2} \sqrt{\sum_{i=2}^n \hat{\varepsilon}_{i-1}^2}}, \quad (4.7)$$

where $\hat{\rho}$ is the empirical correlation coefficient between $\hat{\varepsilon}_i$ and $\hat{\varepsilon}_{i-1}$.

3. Insert $\hat{\rho}$ into the weight matrix W and obtain the estimate $\hat{W}$. Use $\hat{W}$ to re-estimate the regression coefficients using WLS.

To improve the estimates, it is helpful to iterate steps 2 and 3. This method is known as Prais–Winsten estimator. Under quite general conditions, it can be shown that the resulting estimator for β is consistent. In addition to the method by Prais and Winsten, there exist numerous modifications; see [Greene \(2000\)](#).

Table 4.4 Prais–Winsten estimator for the simulated data

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	−0.988	0.308	−3.200	0.002	−1.600	−0.376
<i>x</i>	2.095	0.152	13.74	<0.001	1.792	2.397

Table 4.5 ML estimator for the simulated data

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	−0.982	0.307	−3.200	0.001	−1.583	−0.380
<i>x</i>	2.094	0.139	15.000	<0.001	1.821	2.368

Maximum Likelihood Estimation

For maximum likelihood estimation, we need the additional assumption of normally distributed errors. The likelihood is then given by

$$L(\boldsymbol{\beta}, \sigma^2, \rho) = \frac{1}{(2\pi\sigma^2)^{n/2} |\mathbf{W}^{-1}|^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{W}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right). \quad (4.8)$$

At first we calculate

$$\mathbf{W} = \begin{pmatrix} 1 & -\rho & 0 & \dots & 0 & 0 \\ -\rho & 1 + \rho^2 & -\rho & \dots & 0 & 0 \\ 0 & -\rho & 1 + \rho^2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 + \rho^2 & -\rho \\ 0 & 0 & 0 & \dots & -\rho & 1 \end{pmatrix}.$$

and

$$|\mathbf{W}^{-1}| = \frac{1}{1 - \rho^2}.$$

Then the log-likelihood becomes

$$l(\boldsymbol{\beta}, \sigma^2, \rho) = -\frac{n}{2} \log(\sigma^2) + \frac{1}{2} \log(1 - \rho^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{W}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}),$$

which can not be maximized in closed form. We rather have to compute the ML estimator iteratively. Details are provided in [Greene \(2000\)](#) and in the literature cited therein.

Example 4.8 Simulated Data—ML Estimators

For the simulated data of the preceding examples, Table 4.4 contains the Prais–Winsten estimator and Table 4.5 the maximum likelihood estimator. Both estimators are in close agreement. We used the STATA functions `prais` for the Prais–Winsten estimator and `arima` for the ML estimator.

△

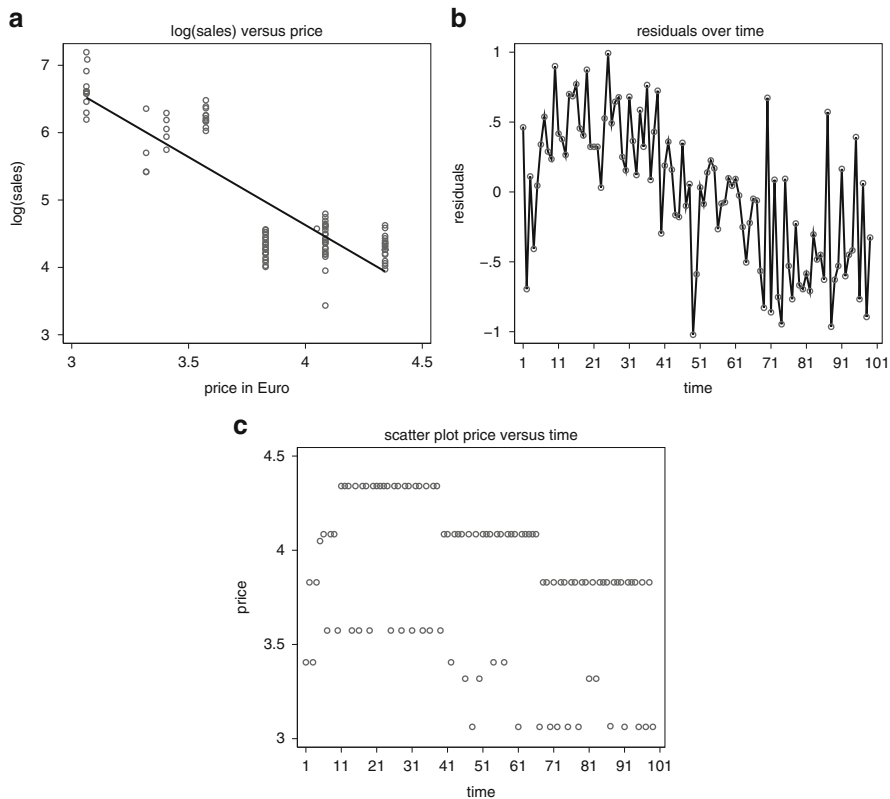


Fig. 4.6 Supermarket scanner data: scatter plot between log-sales and own price including fitted regression line [panel (a)]. Panel (b) shows the residuals over time. Panel (c) displays the price of the coffee brand over time

Example 4.9 Supermarket Scanner Data

For illustration of multiplicative errors, we examined the scanner data in Example 3.2 (p. 86), which are obtained from payments processed at the supermarket check-out counter. The goal is to model the relationship between the weekly sales of a product (in this case a particular coffee brand) and the own price or the price of competing brands (cross price effects); see the scatter plots in Fig. 3.6 (p. 85). In this example, we analyze the data to illustrate the approaches for correlated errors. To simplify matters, we use the data for just one store (Fig. 3.6 on p. 85 contains data of five stores). Due to the multiplicative error structure, we use log-sales rather than original responses. Figure 4.6a shows the scatter plot between log-sales and own price. Additionally, a fitted regression line is included. Panel (b) shows the corresponding residuals over time, which are clearly correlated. The Durbin-Watson test statistic is $d = 1.1751$ (p-value $< 10^{-4}$). The p-value is based on the matlab function `dwtest`. The p-value indicates that the null hypothesis $H_0 : \rho = 0$ can be rejected at the $\alpha = 0.05$ and $\alpha = 0.01$ level. In Fig. 4.7, we additionally examine the empirical autocorrelation and partial autocorrelation function of the residuals. The plots

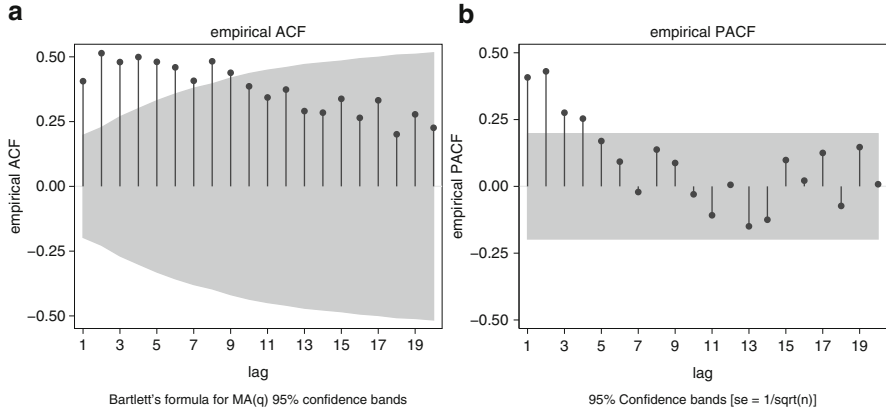


Fig. 4.7 Supermarket scanner data: empirical and partial autocorrelation function of the residuals in the regression between log-sales and price

additionally contain pointwise 95 % confidence intervals (their derivation is omitted at this point). The correlogram provides further evidence for autocorrelated errors, as it decreases slowly with increasing lag. The partial autocorrelation function is relatively small from a lag of 3 onward. Consequently, the correlograms do not point to first-order autocorrelation and a more complex correlation structure should be taken into account (e.g., second-order autocorrelation). We see the benefit here of graphical devices in comparison to formal tests. The Durbin–Watson test provides evidence for autocorrelation, but is not useful to determine the type of correlation.

To correct for autocorrelated errors, we should first examine whether or not an improved model specification eliminates correlations. Estimating a model with autocorrelated errors should be the last resort. In the current example, looking at the prices over time [panel (c) of Fig. 4.6] provides clear evidence how to improve the model. We can identify three time periods with different “regular” sales price of the coffee brand. Initially, the regular price was about 4.35 Euro, then the regular price was further decreased to 4.10 Euro, and finally to 3.8 Euro. During all periods we regularly encounter price reductions to promote the sales of the coffee brand. Clearly, ignoring the temporal development of regular prices is problematic. The marketing literature proposes to replace the price as explanatory variable with the ratio between the actual and the current regular price for the time period; see, for example, [Leeftang, Wittink, Wedel, and Naert \(2000\)](#). We therefore define the new covariate

$$priceq = \frac{\text{actual price}}{\text{regular price}},$$

which takes values between zero and one. If $priceq = 1$, the actual price is equal to the regular price, implying that there was no price reduction during the week. Figure 4.8a shows the scatter plot between log-sales and the ratio resulting from the actual and regular price including the fitted regression line. The residuals over time in panel (b) are now approximately uncorrelated. This is also supported by the Durbin–Watson test with test statistic $d = 1.5946$ (p-value = 0.04559). We can conclude that the Durbin–Watson test does not give us any compelling evidence of autocorrelation.

△

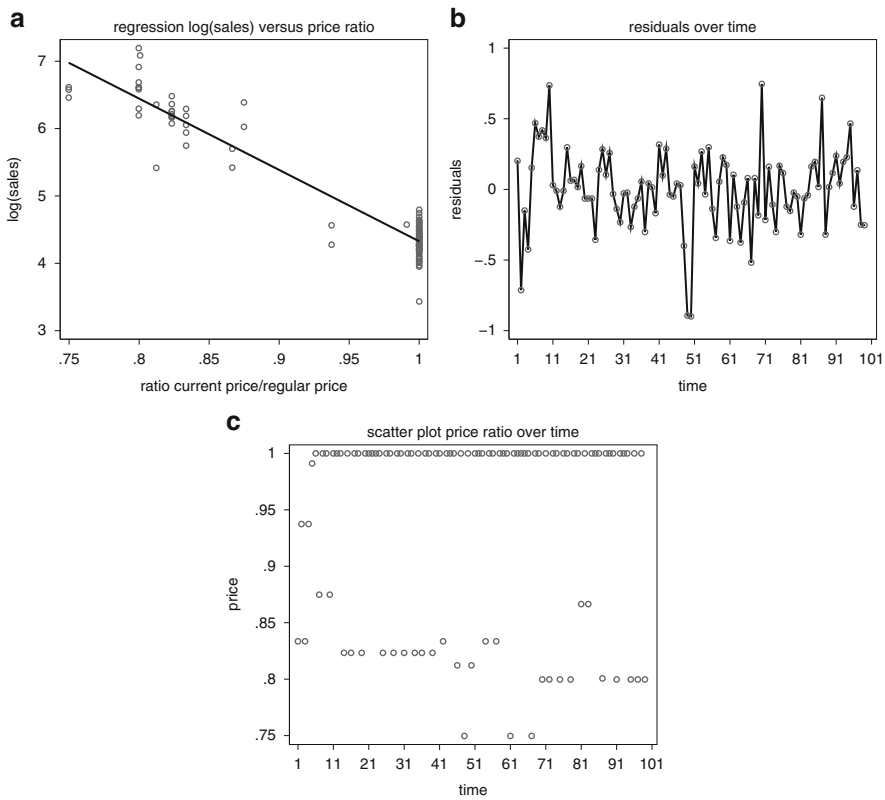


Fig. 4.8 Supermarket scanner data: scatter plot between log-sales and the price ratio actual/regular price including fitted regression line [panel (a)]. Panels (b) and (c) show the residuals and the price ratio over time

4.2 Regularization Techniques

This section deals with another extension of the classical linear model. We present regularization techniques to “regularize” the least squares estimator. This is useful in situations where the least squares estimator is numerically instable leading to (very) high variances. Moreover, regularization techniques provide an alternative to the variable selection procedures discussed in Sect. 3.4.

Throughout this section we assume $\text{Cov}(\epsilon) = \sigma^2 \mathbf{I}$ rather than $\text{Cov}(\epsilon) = \sigma^2 \mathbf{W}^{-1}$ as in the preceding section. However, generalizations of the presented regularization techniques to the general covariance matrix are straightforward.



4.2.1 Statistical Regularization

To compute the least squares estimator in the classical linear model, the system of equations

$$X'X\beta = X'y \quad (4.9)$$

has to be solved with respect to β . Thus far, we have assumed that X has full rank p so that a unique solution exists. However, when X contains columns that are close to collinear or if the number of regression effects p grows large, computing the solution will become numerically instable, even if the coefficients are still identifiable in theory. Moreover, modern applications arising, for example, in genetics often involve situations where the number of covariates exceeds (sometimes by orders of magnitude) the number of observations. In the statistical community, such problems are commonly referred to as “small n , large p ” problems. For example, in studies of gene expression, the expression level of thousands of genes may be measured for only a few dozen individuals. In such situations, regularization techniques are commonly applied to obtain estimates of regression coefficients even in situations where the coefficient matrix $X'X$ in Eq. (4.9) is close to or exactly singular.

The notion of regularization summarizes approaches that make a problem “look nicer than it actually is” by imposing specific restrictions on the set of admissible solutions. In fact, this is closely related to the concept of Bayesian models where prior knowledge on the regression coefficients is incorporated (see Sect. 4.4). Regularization is typically implemented by considering a penalized least squares objective function

$$\text{PLS}(\beta) = (y - X\beta)'(y - X\beta) + \lambda \cdot \text{pen}(\beta) \rightarrow \min_{\beta},$$

where $\text{pen}(\beta)$ is a penalty term measuring the complexity of the vector of regression coefficients and $\lambda \geq 0$ is a smoothing parameter governing the impact of the penalty. In principle, $\text{pen}(\beta)$ will be constructed so that it is large when many of the entries in β are large. Another related principle would be to consider sparsity penalties that, for example, count the number of nonzero entries in β and therefore enforce sparseness of the resulting coefficient vector. The smoothing parameter then determines the trade-off between fidelity to the data as measured by the least squares criterion (λ small) and the impact of the penalty (λ large).

Using mathematical theory for optimization under constraints, one can show that obtaining the penalized least squares estimator

$$\hat{\beta}_{\text{PLS}} = \arg \min_{\beta} [(y - X\beta)'(y - X\beta) + \lambda \cdot \text{pen}(\beta)]$$

is equivalent to seeking the solution

$$\hat{\beta}_{\text{PLS}} = \arg \min_{\beta} (y - X\beta)'(y - X\beta)$$

subject to the constraint

$$\text{pen}(\beta) \leq t,$$

where t is a constant related to the smoothing parameter λ in a one-to-one relationship. A derivation of this well-known fact in optimization theory can be found, e.g., in [Sydsaeter, Hammond, Seierstad, and Strom \(2005\)](#).

4.2.2 Ridge Regression

In Sect. 3.4.4 (p. 159), we already introduced the first statistical regularization approach known as ridge regression. It corresponds to the classical approach of Tikhonov regularization achieved by adding the (squared) L_2 -norm of the solution to an optimality criterion. In case of linear regression, the resulting penalty is simply given by the sum of the squared coefficients so that

$$\text{pen}(\beta) = \sum_{j=0}^k \beta_j^2 = \beta' \beta$$

and

$$\text{PLS}(\beta) = (y - X\beta)'(y - X\beta) + \lambda \beta' \beta.$$

Proceeding in analogy to the derivation of the least squares estimator in Sect. 3.2.1 (p. 105), we first take the derivative of $\text{PLS}(\beta)$ with respect to β and obtain

$$\begin{aligned} \frac{\partial}{\partial \beta} \text{PLS}(\beta) &= \frac{\partial}{\partial \beta} (y'y - 2y'X\beta + \beta'X'X\beta + \lambda \beta' \beta) \\ &= -2X'y + 2X'X\beta + 2\lambda \beta. \end{aligned}$$

Setting the derivative equal to zero and solving for β yields the ridge regularized penalized least squares estimator

$$\hat{\beta}_{\text{PLS}} = (X'X + \lambda I_p)^{-1} X'y,$$

i.e., $\hat{\beta}_{\text{PLS}}$ only differs from $\hat{\beta}_{\text{LS}}$ by the additional term λI_p that arises from the penalty. For values of the smoothing parameter close to zero, the impact of this additional term practically vanishes and we end up with a solution that is close to the usual least squares estimator. However, if λ is large, $X'X + \lambda I_p$ will be invertible, even if $X'X$ does not have full rank. Moreover, the solution $\hat{\beta}_{\text{PLS}}$ will be shrunken towards zero. This is most easily seen from the penalized least squares criterion: When the smoothing parameter λ grows large, the penalty term will completely determine the optimization problem so that the optimal solution has to minimize the penalty term, i.e., $\text{pen}(\beta) = \beta' \beta$. This will clearly be the case for $\beta = 0$.

Typically, penalization of the intercept is not desired in ridge regression so that β_0 should be excluded from the penalty term. This can be achieved in two different ways. A first, popular approach is to center all covariates and the responses so that $\bar{y} = 0$ and $\bar{x} = \mathbf{0}$ which automatically results in $\hat{\beta}_0 = 0$. This implies that the intercept can simply be dropped from the model and therefore is also not penalized. A second approach is to modify the penalty to

$$\text{pen}(\boldsymbol{\beta}) = \sum_{j=1}^k \beta_j^2 = \boldsymbol{\beta}' \mathbf{K} \boldsymbol{\beta},$$

where $\mathbf{K} = \text{diag}(0, 1, \dots, 1) = \text{blockdiag}(0, \mathbf{I}_k)$, i.e., we introduce a *penalty matrix* that excludes the intercept but remains the identity matrix for the rest of the coefficient vector. The ridge regression estimate is then given by

$$\hat{\boldsymbol{\beta}}_{\text{PLS}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1} \mathbf{X}'\mathbf{y},$$

i.e., it has exactly the same structure as before but replaces the identity matrix $\mathbf{I}_p$ with the penalty matrix $\mathbf{K}$. The second approach has the advantage that it can still be used in more complex models when ridge regression shall be combined, for example, with nonlinear regression as introduced in Chap. 8. We therefore follow the second approach in the rest of this section.

The optimality properties derived for the least squares estimator included unbiasedness, i.e.,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}_{\text{LS}}) = \mathbb{E}((\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta},$$

and the covariance matrix of the least squares estimator is given by

$$\text{Cov}(\hat{\boldsymbol{\beta}}_{\text{LS}}) = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}.$$

In contrast, the penalized least squares estimator is biased since

$$\mathbb{E}(\hat{\boldsymbol{\beta}}_{\text{PLS}}) = \mathbb{E}((\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1} \mathbf{X}'\mathbf{y}) = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1} \mathbf{X}'\mathbf{X} \boldsymbol{\beta},$$

as the matrices $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}$ and $\mathbf{X}'\mathbf{X}$ do not cancel, unless $\lambda = 0$. Most of the coefficients in $\boldsymbol{\beta}_{\text{PLS}}$ will typically be shrunk towards zero as compared to the coefficients in $\boldsymbol{\beta}_{\text{LS}}$, i.e.,

$$|\hat{\beta}_{j,\text{PLS}}| \leq |\hat{\beta}_{j,\text{LS}}|, \quad j = 1, \dots, k.$$

This relation holds exactly for orthogonal designs, where $\mathbf{X}'\mathbf{X}$ is a diagonal matrix, and as such all coefficients can be estimated independently (but may be violated for some coefficients in more general settings).

To motivate the shrinkage effect of ridge regression, note first that $\lambda \mathbf{K} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K}) - \mathbf{X}'\mathbf{X}$ is positive semidefinite (for $\lambda > 0$), i.e., “ $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})$ is larger than (or equal to) $\mathbf{X}'\mathbf{X}$ ” in a matrix sense, since a nonnegative amount $\lambda \mathbf{K}$ is added to $\mathbf{X}'\mathbf{X}$. Multiplying two positive semidefinite matrices again yields a positive semidefinite matrix. Therefore, multiplying $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}$ with $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K}) - \mathbf{X}'\mathbf{X}$ yields positive semidefinite $\mathbf{I}_p - (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}\mathbf{X}'\mathbf{X}$. In particular, this implies that the diagonal elements of $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}\mathbf{X}'\mathbf{X}$ are smaller than (or equal to) one because the diagonal elements of a positive semidefinite matrix are always greater than (or equal to) zero (see Theorem A.28.3 of Appendix A.7). Re-expressing the representation of the penalized least squares estimator as

$$\hat{\boldsymbol{\beta}}_{\text{PLS}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{LS}},$$

indicates that we can obtain $\hat{\boldsymbol{\beta}}_{\text{PLS}}$ from $\hat{\boldsymbol{\beta}}_{\text{LS}}$ through premultiplication with a matrix that is “smaller than $\mathbf{I}_p$.” For the special case of an orthogonal design matrix, both $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{X} + \lambda \mathbf{K}$ (and therefore also $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}$) are diagonal matrices so that we can write

$$\hat{\beta}_{j,\text{PLS}} = \frac{(\mathbf{x}^j)'\mathbf{x}^j}{(\mathbf{x}^j)'\mathbf{x}^j + \lambda} \hat{\beta}_{j,\text{LS}}, \quad j = 1, \dots, k,$$

where $\mathbf{x}^j$ denotes the j th column of $\mathbf{X}$. Since the factor $(\mathbf{x}^j)'\mathbf{x}^j / ((\mathbf{x}^j)'\mathbf{x}^j + \lambda)$ is smaller than one, we immediately obtain the shrinkage effect of ridge regression estimates.

The covariance matrix for the ridge regression estimator is given by

$$\text{Cov}(\hat{\boldsymbol{\beta}}_{\text{PLS}}) = \sigma^2(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1}.$$

Proceeding analogously, one can show that

$$\text{Cov}(\hat{\boldsymbol{\beta}}_{\text{LS}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{\text{PLS}})$$

is positive definite (for $\lambda > 0$), i.e., the covariance matrix of the penalized least squares estimator is smaller than the covariance matrix of the least squares estimator. In particular, this implies that

$$\text{Var}(\hat{\beta}_{j,\text{PLS}}) < \text{Var}(\hat{\beta}_{j,\text{LS}}), \quad j = 1, \dots, k,$$

since for a positive definite matrix the diagonal elements are positive.

In summary, we find that the penalized least squares estimator is biased, but has a smaller variance than the least squares estimator. When combining bias and covariance to the mean squared error one hopes to achieve a smaller MSE when choosing an appropriate smoothing parameter λ . In particular, if $\mathbf{X}$ is close to

collinear or if X is high-dimensional, the variance usually dominates the mean squared error and thus regularization is desirable. Often, ridge regression is also considered as a possibility to further shrink small coefficients towards zero. In particular, this may give a clearer indication of which variables are actually required in the model so that variable selection can be achieved.

The smoothing parameter is typically determined using r -fold cross validation; see Sect. 3.4, p. 149. Therefore, the data set is split into r parts of approximately equal size. Estimation is performed for a given grid of smoothing parameters on $r - 1$ parts of the data set, while prediction based on these estimates is obtained for the last remaining part (holdout sample). Cycling through the different splits an average prediction error can be obtained for a given smoothing parameter. The “optimal” smoothing parameter can be defined as the one that minimizes the average prediction error on the hold-out samples.

Finally, the scaling of covariates is important when applying a regularization approach. The penalty formed of squared regression coefficients assumes that all coefficients can be compared in their absolute value. However, the scaling of covariates has immediate impact on the interpretation of these absolute values. For example, the coefficient associated with a covariate measuring a distance will be scaled by a factor of 1,000 when the variable is measured in meters instead of kilometers. Hence it is important to make all variables comparable in their scaling before applying a penalized least squares approach. A common solution is to transform all covariates to have zero mean and unit standard deviation, i.e., to standardize all covariates.

Example 4.10 Price of Used Cars—Ridge Regression



We illustrate the impact of ridge regularization using the data on prices of used cars introduced in Example 3.19 (p. 152). Instead of performing a formal variable selection, we will use a relatively complex model comprising all covariates and also higher-order polynomials. Often such a model can yield rather large uncertainty in the coefficient estimates. As a remedy, we apply ridge regularization that effectively reduces the variance while introducing some bias in the coefficient estimates. Estimation is based on the R package `penalized` and in particular function `optL2`.

For the variables representing the age of the car in months (*age*) and kilometer reading in 1,000 km (*kilometer*), we consider orthogonal polynomials of degree three (see Example 3.5 on p. 90) to allow for potential nonlinearity while the remaining covariates, corresponding to months until the next appointment with the technical inspection agency (*TIA*), ABS brake (*extras1*), and sunroof (*extras2*), are included with linear effects. The second column in Table 4.6 represents the least squares estimates obtained for this model specification. Note that our analyses in Example 3.19 have shown that some of the variables may not actually be needed to adequately predict the prices of used cars. As a consequence, regularization in terms of a ridge penalty may be applied with the aim to shrink such redundant parameters towards zero.

Figure 4.9 illustrates the impact of the ridge penalty for a variety of smoothing parameter values λ . On the left-hand side, the graph starts with a rather large value of the smoothing parameter yielding strong shrinkage towards zero. When moving to the right (corresponding to a decrease in the smoothing parameter), the coefficient estimates approach the least squares estimate for $\lambda \rightarrow 0$. The surprising scaling of the x -axis with large values on the left and small values on the right is motivated by the fact that many graphics represent coefficient paths as functions of the constraint parameter t instead of the smoothing

Table 4.6 Prices of used cars: estimated coefficients for different models

Variable	LS	Ridge $\lambda = 6.43$	Ridge $\lambda \approx 250$	LASSO $\lambda = 10.34$
<i>intercept</i>	3.585	3.580	3.569	3.421
<i>ageop1</i>	−0.709	−0.672	−0.594	−0.682
<i>ageop2</i>	0.172	0.164	0.146	0.150
<i>ageop3</i>	0.016	0.015	0.013	−
<i>kilometerop1</i>	−0.437	−0.425	−0.393	−0.412
<i>kilometerop2</i>	0.142	0.138	0.128	0.110
<i>kilometerop3</i>	0.009	0.010	0.010	−
<i>TIA</i>	−0.005	−0.005	−0.004	−
<i>extras1</i>	−0.114	−0.104	−0.086	−0.036
<i>extras2</i>	−0.031	−0.042	−0.060	−

Second column: least squares estimate; third column: ridge regression with estimated smoothing parameter based on cross validation; fourth column: ridge regression with fixed, large smoothing parameter; fifth column: LASSO with estimated smoothing parameter based on cross validation

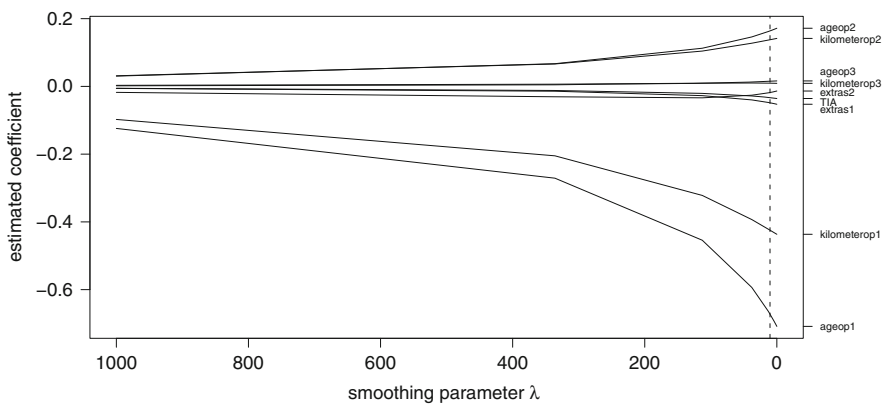


Fig. 4.9 Prices of used cars: estimated ridge regression coefficients as a function of the smoothing parameter λ . The *dashed vertical line* indicates the optimal smoothing parameter determined by tenfold cross validation. All effects refer to standardized versions of the covariates

parameter. In order to comply with the resulting visual impression of coefficient paths evolving from zero to the least squares estimate, we chose a reverse scaling with respect to the smoothing parameter. Note that Fig. 4.9 shows standardized coefficients, i.e., all covariates have been standardized to zero mean and unit variance prior to the analysis to make them comparable in their magnitude. In contrast, Table 4.6 contains estimates transformed back to the original scaling. In fact, the scaling only impacts the categorical covariates since the orthonormal polynomials constructed for continuous covariates are already standardized.

To obtain an optimal amount of shrinkage, the smoothing parameter λ was determined based on tenfold cross validation (the optimal smoothing parameter is indicated as a dashed line in Fig. 4.9). Clearly, only a very small amount of smoothness is needed, so that estimates remain almost unchanged after applying ridge regularization. The third column of Table 4.6 contains the corresponding precise numerical values while the fourth column

contains estimates achieved with a rather large smoothing parameter. In this latter case, the impact of the penalty can be seen much more clearly with shrinkage towards zero. Note, however, that single coefficients may actually also increase after applying a penalty as can be seen for the coefficient of *extras2*. While this may seem to be surprising at first sight, it is a result of (negative) correlations in the covariates. If a certain effect is reduced, the effect of a closely related variable may increase. For very large values of the smoothing parameter, however, all coefficient estimates will eventually approach a limiting value of zero. We will provide a geometric explanation later in this section. $\triangle$

4.2.3 Least Absolute Shrinkage and Selection Operator

While ridge regression enables estimation of regression coefficients in high-dimensional covariate settings or with design matrices that are close to collinear, it does not yield a sparse solution: all estimated regression coefficients will still be different from zero (with probability one). For interpretational purposes, it would, however, be desirable not only to shrink small coefficients towards zero but to have the possibility to set some effects exactly to zero. In particular, this would allow us to combine model estimation with variable selection in one single model estimation step. Such an approach can be obtained by replacing the penalty of squared regression coefficients with the penalty of absolute values yielding

$$\text{pen}(\boldsymbol{\beta}) = \sum_{j=1}^k |\beta_j|$$

and

$$\hat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg \min_{\boldsymbol{\beta}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^k |\beta_j|, \quad (4.10)$$

where again we left the intercept unpenalized. As for ridge regression, the penalized least squares criterion balances between fit to the data as measured by the least squares criterion and regularized solutions as determined by the penalty. The trade-off between these two goals is governed by the smoothing parameter λ . Since $\hat{\boldsymbol{\beta}}_{\text{LASSO}}$ is defined in terms of an absolute value penalty and allows to select covariates in a variable selection type fashion (as we will see in the following), it is referred to as the least absolute shrinkage and selection operator (LASSO, Tibshirani, 1996).

The difference between ridge regression and the LASSO can be seen from the different form of the penalty functionals depicted in Fig. 4.10. Ridge regression imposes a quadratic penalty that has very strong impact on large coefficient values but small penalty for values close to zero (compare also the coefficient path depicted in Fig. 4.9). In contrast, the absolute value penalty for the LASSO increases at a slower rate for large coefficient values, but moves away from zero faster for coefficients that are close to zero. Consequently, we expect the desirable behavior that small coefficients will be more strongly shrunk towards zero, while larger coefficients will be less affected by the penalty.

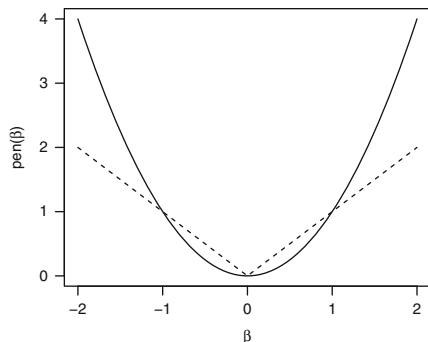


Fig. 4.10 Penalties for ridge regression (*dashed line*) and the LASSO (*solid line*)

In contrast to ridge regression, no closed-form solution for the LASSO-regularized estimate is available. While the penalized least squares criterion (4.10) is not differentiable, because of the inclusion of the absolute value penalty, one can still obtain “estimation equations” similar to the normal equations in the classical linear model. Omitting their derivation they are given by

$$2X'X\beta + 2X'y + \lambda \sum_{j=1}^k \text{sign}(\beta_j) = \mathbf{0}.$$

However, no explicit solution can be computed because of the sign function that indicates that the LASSO estimate, in contrast to the ridge estimator, is nonlinear in the data. Stated differently, it cannot be expressed as a linear estimator $\hat{\beta} = A y$ with a $p \times n$ -matrix A (compare also Sect. 3.2.3, p. 117). This also implies that statistical properties of $\hat{\beta}_{\text{LASSO}}$ are more difficult to derive than in case of ridge regression. Still, these are in principle analogously to ridge regression, i.e., $\hat{\beta}_{\text{LASSO}}$ is biased, but has smaller variance than the least squares estimate. A comparison between ridge and LASSO is more complicated and there will be no general answer on which of the two is preferable in terms of mean squared error.

The LASSO criterion (4.10) can equivalently be rewritten as

$$\hat{\beta}_{\text{PLS}} = \arg \min_{\beta} (y - X\beta)'(y - X\beta)$$

subject to the constraint

$$\sum_{j=1}^k |\beta_j| \leq t,$$

where again t and the smoothing parameter λ are connected in a one-to-one relationship; see, e.g., Sydsaeter et al. (2005) in Sect. 3.5 for a proof of the equivalence. For interpretational purposes and graphical visualization, one often

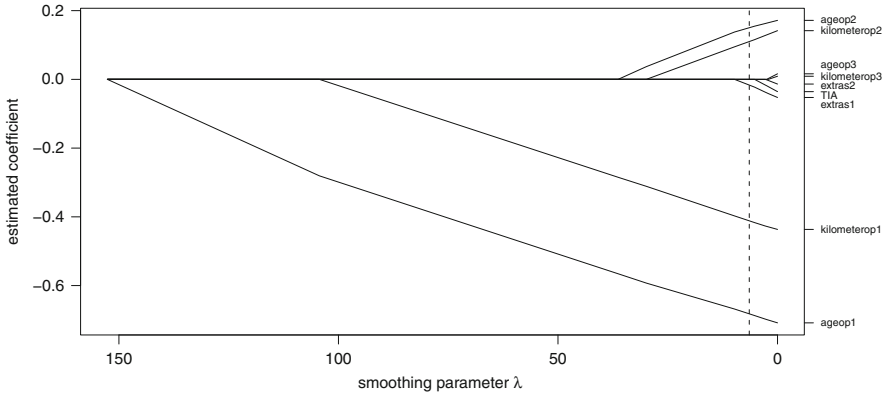


Fig. 4.11 Prices of used cars: estimated LASSO-regularized coefficients as a function of the smoothing parameter λ . The *dashed vertical line* indicates the optimal smoothing parameter determined by tenfold cross validation. All effects refer to standardized versions of the covariates

varies t between the two extremal values 0 and t_{LS} , corresponding to extreme and no regularization, where

$$t_{LS} = \sum_{j=1}^k |\hat{\beta}_{LS,j}|.$$

Estimation of the LASSO relies on numerical optimization approaches. In the original proposal of the LASSO (Tibshirani, 1996), a quadratic programming approach has been applied to optimize the penalized least squares criterion subject to the inequality constraint on the sum of absolute values. Most current implementations use least angle regression (LARS) to determine the LASSO estimates (Efron, Hastie, Johnstone, & Tibshirani, 2004). The main advantage of LARS is that it allows to determine the complete coefficient paths in one estimation run. We used an alternative approach developed in Goeman (2010) and implemented in the R package `penalized` to compute both LASSO-regularized estimates and ridge regression estimates. This package also implements the determination of optimal smoothing parameters based on cross validation.

Example 4.11 Price of Used Cars—LASSO



We repeat the analysis conducted in Example 4.10, now replacing the ridge penalty with the LASSO penalty (estimated via function `optL1` also contained in the R package `penalized`). The resulting coefficient paths are shown in Fig. 4.11. The first striking difference compared to the results achieved with ridge regression is that for large values of the smoothing parameter several coefficients are actually set to zero. At a sequence of threshold values, one covariate at a time enters the model until all coefficients are nonzero for values of the smoothing parameter approaching zero and finally achieve the values of the least squares estimate. Moreover, the paths are all linear in the smoothing parameter and only change their slope once a new variable enters the model.

More specifically, several of the covariates associated with small least squares estimates enter the model very late, with rather small values of the smoothing parameter. When

determining a data-driven, optimal value of the smoothing parameter through tenfold cross validation, these coefficients actually drop out of the model so that a sparse solution is obtained. In fact, the resulting model is quite close to the model chosen by AIC in Example 3.19 with quadratic effects of age and kilometer reading and no effects of *TIA* and *extras2*. With AIC, *extras1* was also deleted from the model which is not the case with LASSO regularization. Note, however, that the estimated coefficient is very small especially compared to the least squares estimate (Table 4.6).

When comparing the cross validation criterion for the least squares estimate, ridge regression, and LASSO-regularized estimation, we obtain values of -202.519 (least squares), -204.733 (ridge), and -200.746 (LASSO). The best fit is obtained with the LASSO, while ridge regression actually leads to a larger cross validation criterion in comparison to least squares. Therefore, in this particular example, it is not per se important to regularize estimates, but we have to use a specific form of regularization to achieve an improvement in terms of model fit. This may be explained by the specific properties of the regularization approaches in combination with the settings in our example. While ridge regression shrinks large, important coefficients more strongly than the LASSO, we have the reverse behavior for small coefficients (again see Fig. 4.10). To obtain good prediction for the price of used cars, the type of regularization imposed by the LASSO seems to be more plausible. Note also that we have only a moderate number of covariates compared to the sample size and therefore the need for regularization may not be so great in this example. $\triangle$

4.2.4 Geometric Properties of Regularized Estimates

As we have seen in the application, LASSO regularization yields estimated coefficient vectors such that some of the parameters are estimated to be exactly zero. To understand this behavior, we have to investigate the geometric properties of penalized least squares estimation. For illustration purposes, we will consider a bivariate coefficient vector $\boldsymbol{\beta} = (\beta_1, \beta_2)'$, but all results are easily generalized to the multivariable setup. Note that we have not included an intercept but consider standardized covariates and a centered response.

First, the least squares criterion $LS(\boldsymbol{\beta})$ can be rewritten as

$$\begin{aligned} LS(\boldsymbol{\beta}) &= (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})' X' X (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \mathbf{y}' (\mathbf{I}_n - X(X'X)^{-1}X') \mathbf{y} \\ &= (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})' X' X (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} \end{aligned} \quad (4.11)$$

and is therefore (up to an additive constant) equivalent to the quadratic form

$$(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})' X' X (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$$

in $\boldsymbol{\beta}$. A proof can be found in the appendix of this chapter on p. 259.

Since Eq. (4.11) defines a quadratic form in $\boldsymbol{\beta}$, the contour lines of $LS(\boldsymbol{\beta})$, i.e., the values of $\boldsymbol{\beta}$ resulting from solving $LS(\boldsymbol{\beta}) = c$ for a constant c , are ellipses with the specific shape determined by the matrix $X'X$. On the other hand, for two dimensions, the constraint

$$|\beta_1| + |\beta_2| = t$$

4.3 Regularized Estimation

Penalized Least Squares Criteria

Regularized estimation in the linear model relies on penalized least squares criteria

$$\text{PLS}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \cdot \text{pen}(\boldsymbol{\beta})$$

with smoothing parameter $\lambda \geq 0$ and $\text{pen}(\boldsymbol{\beta})$ penalizing model complexity.

Ridge Regression

For ridge regression, the penalty is given by the sum of squared coefficients, i.e.,

$$\text{pen}(\boldsymbol{\beta}) = \sum_{j=1}^k \beta_j^2 = \boldsymbol{\beta}' \mathbf{K} \boldsymbol{\beta},$$

with penalty matrix $\mathbf{K} = \text{diag}(0, 1, \dots, 1)$. The resulting penalized least squares estimate is

$$\hat{\boldsymbol{\beta}}_{\text{PLS}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{K})^{-1} \mathbf{X}'\mathbf{y}.$$

LASSO

For the LASSO, the penalty is given by the sum of absolute coefficients, i.e.,

$$\text{pen}(\boldsymbol{\beta}) = \sum_{j=1}^k |\beta_j|.$$

The resulting estimate is not available in closed form and has to be determined numerically (for example based on quadratic programming).

Choice of the Smoothing Parameter

The optimal smoothing parameter λ can be determined based on r -fold cross validation.

Software

R package `penalized`.

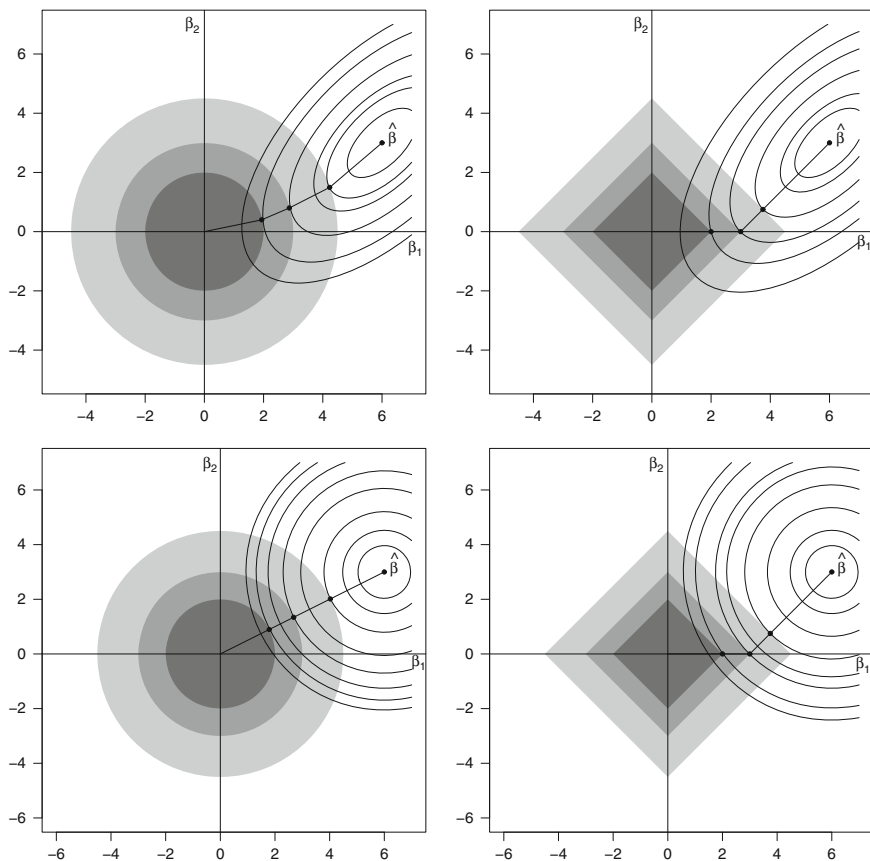


Fig. 4.12 Geometrical interpretation of the penalized least squares criteria for ridge regression (*left panel*) and LASSO (*right panel*). In the *upper panel*, a non-diagonal matrix $X'X$ is considered while the *lower panel* corresponds to an orthonormal design with $X'X = I_2$

defines diamond-shaped contour lines with side lengths $\sqrt{2}t$. Therefore the LASSO-regularized estimate for a given t is the contact point between the two geometrical regions defined by the constraint and the least squares criterion. If this contact point is located in one of the corners of the diamond, some of the coefficients will be estimated to be zero.

This geometric definition of LASSO-regularized regression estimates is visualized in Fig. 4.12 for artificial data and a hypothetical value of $\hat{\beta} = (6, 3)'$ for the least squares estimate. The contour lines of the least squares criterion are centered around this least squares estimate since it is the corresponding unique minimum. In the upper panel of Fig. 4.12, usual data with correlated design vectors have been considered so that the contour lines of the least squares criterion are ellipsoids. In comparison, the lower panel considers the special case of orthonormal design

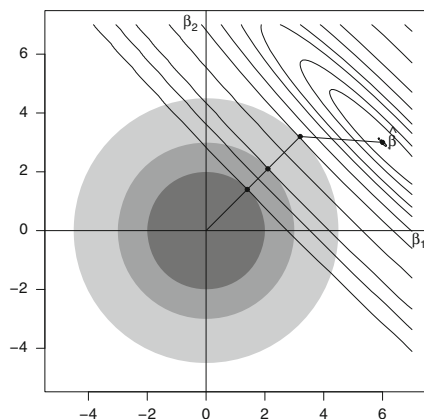


Fig. 4.13 Ridge regression estimates in case of a design matrix with large negative correlation

matrix with $X'X = I_2$ so that the contour lines are exact circles. The contours of the penalty are depicted as shaded areas centered about the minimum of the penalty functional given by $(0, 0)'$. Depending on the value of the smoothing parameter, a specific contact point of the contour lines of the fit and penalty criterion is chosen that determines the regularized estimated coefficients. If the smoothing parameter is large enough, the contact point will be forced to lie on one (or several) of the coefficient axes, and therefore one (or more) coefficients are shrunk to zero.

For ridge regression, the contour lines are circular, defined by the constraint

$$\beta_1^2 + \beta_2^2 = t.$$

When considering candidate contact points within the geometric regions that are defined by the constrained least squares criterion, no possibility arises to set coefficients equal to zero, as illustrated in the left panel of Fig. 4.12. In contrast to the LASSO, an increasing value for the ridge constraint parameter, t , provides a smooth transition of the coefficients towards zero.

Figure 4.13 provides a geometric explanation for the surprising behavior that we observed in Example 4.10, i.e., where ridge regression in fact yielded an increased, instead of shrunk estimates, for some coefficients. We previously argued that negative correlations between covariates are the reason for this observation. In Fig. 4.13, we have generated artificial data with a rather high negative correlation, which is indicated by the negative orientation of the ellipsoids representing the least squares criterion. Seeking the contact point between the outer contour line for the penalty and the least squares criterion yields a shrunk estimate for β_1 but a somewhat increased estimate for β_2 due to the extreme orientation of the least squares contours. When increasing the smoothing parameter (i.e., moving towards the inner contours of the penalty), we again find the expected shrinkage effect and finally approach zero for both regression coefficients.

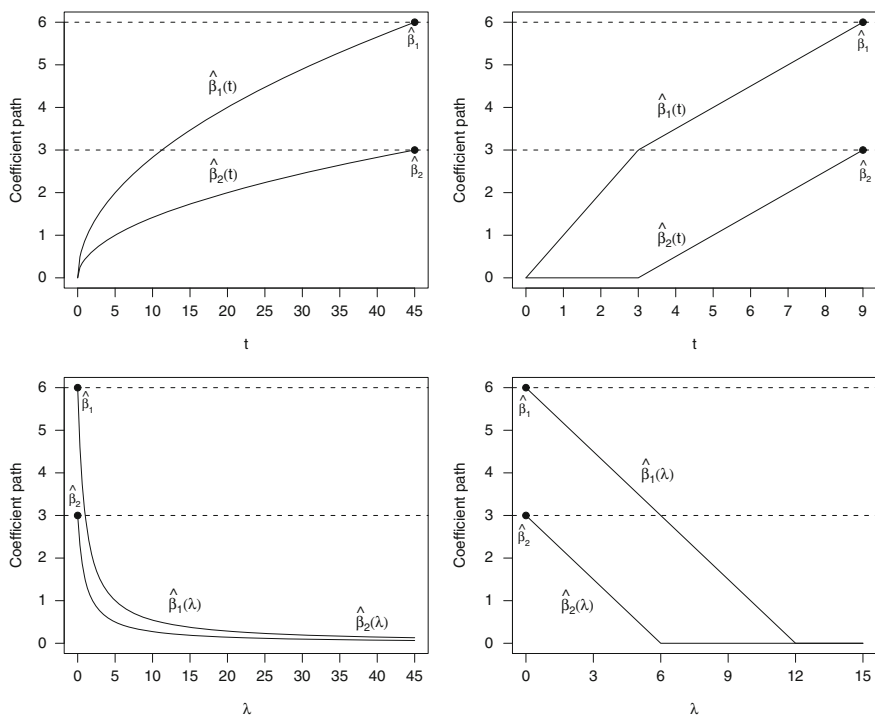


Fig. 4.14 Coefficient paths for ridge regression (*left panel*) and the LASSO (*right panel*) as functions of either the complexity parameter t (*upper panel*) or the smoothing parameter λ (*lower panel*). The *dashed lines* indicate the values of the least squares estimates

Additional insights can be gained when considering situations with orthonormal design matrices, i.e., with $X'X = I$. In this case, the contour lines of the least squares criterion are circular and explicit formulae for both ridge and LASSO-regularized estimates can be derived either in terms of the smoothing parameter λ or the constraint parameter t . For the LASSO, we obtain

$$\hat{\beta}_{\text{LASSO},j}(\lambda) = \text{sign}(\hat{\beta}_{\text{LS},j}) \left[|\hat{\beta}_{\text{LS},j}| - \frac{\lambda}{2} \right]_+,$$

where $[x]_+ = \max\{0, x\}$. For ridge regression, we have

$$\hat{\beta}_{\text{PLS},j}(\lambda) = \frac{1}{1 + \lambda} \hat{\beta}_{\text{LS},j}.$$

A comparison of these expressions again reveals the distinct, yet different, behavior of ridge regression and the LASSO (also see the graphical representation in Fig. 4.14). For the LASSO, the restriction to positive parts via the function $[x]_+$ allows to set coefficients to zero if they are smaller than 0.5λ . Moreover, the LASSO-regularized estimates move away from zero at a constant slope until a

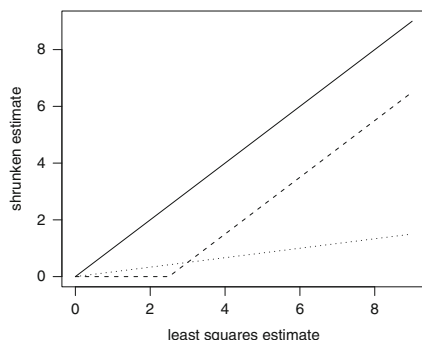


Fig. 4.15 Ridge (*dotted line*) and LASSO (*dashed line*) estimates as a function of the least squares estimate in case of an orthonormal design and for smoothing parameter $\lambda = 5$. The *solid line* represents the unregularized least squares estimate

further nonzero regression coefficient enters the model (top right panel of Fig. 4.14). In contrast, estimates from ridge regression are shrunk towards zero for a large penalty, but actually never reach zero. Specifically, they are simply shrunk by the factor $1/(1 + \lambda)$. However, note again that this only holds in case of orthonormal design matrices.

Figure 4.15 illustrates this shrinkage behavior from a different perspective, which shows a regularized estimate as a function of the least squares estimate for a fixed value of the smoothing parameter λ ($\lambda = 5$ in this case). While the LASSO-regularized estimate equals zero for a small coefficient (smaller than 0.5λ), the ridge regression estimate is simply a scaled version of the least squares estimate with the amount of scaling determined by λ .

4.2.5 Partial Regularization

Thus far, we have exclusively dealt with the situation that all coefficients of a regression model (except the intercept) should be subject to regularization. However, it is also possible to penalize only parts of the coefficient vector, yielding, for example,

$$(y - X_1\beta_1 - X_2\beta_2)'(y - X_1\beta_1 - X_2\beta_2) + \lambda \text{pen}(\beta_2),$$

where $X = (X_1, X_2)$ is a partition of the design matrix and $\beta = (\beta_1', \beta_2')'$ analogously partitions the coefficients. The penalty now only applies to the part of the model defined in $X_2\beta_2$, while the remaining covariates in X_1 are left untouched. Such a distinction can, for example, be relevant when including clinical covariates and gene expression levels in a simultaneous regression model. While the former will typically be included due to prior knowledge and are of a much smaller number, gene expression often contains diverse, heterogeneous, and high-dimensional

information. Therefore it is sensible to regularize only the regression coefficients of the latter, while coefficients associated with information such as age or disease status remain unpenalized.

4.3 Boosting Linear Regression Models

In this section, we will describe an alternative regularization approach called boosting where regularization is implicitly achieved through early stopping of an iterative stepwise algorithm. Boosting turns out to be a rather general and versatile regularization approach since estimation problems are described in terms of a loss function and, as a consequence, boosting can also be applied beyond the linear model, e.g., in robust regression or generalized linear models for non-Gaussian responses. Moreover, the implicit regularization performed with boosting allows for automatic model choice and variable selection, similar to the LASSO. A disadvantage of boosting, similar to ridge regression and the LASSO, is that standard inference tools like standard errors or confidence intervals are not easily available. We introduce the general ideas of boosting applied to classical linear models and further present the general framework at the end of the section.

4.3.1 Basic Principles

Consider the linear model

$$y_i = \eta_i + \varepsilon_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n,$$

with the standard predictor $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$. Similar to the cases of ridge regression and the LASSO, we are interested in situations where the ordinary least squares estimate is not optimal. This especially may be the case when the number of regression coefficients is large, when the design matrix is close to collinear, or when we are interested in determining a suitable subset of covariates. In particular in the last case, using either too many or using too few covariates may induce suboptimal predictions (as discussed in Sect. 3.4), and regularized estimation is often a useful alternative to least squares estimation.

The principle idea of boosting can be explained along the following basic algorithm:

1. Choose an initial estimate $\hat{\boldsymbol{\beta}}^{(0)}$ and set $t = 1$.
2. Compute the current residuals $\mathbf{u} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}^{(t-1)}$ and the corresponding least squares estimate $\hat{\mathbf{b}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{u}$. Perform the update

$$\hat{\boldsymbol{\beta}}^{(t)} = \hat{\boldsymbol{\beta}}^{(t-1)} + \nu \hat{\mathbf{b}}$$

with $0 < \nu < 1$ and set $t = t + 1$.

3. Iterate step 2 for a fixed number of iterations m_{stop} .

Starting from the initial value $\hat{\beta}^{(0)}$, the algorithm iteratively proceeds towards the final estimate by updating $\hat{\beta}^{(t-1)}$ with *small portions* of the least squares estimate obtained from current residuals. Instead of making large steps towards the least squares estimate, we multiply $\hat{\mathbf{b}}$ with the step length factor ν to implicitly implement regularization since—when stopping early enough—the multiplication with ν yields a proportional shrinkage of the estimates analogous to ridge regression. Usually ν will be chosen to be rather small, for example, $\nu = 0.1$ or even $\nu = 0.01$.

In boosting terminology, the least squares estimate used in step 2. of the algorithm is a *base-learning procedure* that provides a means of fitting the model of interest. Often this single estimation step will be replaced with componentwise fits in the following. We then actually fit a sequence of base-learning procedures (for example corresponding to the different covariates in the linear model) and only update the best-fitting one. Yet the application of the step length factor induces regularization.

We now consider a somewhat different interpretation of boosting that is useful to generalize the approach in particular in terms of the available class of base-learning procedures.

When starting with initial guesses $\hat{\beta}^{(0)}$, we can compute the lack of fit information associated with this starting values as the corresponding derivative of the least squares criterion, i.e.,

$$\left. \frac{\partial}{\partial \beta} \text{LS}(\beta) \right|_{\beta=\hat{\beta}^{(0)}} = -2X' (y - X\hat{\beta}^{(0)}).$$

This derivative indicates the direction towards the least squares estimate and would be zero if the initial value is already the least squares estimate since it then coincides with the normal equations. In fact, here the minimizer of the least squares criterion could be obtained in one step by setting the derivative to zero and solving with respect to the regression coefficients. However, to achieve regularization, we only do small steps towards the solution so that

$$\hat{\beta}^{(1)} = \hat{\beta}^{(0)} + \nu \hat{\mathbf{b}},$$

where $\hat{\mathbf{b}} = (X'X)^{-1}X'u$ is the least squares estimate obtained from the residuals $u = y - X\hat{\beta}^{(0)}$.

4.3.2 Componentwise Boosting

After introducing the basic principles, we will next move towards more interesting versions of boosting algorithms. Figure 4.16 shows how the boosting approach moves from the initial guess $[(0, 0)']$ in this case] towards the minimum of the least squares criterion. If we use $\hat{\mathbf{b}} = (X'X)^{-1}X'u$ to update the coefficient vector, we always move along the direction of the steepest descent directly towards the

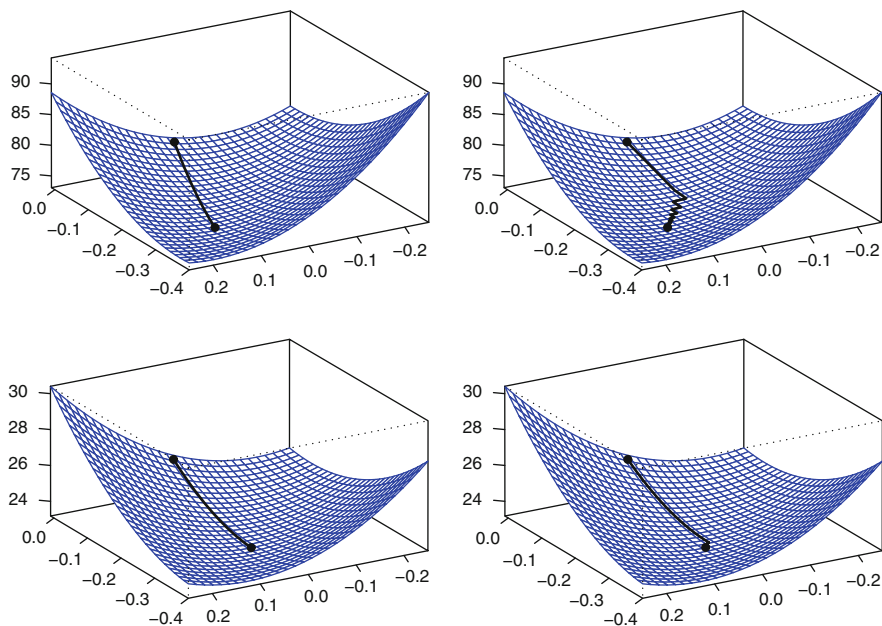


Fig. 4.16 Surface plots of the least squares criterion together with the coefficient path from the starting value $(0, 0)'$ towards the least squares estimate. The *left panels* show direct paths along the steepest descent; the *right panels* restrict the directions to be parallel to the coefficient axes

least squares estimate. In the upper left panel of Fig. 4.16, this is indicated for a least squares estimate where both components are quite different from zero. In the bottom left panel, in contrast, one of the coefficients is estimated to be rather close to zero. In this case, the path along steepest descent is almost parallel to one of the coefficient axes.

The basic idea of componentwise boosting now is to restrict updates in the iterative fitting process to directions that are *exactly parallel* to the coefficient axes. This is implemented as follows: We first compute least squares estimates for all covariates separately, i.e., estimate the models $u_i = b_j x_{ij} + \varepsilon_{ij}$ and obtain coefficient estimates

$$\hat{b}_j = ((\mathbf{x}^j)' \mathbf{x}^j)^{-1} (\mathbf{x}^j)' \mathbf{u} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j) u_i}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}, \quad j = 0, \dots, k$$

based on the current residuals $\mathbf{u}$ where $\mathbf{x}^j = (x_{1j}, \dots, x_{nj})'$ is the column of $\mathbf{X}$ corresponding to the j th covariate and $\bar{x}_j$ is the average of the covariate values. We then do not update all coefficients, but rather only the one that leads to the largest reduction in the least squares criterion, i.e., we determine

$$j^* = \arg \min_{j=0,\dots,k} \sum_{i=1}^n (u_i - x_{ij} \hat{b}_j)^2,$$

and then update the coefficients using

$$\begin{aligned} \hat{\beta}_{j^*}^{(1)} &= \hat{\beta}_{j^*}^{(0)} + v \hat{b}_{j^*}, \\ \hat{\beta}_j^{(1)} &= \hat{\beta}_j^{(0)}, \quad j \neq j^*. \end{aligned}$$

Such an approach is illustrated in the right panel of Fig. 4.16. The upper right panel shows that in the beginning the algorithm moves along one of the coefficient axes for a longer time until a certain decrease in the least squares criterion is achieved. From this, it then alternates between steps in the two different directions. For the lower right panel this implies that the algorithm moves along one direction for many steps (as long as the corresponding covariate is “more informative” about the variability in the response). Finally at the very end of the algorithm, a step is made towards the coefficient that is close to zero.

This finding enables to implement implicit regularization or shrinkage of the estimated coefficients by stopping the algorithm not at the minimum of the least squares criterion, but after a fixed number of iterations. In this case of early stopping, covariates associated with small least squares estimates will drop out of the model, while all other least squares estimates will be shrunk towards the initial guesses (which typically will be $\beta^{(0)} = \mathbf{0}$). In principle, this achieves a similar shrinkage as induced by the LASSO presented in the last section. Recall in the case of an orthonormal design, LASSO-regularized estimates were given by

$$\hat{\beta}_{\text{LASSO},j}(\lambda) = \text{sign}(\hat{\beta}_{\text{LS},j}) \left[|\hat{\beta}_{\text{LS},j}| - \frac{\lambda}{2} \right]_+;$$

also see Fig. 4.15. The quantity that plays a similar role to the smoothing parameter λ for the LASSO, in case of boosting, is the number of iterations. In particular, a small number of iterations will induce large regularization and will yield a sparse model with only a few coefficients differing from zero. On the other hand, choosing a large number of iterations will yield estimates that are close to the least squares estimate. As with the smoothing parameter, the optimal number of iterations will typically be determined via r -fold cross validation to minimize prediction error.

Example 4.12 Price of Used Cars—Boosting



To illustrate the application of componentwise boosting, we again consider the data set on prices of used cars with the same model specification as in Example 4.10 (p. 206). Our computations are based on a generic boosting implementation for a wide range of regression models available in the R package `mboost` (function `glmboost`). Figure 4.17 shows the resulting coefficient path as a function of the number of boosting iterations. While the upper plot contains the paths for a large number of iterations, the lower panel shows the path restricted to the optimal number of iterations (determined by tenfold cross validation).

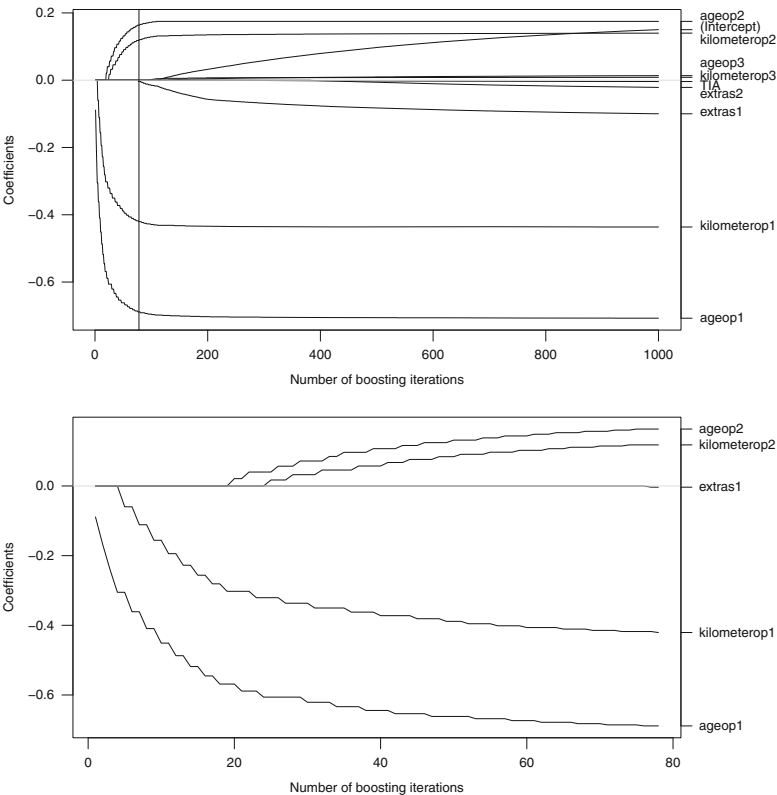


Fig. 4.17 Prices of used cars: estimated regression coefficients as a function of the boosting iterations. The *upper panel* shows the path for 1,000 boosting iterations; the *horizontal line* indicates the optimal stopping iteration determined by tenfold cross validation. The *lower panel* shows the path only until this optimal iteration

Figure 4.18 illustrates the process of determining the optimal number of iterations. For each of the cross validation folds, the prediction error is shown as a function of the boosting iterations (in grey). Averaging over the folds gives the cross validation score shown as a black line. During the first iteration the score strongly decreases but reaches a plateau after about 50 iterations. In fact, the cross validation score reaches a minimum at 78 iterations and then starts to increase again (although very slowly in our example). This indicates that the resulting fit will be rather insensitive to the choice of the boosting iterations. However, since we are interested in a sparse model with not too many nonzero coefficients, a small value will usually be preferable.

Similar to the LASSO, some variables drop out of the model when using the optimal number of boosting iterations, in particular the cubic effects of the age of the car and kilometer reading as well as the effects of *TIA* and *extras2*. These effects are also estimated to be rather small in a least squares approach, as shown in Table 4.7, and directly visible from the overplotting of several variable names that are associated with small effects in the upper panel of Fig. 4.17. Table 4.7 also contains the boosting estimates obtained after 1,000 iterations and for the optimal number of boosting iterations ($m_{\text{stop}} = 78$). While the

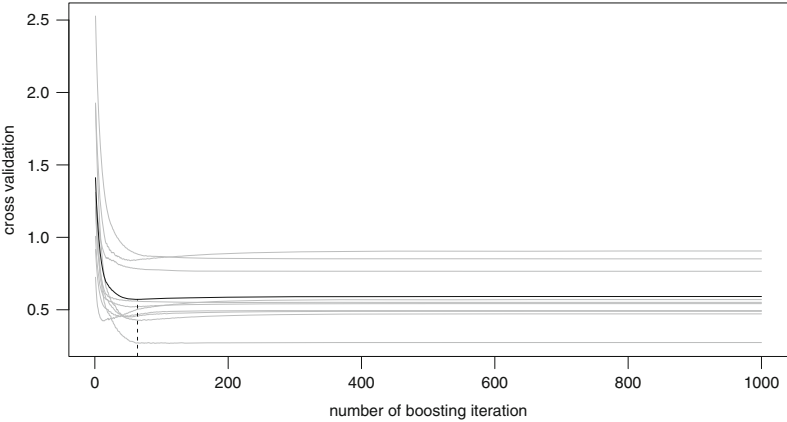


Fig. 4.18 Prices of used cars: cross validation criterion (*black line*) as a function of the boosting iterations. The *grey lines* indicate the prediction error obtained within a single cross validation fold. The *dashed vertical line* shows the optimal boosting iteration, i.e., the minimal cross validation criterion

Table 4.7 Prices of used cars: estimated coefficients for different models

Variable	LS	$m_{\text{stop}} = 1,000$	$m_{\text{stop}} = 78$
<i>intercept</i>	3.585	3.547	3.397
<i>ageop1</i>	−0.709	−0.707	−0.689
<i>ageop2</i>	0.172	0.175	0.163
<i>ageop3</i>	0.016	0.014	–
<i>kilometerop1</i>	−0.437	−0.436	−0.421
<i>kilometerop2</i>	0.142	0.140	0.118
<i>kilometerop3</i>	0.009	0.009	–
<i>TIA</i>	−0.005	−0.004	–
<i>extras1</i>	−0.114	−0.100	−0.003
<i>extras2</i>	−0.031	−0.021	–

Second column: least squares estimate, third column: boosting with large stopping iteration, fourth column: boosting with optimal stopping iteration determined by tenfold cross validation

former are rather close to the least squares estimates (and would be even closer for a further increase in the iterations), the latter are considerably shrunken towards zero.
Overall the results are again very close to the AIC optimal model of Example 3.19 and the LASSO in Example 4.11. △

4.3.3 Generic Componentwise Boosting

To enable the generalization of boosting to more general model classes, it is advantageous to reformulate it in a somewhat more general context. Instead of relating estimation of a linear model to the determination of regression coefficients,

the aim of fitting a linear model can also be interpreted as minimizing the sum of squared errors

$$\text{LS}(\boldsymbol{\eta}) = \sum_{i=1}^n (y_i - \eta_i)^2$$

with respect to the predictor vector $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)$. If a candidate predictor value $\hat{\boldsymbol{\eta}}^{(0)}$ is given, the corresponding lack of fit can then be evaluated with $\text{LS}(\hat{\boldsymbol{\eta}}^{(0)})$, but more detailed information is contained in the unit-specific derivatives

$$u_i = \left. \frac{\partial}{\partial \eta_i} \text{LS}(\boldsymbol{\eta}) \right|_{\eta_i = \hat{\eta}_i^{(0)}} = 2 \left(y_i - \hat{\eta}_i^{(0)} \right).$$

For a perfect fit, all these derivatives will be zero while large derivatives point towards observations where the fit could be substantially improved. In fact, the derivatives are basically the residuals obtained by plugging in the candidate predictor (multiplied with a factor of 2).

Now, an improved predictor $\hat{\boldsymbol{\eta}}^{(1)}$ is obtained by fitting a linear model to the derivatives u_i , which correspond to the residuals. This can be conceptualized by multiplying the vector of derivatives $\mathbf{u} = (u_1, \dots, u_n)'$ with the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

of the linear model. Similar as in the basic algorithm discussed before, we do not update the predictor $\hat{\boldsymbol{\eta}}^{(0)}$ with the full fit obtained by applying the hat matrix but only with a fraction $0 < \nu < 1$, i.e.,

$$\hat{\boldsymbol{\eta}}^{(1)} = \hat{\boldsymbol{\eta}}^{(0)} + \nu \mathbf{H} \mathbf{u}.$$

Based on the updated predictor $\hat{\boldsymbol{\eta}}^{(1)}$, we can obtain updated derivatives, fit the linear model again, and iterate this process until finally ending up with the least squares fit for the given linear model. The advantage of the general formulation in terms of predictors instead of regression coefficients is that we can replace the least squares base-learners with any fitting routine that provides predictions for the linear model. In particular, we do not have to rely on parametric models but can instead consider regression trees or semiparametric regression techniques as described in Chap. 8.

Figure 4.19 illustrates predictor-based boosting for the model $y_i = 2x_i + \varepsilon_i$, ε_i i.i.d. $N(0, 0.25)$. As starting predictor, we chose $\hat{\boldsymbol{\eta}}^{(0)} = \mathbf{0}$; an alternative popular choice would be the average response value, i.e., $\hat{\boldsymbol{\eta}}^{(0)} = \bar{y}\mathbf{1}_n$. The top left plot of Fig. 4.19 shows the data set together with the starting predictor. The top right plot shows the resulting residuals which in the first iteration are the original data. The middle row shows similar information obtained in the fourth iteration when applying a step length factor of $\nu = 0.1$ (with one intermediate fit shown in addition). The bottom row shows the results after 60 iterations. The estimated function has approximately reached the least squares fit and the residuals do no longer show any specific structure.

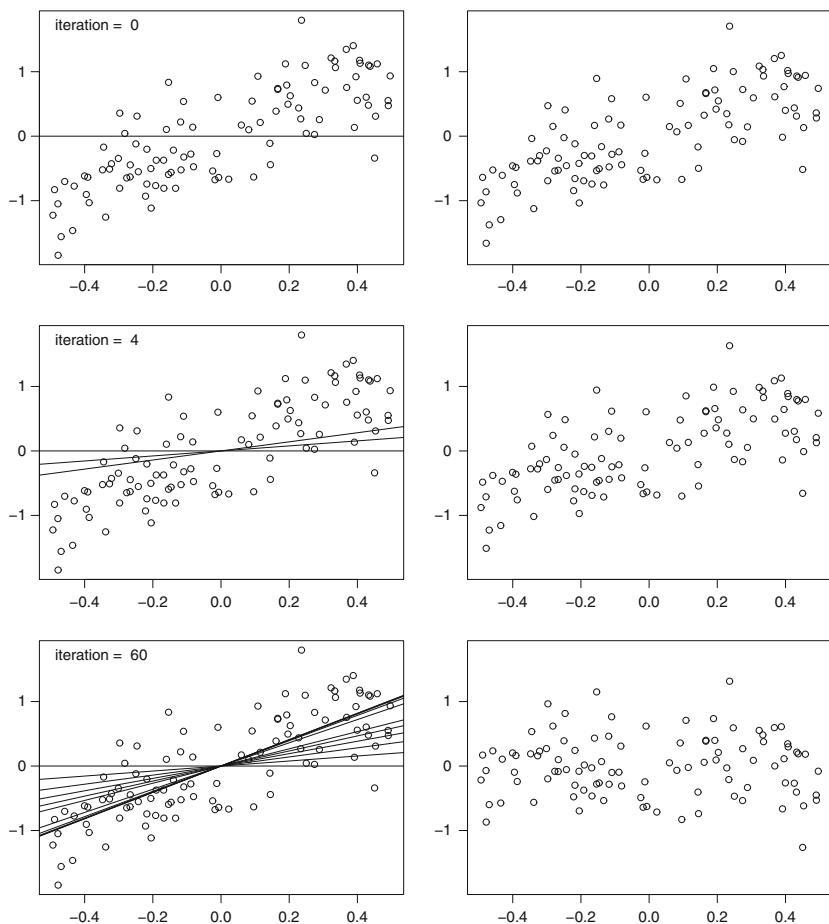


Fig. 4.19 Iterative update of model fits with boosting. The *left panel* shows the current fit (along with some intermediate *regression lines*); the *right panel* shows the corresponding residuals

If boosting only implements an alternative method for regularized estimation in linear models, why should we be interested in using it? There are actually a number of advantages:

- It is possible to replace the least squares criterion $LS(\eta)$ with more general loss functions $\rho(\eta)$ such as the absolute value loss

$$\rho(\eta) = \sum_{i=1}^n |y_i - \eta_i|$$

corresponding to median regression. Therefore, boosting does not only allow to fit linear models with regularization, but also more general types of models. For example, boosting can also be used in generalized linear models for non-

Gaussian models as will be discussed in the following chapter when the negative log-likelihood is used as the loss function; see Sect. 5.7.

- We can also use more complex model components and replace the least squares base-learners with more general learning approaches such as regression trees or penalized splines. We will return to such more complex model structures in Chap. 9.
- Boosting combines regularization with implicit variable selection. This again is of particular interest in more complex regression models comprising, for example, nonlinear effects where the LASSO can no longer be used (at least not without suitable modifications).
- Boosting can also be used in situations with high-dimensional covariate vectors, where p may even exceed the sample size n . Since we only fit one linear effect at a time, the implicit regularization achieved by early stopping also allows to fit models for p large compared to n .

Finally, we want to further clarify the role of the step length factor ν in the boosting algorithm. When using a large step length, the boosting algorithm would make large steps towards the least squares estimates in each iteration. While at first glance this may seem to be desirable since it would ensure fast convergence, it actually could be problematic for two reasons: On the one hand, the regularization property of boosting would work very unevenly on the different coefficients while small steps allow to regularize all effects with a comparable strength. On the other hand, in case of correlated covariates, boosting would only be able to pick one covariate, make a large step towards the least squares estimate of the covariate, and would never include the remaining, correlated covariates. Making smaller steps in each iteration avoids this overshooting in specific directions and also allows the simultaneous inclusion of several correlated variables in the final model.

4.4 Bayesian Linear Models

This section covers linear models from a Bayesian point of view. Readers who are unfamiliar with the basic concepts of Bayesian inference should first consult Sect. B.5 in Appendix B. One advantage of the Bayesian approach, in relation to classical inference, is the possibility to consider *prior information* on the parameters in the model. Bayesian models are particularly useful to *regularize* regression problems where data information is limited. We will see, that some of the frequentist regularization techniques in Sect. 4.2, such as ridge regression and the LASSO, have a Bayesian interpretation. Another advantage of the Bayesian approach is that model choice and variable selection can be an integral part of the estimation process.

The starting point of this section is the classical linear model equipped with the conjugate normal-inverse gamma prior. This standard model described in all textbooks on Bayesian analysis will be treated in Sect. 4.4.1. Section 4.4.2 discusses Bayesian regularization techniques as a counterpart to the frequentist techniques encountered in Sect. 4.2. In Sects. 4.4.3 and 4.4.4, we discuss model choice and variable selection from a Bayesian point of view.

4.4 Generic Componentwise Boosting for Linear Models

Model

Given are observations for the model

$$y_i = \mathbf{x}_i \boldsymbol{\beta} + \varepsilon_i = \sum_{j=0}^k x_{ij} \beta_j + \varepsilon_i$$

with $x_{i0} = 1$ for the intercept.

Algorithm

1. Initialize the regression coefficients, e.g., as

$$\beta_0^{(0)} = \bar{y} \quad \text{and} \quad \beta_j^{(0)} = 0, \quad j = 1, \dots, k,$$

choose a number of iterations m_{stop} and set $t = 0$.

2. Increase t by 1. Compute the negative gradients (“residuals”)

$$u_i = -\frac{\partial}{\partial \eta} \rho(y_i, \eta) \Big|_{\eta = \hat{\eta}_i^{(t-1)}}, \quad i = 1, \dots, n,$$

where ρ is a general loss function, e.g., the least squares criterion, describing the estimation problem.

3. Fit separate linear models for all covariates, i.e., obtain

$$\hat{b}_j = ((\mathbf{x}^j)' \mathbf{x}^j)^{-1} (\mathbf{x}^j)' \mathbf{u}, \quad j = 0, \dots, k$$

and determine the best-fitting variable via

$$j^* = \arg \min_{j=0, \dots, k} \sum_{i=1}^n (u_i - x_{ij} \hat{b}_j)^2.$$

4. Update the coefficients and the predictor:

$$\eta_i^{(t)} = \eta_i^{(t-1)} + v \cdot x_{ij^*} \hat{b}_{j^*}$$

$$\hat{\beta}_{j^*}^{(1)} = \hat{\beta}_{j^*}^{(0)} + v \hat{b}_{j^*}$$

$$\hat{\beta}_j^{(1)} = \hat{\beta}_j^{(0)}, \quad j \neq j^*.$$

5. Iterate steps 2–4 until $t = m_{\text{stop}}$.

An optimal value of m_{stop} can be determined via cross validation.

Software

R packages `mboost` and `gbm`.

4.4.1 Standard Conjugate Analysis

Our starting point is the classical linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. In contrast to classical inference, the Bayesian approach considers the unknown parameters $\boldsymbol{\beta}$ and σ^2 as random variables. Thus, the distribution of the response $\mathbf{y}$ can be understood as conditional on the parameters $\boldsymbol{\beta}$ and σ^2 , and we obtain the observation model

$$\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}).$$

We next introduce the classical conjugate prior distribution in Bayesian linear models.

Normal-Inverse Gamma Prior

The standard conjugate prior for linear models described in all introductory textbooks on Bayesian inference is obtained by assuming a multivariate normal prior for the regression coefficients

$$\boldsymbol{\beta} \mid \sigma^2 \sim \mathcal{N}(\mathbf{m}, \sigma^2 \mathbf{M}),$$

with known expectation $\mathbf{m}$ and covariance matrix $\mathbf{M}$, e.g., $\mathbf{m} = \mathbf{0}$ and $\mathbf{M} = \mathbf{I}$. A normal distribution seems a natural choice as the distribution of the estimated regression coefficients in the classical linear model is (approximately) multivariate normal. For σ^2 , we specify an inverse gamma distribution with hyperparameters a and b , i.e.,

$$\sigma^2 \sim \text{IG}(a, b). \quad (4.12)$$

To shed light on the specific form of the inverse gamma prior for σ^2 , Fig. 4.20 shows the prior density for various choices of a and b . Of particular interest is the case $a = b$ and both values approaching zero. Then the distribution of $\log \sigma^2$ tends to a uniform distribution as can be shown analytically through the change in variables theorem; see Theorem B.1 of Appendix B.1. This is why small values for a and b are identified with a weakly informative or noninformative prior. We will further elaborate on this point below when we discuss noninformative priors in the linear model.

The joint prior for $\boldsymbol{\beta}$ and σ^2 is a normal-inverse gamma distribution with density

$$\begin{aligned} p(\boldsymbol{\beta}, \sigma^2) &= p(\boldsymbol{\beta} \mid \sigma^2) p(\sigma^2) \\ &= \frac{1}{(2\pi)^{\frac{p}{2}} (\sigma^2)^{\frac{p}{2}} |\mathbf{M}|^{\frac{1}{2}}} \exp\left(-\frac{1}{2\sigma^2} (\boldsymbol{\beta} - \mathbf{m})' \mathbf{M}^{-1} (\boldsymbol{\beta} - \mathbf{m})\right) \\ &\quad \frac{b^a}{\Gamma(a)} \frac{1}{(\sigma^2)^{a+1}} \exp\left(-\frac{b}{\sigma^2}\right) \end{aligned} \quad (4.13)$$

and parameters $\mathbf{m}$, $\mathbf{M}$, a , and b ; see also Definition B.3.5 in Appendix B.3. We write $\boldsymbol{\beta}, \sigma^2 \sim \text{NIG}(\mathbf{m}, \mathbf{M}, a, b)$. Ignoring all factors in Eq. (4.13) that are independent of $\boldsymbol{\beta}$ and σ^2 , we obtain

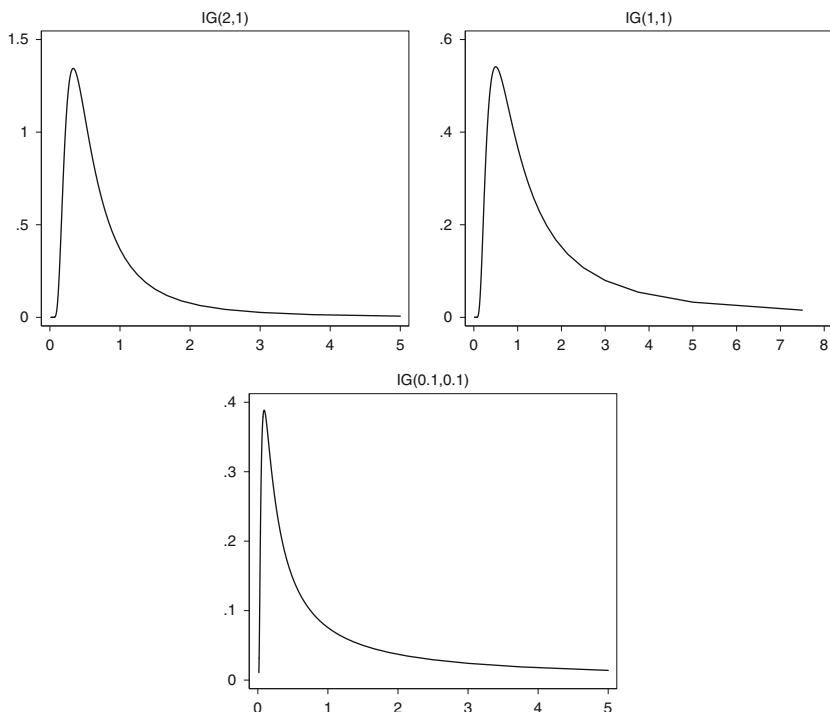


Fig. 4.20 Inverse gamma prior density for σ^2 for various values of the hyperparameters a and b

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{(\sigma^2)^{\frac{p}{2}+a+1}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{m})' \mathbf{M}^{-1}(\boldsymbol{\beta} - \mathbf{m}) - \frac{b}{\sigma^2}\right) \quad (4.14)$$

for the density of the normal-inverse gamma prior.

We now derive some useful properties of the general NIG($\mathbf{m}$, $\mathbf{M}$, a , b) prior. For convenience, these are summarized in Definition B.3.5 (p. 652) of Appendix B. Readers who are not interested in the technical details can safely skip this more difficult part.

From the properties of the multivariate normal distribution and the inverse gamma distribution (see Definition B.12 in Appendix B), it follows that

$$\mathbf{E}(\boldsymbol{\beta} \mid \sigma^2) = \mathbf{m} \quad \text{Cov}(\boldsymbol{\beta} \mid \sigma^2) = \sigma^2 \mathbf{M}$$

and

$$\mathbf{E}(\sigma^2) = \frac{b}{a-1} \quad \text{if } a > 1 \quad \text{Var}(\sigma^2) = \frac{b^2}{(a-1)^2(a-2)} \quad \text{if } a > 2.$$

We next derive the unconditional prior distribution of β . Since the conditional mean of β is independent of σ^2 , we obtain the unconditional mean

$$E(\beta) = E(E(\beta | \sigma^2)) = m,$$

using the law of iterated expectations (see Theorem B.2.9 in Appendix B.2). For the unconditional covariance matrix, we have

$$\text{Cov}(\beta) = E(\text{Cov}(\beta | \sigma^2)) + \text{Cov}(E(\beta | \sigma^2)) = E(\sigma^2)M = \frac{b}{a-1}M;$$

see Theorem B.2.10 in Appendix B.2.

In order to derive the specific distribution of β , it is useful to determine the conditional distribution of $\sigma^2 | \beta$ first. The density of this distribution is proportional to the joint distribution (4.14) of β, σ^2 , which yields

$$\begin{aligned} p(\sigma^2 | \beta) &\propto p(\beta, \sigma^2) \\ &\propto \frac{1}{(\sigma^2)^{a+\frac{p}{2}+1}} \exp\left(-\frac{1}{\sigma^2} \left(\frac{1}{2}(\beta - m)'M^{-1}(\beta - m) + b\right)\right). \end{aligned} \quad (4.15)$$

This is the form of an inverse gamma distribution, more specifically

$$\sigma^2 | \beta \sim \text{IG}\left(a + \frac{p}{2}, b + \frac{1}{2}(\beta - m)'M^{-1}(\beta - m)\right).$$

Hence the normalizing constant in Eq. (4.15) is given by

$$c = \frac{(b + \frac{1}{2}(\beta - m)'M^{-1}(\beta - m))^{a+\frac{p}{2}}}{\Gamma(a + \frac{p}{2})}.$$

Turning again our attention to the unconditional distribution of β , we can integrate Eq. (4.14) with respect to σ^2 . Since the integral of the density of $\sigma^2 | \beta$ must equal one, it follows that

$$\begin{aligned} &\int \frac{1}{(\sigma^2)^{a+\frac{p}{2}+1}} \exp\left(-\frac{1}{2\sigma^2}(\beta - m)'M^{-1}(\beta - m)\right) \exp\left(-\frac{b}{\sigma^2}\right) d\sigma^2 \\ &= \int c^{-1} p(\sigma^2 | \beta) d\sigma^2 \\ &= (b + \frac{1}{2}(\beta - m)'M^{-1}(\beta - m))^{-(a+\frac{p}{2})} \Gamma\left(a + \frac{p}{2}\right). \end{aligned}$$

We now have

$$\begin{aligned} p(\boldsymbol{\beta}) &\propto \left(b + \frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})' \mathbf{M}^{-1}(\boldsymbol{\beta} - \mathbf{m})\right)^{-(a + \frac{p}{2})} \Gamma\left(a + \frac{p}{2}\right) \\ &\propto \left(1 + \frac{1}{2a}(\boldsymbol{\beta} - \mathbf{m})' \left(\frac{b}{a} \mathbf{M}\right)^{-1}(\boldsymbol{\beta} - \mathbf{m})\right)^{-(a + \frac{p}{2})}. \end{aligned}$$



This is the density of a multivariate t-distribution with $2a$ degrees of freedom, having location parameter $\mathbf{m}$, and dispersion matrix $b/a\mathbf{M}$, i.e., $\boldsymbol{\beta} \sim t(2a, \mathbf{m}, b/a\mathbf{M})$. Details of the multivariate t-distribution can be found in Appendix B.3.4 (p. 651).

Eliciting the $\text{NIG}(\mathbf{m}, \mathbf{M}, a, b)$ prior is a difficult task in practice as there are four parameters to choose (including the prior covariance matrix). Full specification of $\mathbf{m}$ and $\mathbf{M}$ is typically only possible if there is prior knowledge, e.g., in the form of past results. One then could use the posterior parameters of the previous analysis as the prior values for the new analysis; see Example 4.13 (p. 235) below.

Zellner's g -Prior

In the absence of valid prior knowledge, the choice of the prior parameters is more difficult, if not impossible. For the prior mean a possible choice is $\mathbf{m} = (m_1, 0, \dots, 0)'$ where m_1 is a prior guess for the overall level. Assuming zero means for the other components of $\boldsymbol{\beta}$ reflects the assumption that a priori the effect of the covariates is centered around zero. More difficult is the specification of the prior correlation structure. A widely used choice is Zellner's g -prior (Zellner, 1986)

$$\boldsymbol{\beta} \mid \sigma^2 \sim \text{N}(\mathbf{m}, \sigma^2(g\mathbf{X}'\mathbf{X})^{-1}),$$

where $g > 0$ is a hyperparameter. This is a special case of the $\text{NIG}(\mathbf{m}, \mathbf{M}, a, b)$ prior with $\mathbf{M} = (g\mathbf{X}'\mathbf{X})^{-1}$. The prior assumes that the prior precision $\mathbf{M}^{-1}$ is a fraction of the precision $\mathbf{X}'\mathbf{X}$ in the data. Choices for g are discussed, e.g., in Fernandez, Ley, and Steel (2001), and include the following:

- The choice $g = 1/n$ corresponds to assigning roughly the same information as is contained in one observation.
- The choice $g = 1/k^2$ (where k is the number of regressors) is suggested by the risk inflation criterion of Foster and George (1994). They consider the predictive risk function $R(\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}) = \text{E}((\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}))$, i.e., the expected squared difference between the true, but unknown, mean $\mathbf{X}\boldsymbol{\beta}$ and the estimate $\mathbf{X}\hat{\boldsymbol{\beta}}$. The risk inflation criterion is defined as

$$\text{RIC} = \sup_{\boldsymbol{\beta}} \frac{R(\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}_{\text{working}})}{R(\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}_{\text{true}})},$$

where $\hat{\boldsymbol{\beta}}_{\text{true}}$ is the least squares estimator obtained if we estimate the correctly specified model and $\hat{\boldsymbol{\beta}}_{\text{working}}$ is the least squares estimator of a working model

that is not necessarily correct (the typical situation in practice). Thus the risk inflation criterion is the maximum of the ratio between the risk of the working model and the risk of the correct model. [George and Foster \(2000\)](#) showed that in a model with Zellner's g -prior and known variance σ^2 , the selection of the model with highest posterior probability is equivalent to maximizing the RIC provided that we choose $g = 1/k^2$.

- A compromise between the two choices is given by $g = 1/\max\{n, k^2\}$. This choice is proposed in [Fernandez et al. \(2001\)](#) based on theoretical reasoning and simulation results.

Noninformative Prior

An alternative to Zellner's g -prior is to use fully noninformative priors. As pointed out in Appendix [B.5.1](#), the construction of noninformative priors is a quite delicate task. The widely accepted noninformative prior in the linear model is given by

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}, \quad (4.16)$$

which is the reference prior that maximizes the expected Kullback–Leibler distance of the posterior distribution relative to the prior. Informally, the reference prior can be characterized as the distribution that maximizes the influence of the data on the posterior. Since the density (4.16) cannot be normalized such that it integrates to one, it is an improper prior. The prior can be expressed as the product between a uniform prior $p(\boldsymbol{\beta}) \propto 1$ for $\boldsymbol{\beta}$ and the prior $p(\sigma^2) \propto 1/\sigma^2$ for σ^2 so that $\boldsymbol{\beta}$ and σ^2 are a priori stochastically independent. Note that the prior for σ^2 is equivalent to a uniform prior for $\log(\sigma^2)$. Technically, we can identify the noninformative prior (4.16) with the conjugate $NIG(\mathbf{m}, \mathbf{M}, a, b)$ prior by setting $\mathbf{m} = \mathbf{0}$, $\mathbf{M}^{-1} = \mathbf{0}$, $a = -p$, and $b = 0$. This is useful for posterior analysis because we can treat the noninformative case within the standard prior. Note, however, that we have to be very careful when proceeding this way. When dealing with improper priors, it is important to check whether the resulting posterior is truly proper. For the noninformative prior (4.16) this is indeed the case.

Another approach to define a noninformative prior is described in [O'Hagan \(1994\)](#). Here we start with the marginal $IG(a, b)$ distribution for σ^2 . If $a \rightarrow 0$ and $b \rightarrow 0$ tend to zero we obtain

$$p(\sigma^2) \propto \frac{1}{\sigma^2}.$$

The prior corresponds to a uniform distribution for $\log \sigma^2$. For the joint $NIG(\mathbf{m}, \mathbf{M}, a, b)$ prior, we then have

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{(\sigma^2)^{\frac{p}{2}+1}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{m})' \mathbf{M}^{-1}(\boldsymbol{\beta} - \mathbf{m})\right).$$

If rather $\mathbf{M}^{-1} = \mathbf{0}$, we arrive at the alternative noninformative prior

$$p(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-(p+2)}.$$

This can be shown to be Jeffreys' prior. Although Jeffreys' prior is usually not used in multiparameter settings, our derivation justifies the widely used choice of a and b as equal and near zero as a weakly informative choice for the prior of σ^2 (and more generally variance parameters).

Posterior Analysis

Bayesian inference is based on properties of the posterior distribution, i.e., on the conditional distribution of the unknown parameters $\boldsymbol{\beta}$ and σ^2 given the data $\mathbf{y}$. The density of the posterior distribution is proportional to the product of the likelihood and the prior distribution. Hence, we obtain

$$\begin{aligned} p(\boldsymbol{\beta}, \sigma^2 | \mathbf{y}) &\propto L(\boldsymbol{\beta}, \sigma^2) p(\boldsymbol{\beta} | \sigma^2) p(\sigma^2) \\ &\propto \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right) \\ &\quad \frac{1}{(\sigma^2)^{\frac{p}{2}}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \mathbf{m})'\mathbf{M}^{-1}(\boldsymbol{\beta} - \mathbf{m})\right) \\ &\quad \frac{1}{(\sigma^2)^{a+1}} \exp\left(-\frac{b}{\sigma^2}\right). \end{aligned} \tag{4.17}$$

Technically, we can determine the properties of the posterior distribution mainly in two ways. On the one hand, we can derive the posterior distribution and its properties analytically. Alternatively, we can draw a random sample from the posterior distribution based on Markov chain Monte Carlo (MCMC) simulation methods. In most cases, an analytical derivation of the posterior distribution is impossible and only the time-consuming sampling-based approach is possible. The linear model is one of a few examples, in which the posterior distribution is analytically tractable, at least for the standard NIG($\mathbf{m}$, $\mathbf{M}$, a , b) prior. We can show that the posterior distribution, like the prior distribution, is a normal-inverse gamma distribution. The parameters $\tilde{\mathbf{m}}$, $\tilde{\mathbf{M}}$, $\tilde{a}$, and $\tilde{b}$ of the distribution can be found in Box 4.5; see Sect. 4.5.2, p. 260, for a derivation. Additionally, the box contains conditional and marginal posteriors of the parameters as well as other quantities of interest. They can be obtained directly from the properties of the normal-inverse gamma distribution derived on p. 228 and summarized in Definition B.3.5 (p. 652) of Appendix B.

Of particular interest is the posterior mean

$$\hat{\boldsymbol{\beta}}_B = \mathbf{E}(\boldsymbol{\beta} | \mathbf{y}) = \tilde{\mathbf{m}} = (\mathbf{X}'\mathbf{X} + \mathbf{M}^{-1})^{-1}(\mathbf{M}^{-1}\mathbf{m} + \mathbf{X}'\mathbf{y})$$

4.5 Bayesian Linear Model with Conjugate Prior

Observation Model and Prior Distribution

1. *Observation model:* $y \mid \beta, \sigma^2 \sim N(X\beta, \sigma^2 I)$.
2. *Prior distribution:* $\beta \mid \sigma^2 \sim N(m, \sigma^2 M)$ and $\sigma^2 \sim \text{IG}(a, b)$.

Posterior

$\beta, \sigma^2 \mid y \sim \text{NIG}(\tilde{m}, \tilde{M}, \tilde{a}, \tilde{b})$ with parameters

$$\tilde{M} = (X'X + M^{-1})^{-1} \quad \tilde{m} = \tilde{M}(M^{-1}m + X'y),$$

and

$$\tilde{a} = a + \frac{n}{2} \quad \tilde{b} = b + \frac{1}{2} (y'y + m'M^{-1}m - \tilde{m}'\tilde{M}^{-1}\tilde{m}).$$

- The conditional posterior distribution of β given σ^2 is $\beta \mid \sigma^2, y \sim N(\tilde{m}, \sigma^2 \tilde{M})$.
- The marginal posterior of β is $\beta \mid y \sim t(2\tilde{a}, \tilde{m}, \tilde{b}/\tilde{a}\tilde{M})$.

Posterior with Noninformative Prior

In case of a noninformative prior with $m = 0$, $M^{-1} = 0$, $a = -p$, and $b = 0$, we obtain

$$\tilde{M} = (X'X)^{-1} \quad \tilde{m} = (X'X)^{-1}X'y = \hat{\beta}_{\text{LS}}$$

and

$$\tilde{a} = -p + \frac{n}{2} \quad \tilde{b} = \frac{1}{2} (y'y - \hat{\beta}'_{\text{LS}}X'X\hat{\beta}_{\text{LS}}).$$

- The conditional posterior distribution of β given σ^2 is $\beta \mid \sigma^2, y \sim N(\hat{\beta}_{\text{LS}}, \sigma^2(X'X)^{-1})$.
- The marginal posterior of β is $\beta \mid y \sim t(2\tilde{a}, \hat{\beta}_{\text{LS}}, \tilde{b}/\tilde{a}\tilde{M})$.

Posterior Mean

$$\hat{\beta}_B = E(\beta \mid y) = \tilde{m} = (X'X + M^{-1})^{-1}(M^{-1}m + X'y).$$

For a noninformative prior the posterior mean coincides with the least squares estimator.

Software

- Functions `bayesLMRef` and `bayesLMConjugate` of the R package `spBayes`.
- Software package `BayesX` for noninformative priors only (see also the R interface `R2BayesX`).
- Function `zlm` of the R package `BMS` (Zellner's g -prior only).

as a point estimate of β . Using the matrix $A = (X'X + M^{-1})^{-1}X'X$, we can write the Bayes estimator as a weighted average of the prior expectation m and the least squares estimator $\hat{\beta}$:

$$\hat{\beta}_B = (I - A)m + A\hat{\beta}.$$

To interpret the Bayes estimator, note that the diagonal elements of M contain (up to the factor σ^2) the prior variances of β . The greater the diagonal elements of M (i.e., the variances of β), the smaller are the elements of M^{-1} . In the limit $M^{-1} \rightarrow 0$ the matrix A approaches the identity matrix and $\hat{\beta}_B$ the ordinary least squares estimator. On the contrary, small elements in M (corresponding to small variances of β) imply that the matrix A approaches the zero matrix and $I - A$ the identity matrix. The Bayes estimator is then identical with the prior mean m . This gives us the following interpretation of $\hat{\beta}_B$: The smaller the prior information about β , i.e., the greater the diagonal elements of M , the closer is $\hat{\beta}_B$ to the least squares estimator. The larger the prior information, i.e., the smaller the diagonal elements of M , the more the prior mean m dominates $\hat{\beta}_B$. For the noninformative prior with $m = 0$ and $M^{-1} = 0$ the Bayes estimator coincides with the least squares estimator, i.e.,

$$\hat{\beta}_B = (X'X)^{-1}X'y = \hat{\beta}_{LS}.$$

Full Conditional Densities and MCMC Inference

Since the linear model is often a building block for more complex models where analytical posterior analysis is impossible, we additionally discuss simulation-based inference using MCMC methods. We develop a Gibbs sampler (see Appendix B.5.3 for an introduction), that consecutively draws random numbers from the full conditional distributions of β and σ^2 .

We start by deriving the full conditional of β . The density is proportional to the density (4.17) of the posterior distribution, and we can disregard all factors which are independent of β . In the appendix of this chapter on p. 211, we derive the identity

$$(y - X\beta)'(y - X\beta) = (\beta - \hat{\beta})'X'X(\beta - \hat{\beta}) + y'(I_n - X(X'X)^{-1}X')y \quad (4.18)$$

for the least squares criterion. Since the second summand does not depend on β , we obtain

$$\begin{aligned} p(\beta | \cdot) &\propto \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right) \exp\left(-\frac{1}{2\sigma^2}(\beta - m)'M^{-1}(\beta - m)\right) \\ &\propto \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})'X'X(\beta - \hat{\beta})\right) \exp\left(-\frac{1}{2\sigma^2}(\beta - m)'M^{-1}(\beta - m)\right) \end{aligned}$$

for the full conditional of β . This is the form of a multivariate normal distribution; see Appendix B, Theorem B.4. According to the theorem, the covariance matrix is given by

$$\Sigma_{\beta} = \left(\frac{1}{\sigma^2} X'X + \frac{1}{\sigma^2} M^{-1} \right)^{-1}. \quad (4.19)$$

For the mean μ_{β} , we obtain

$$\mu_{\beta} = \Sigma_{\beta} \left(\frac{1}{\sigma^2} X'y + \frac{1}{\sigma^2} M^{-1}m \right). \quad (4.20)$$

In summary, we have $\beta \mid \cdot \sim N(\mu_{\beta}, \Sigma_{\beta})$.

Similarly, the full conditional distribution of σ^2 can be derived. We have

$$\begin{aligned} p(\sigma^2 \mid \cdot) &\propto \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp\left(-\frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta)\right) \\ &\quad \frac{1}{(\sigma^2)^{\frac{p}{2} + a + 1}} \exp\left(-\frac{1}{2\sigma^2} (\beta - m)'M^{-1}(\beta - m) - \frac{b}{\sigma^2}\right) \\ &= \frac{1}{(\sigma^2)^{a + \frac{n}{2} + \frac{p}{2} + 1}} \\ &\quad \exp\left(-\frac{1}{\sigma^2} \left(b + \frac{1}{2}(y - X\beta)'(y - X\beta) + \frac{1}{2}(\beta - m)'M^{-1}(\beta - m)\right)\right). \end{aligned}$$

This is the form of an inverse gamma distribution with parameters

$$a' = a + \frac{n}{2} + \frac{p}{2} \quad (4.21)$$

and

$$b' = b + \frac{1}{2}(y - X\beta)'(y - X\beta) + \frac{1}{2}(\beta - m)'M^{-1}(\beta - m). \quad (4.22)$$

Summarizing, we obtain the following Gibbs sampler:

1. Define initial values $\beta^{(0)}$ and $(\sigma^2)^{(0)}$. Set $t = 1$.
2. Sample $\beta^{(t)}$ by drawing from the Gaussian full conditional with covariance matrix (4.19) and mean (4.20). Replace in (4.19) and (4.20) σ^2 by the current state of the chain $(\sigma^2)^{(t-1)}$.
3. Sample $(\sigma^2)^{(t)}$ by drawing from the inverse gamma full conditional with parameters a' and b' given by Eqs. (4.21) and (4.22). Replace in Eqs. (4.21) and (4.22) β by the current state of the chain $\beta^{(t)}$.
4. Stop if $t = T$, otherwise set $t = t + 1$ and go to 2.

Example 4.13 Munich Rent Index—Quality of Kitchen

In Example 3.7 (p. 100), we were confronted with the problem to update the Munich rent index 1999 with new data collected in 2001. Recall that we constructed a two period model with an interaction between the quality of kitchen and the data collection period. In this example, we follow a Bayesian approach to cope with the update problem. For illustration, we again use model (3.7) of Example 3.5 (p. 90) augmented by the kitchen dummy variables. Hence, the model consists of the transformed living area $1/area$, a cubic polynomial for year of construction, and the two kitchen dummies $nkitchen$ (“normal kitchen”) and $pkitchen$ (“premium kitchen”). We first develop a Bayesian version of the

Table 4.8 Munich rent index: estimation results based on the noninformative prior (4.16) for the parameters

Variable	Coefficient	Standard deviation	2.5 % Quantile	97.5 % Quantile
<i>invarea</i>	122.5,417	5.5,877	−111.5,955	−133.7,277
<i>yearc</i>	−0.0861	0.0351	−0.1,549	−0.0174
<i>yearc2</i>	0.0015	0.0007	0.0002	0.0028
<i>yearc3</i>	0.0000	0.0000	0.0000	0.0000
<i>nkitchen</i>	0.9274	0.1258	0.6770	1.1840
<i>pkitchen</i>	1.1022	0.1873	−0.7410	1.4718

Results are based on the data collected in 1999

model based solely on the data for 1999. At that time we have no prior information regarding the unknown parameters. We therefore use the noninformative prior (4.16). Although the model could in principle be estimated analytically using the results of Box 4.5, we used the Gibbs sampler outlined on p. 234 for inference. The reason is that the available software packages usually do not support the analytical solution. Using the function `bayesLMRef` of the R package `spBayes`, we obtained the results of Table 4.8. Up to sampling imprecision, the posterior mean is identical to the least squares estimator (as suggested by the analytically derived posterior). The posterior standard deviations and the quantiles are also very close to the respective least squares standard errors and the 95 % confidence intervals. Note that the analytical posterior mean (and mode) coincides exactly with the least squares estimator, while the posterior standard deviation and quantiles are slightly different from their least squares counterpart.

Turning our attention to the new data collected in 2001 for the rent index update, it seems natural to use the posterior values obtained with the data from 1999 as prior information for the new analysis. That is we estimate a Bayesian linear model using the data collected in 2001 together with a $\text{NIG}(\mathbf{m}, \mathbf{M}, a, b)$ prior with parameters derived from the posterior obtained with the 1999 data. Clearly, $\mathbf{m}$ should be the empirical mean vector of the sampled regression coefficients in the MCMC sampler for the 1999 data. Since the prior covariance matrix is given by $\sigma^2 \mathbf{M}$, we set $\mathbf{M} = 1/\hat{\sigma}^2 \mathbf{S}$, where $\mathbf{S}$ is the empirical covariance matrix of the MCMC samples for the 1999 data and $\hat{\sigma}^2$ the empirical mean of the samples for σ^2 . To obtain prior values for a and b , we note that the mean and variance of the $\text{IG}(a, b)$ prior for σ^2 are given by

$$\begin{aligned} E(\sigma^2) &= \frac{b}{a-1}, \\ \text{Var}(\sigma^2) &= \frac{b^2}{(a-1)^2(a-2)} = E(\sigma^2)^2 \frac{1}{a-2}. \end{aligned}$$

Solving for a and b yields

$$\begin{aligned} a &= \frac{E(\sigma^2) + 2\text{Var}(\sigma^2)}{\text{Var}(\sigma^2)}, \\ b &= (a-1)E(\sigma^2). \end{aligned}$$

Based on the analysis for the 1999 data, we can now replace $E(\sigma^2)$ and $\text{Var}(\sigma^2)$ by their posterior estimates $\hat{\sigma}^2$ and $s_{\hat{\sigma}^2}^2$ to obtain prior values for a and b .

Using these values for the $\text{NIG}(\mathbf{m}, \mathbf{M}, a, b)$ prior, we arrive at the results given in Table 4.9. The table is obtained from the function `bayesLMConjugate` in the R package `spBayes`. For comparison, we additionally included the least squares estimates for the

Table 4.9 Munich rent index: comparison of the Bayes estimate with informative prior and the least squares estimate for the data collected in 2001

Variable	LS 2001		Bayes 2001	
	Coeff.	Std	Coeff.	Std
<i>invarea</i>	125.8373	7.2360	124.3540	3.3618
<i>yearc</i>	−0.0335	0.0480	−0.0631	0.0208
<i>yearc2</i>	0.0004	0.0009	0.0010	0.0004
<i>yearc3</i>	0.0000	0.0000	0.0000	0.0000
<i>nkitchen</i>	1.2944	0.1701	1.0327	0.0795
<i>pkitchen</i>	1.7935	0.2714	1.2910	0.1193

2001 data. In particular for the kitchen dummies, the least squares estimates for the 2001 data differ considerably from those for the 1999 data (see Table 4.8). The Bayes estimator is a compromise between the least squares results for the 1999 and the 2001 data. Although the new data based on 2001 have an impact on the estimates for *nkitchen* and *pkitchen*, the Bayes estimator is closer to the least squares estimate for the 1999 data. This is a clear result of the prior which pulls the posterior mean to a certain extent towards the prior mean, which is identical to the 1999 least squares estimate. We also observe that the posterior standard deviations of the regression coefficients are considerably lower than the least squares standard errors. This is again a result of the use of additional prior information. $\triangle$

4.4.2 Regularization Priors

In recent years, a number of “regularization priors” have been proposed in the literature. The main idea is to define priors that have a similar effect as the penalties in the penalized least squares approaches of Sect. 4.2. In this section we will develop some of the most widely used Bayesian regularization priors. As this topic is subject to extensive current research, our presentation cannot be exhaustive (but see Sect. 4.5.1 for references). We start with a Bayesian version of ridge regression, followed by Bayesian LASSO.

Throughout this section, the observation model is a classical linear model given by

$$\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2 \sim N(\beta_0 \mathbf{1} + \tilde{\mathbf{X}} \tilde{\boldsymbol{\beta}}, \sigma^2 \mathbf{I}),$$

where $\tilde{\mathbf{X}}$ is the $n \times k$ -design matrix excluding the column of ones for the intercept and $\tilde{\boldsymbol{\beta}}$ is the corresponding vector of regression coefficients excluding β_0 . Since the intercept is not subject to regularization, we assume a noninformative (diffuse) prior, i.e.,

$$p(\beta_0) \propto \text{const.}$$

We also assume that the intercept β_0 is independent of the other regression coefficients $\tilde{\boldsymbol{\beta}}$. For the variance parameter σ^2 we specify the usual inverse gamma prior with hyperparameters a and b , i.e., $\sigma^2 \sim \text{IG}(a, b)$.

Bayesian Ridge Regression

Ridge regression minimizes the penalized least squares criterion

$$\text{PLS}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \tilde{\boldsymbol{\beta}}' \tilde{\boldsymbol{\beta}}; \quad (4.23)$$

see Sect. 4.2.2. Usually the intercept is not penalized so that here the penalty is restricted to $\tilde{\boldsymbol{\beta}} = (\beta_1, \dots, \beta_k)'$. The penalty shrinks the parameters towards zero in order to reduce the variance of the least squares estimator at the cost of a (typically small) bias. The amount of penalization is governed by the parameter λ . Small values for λ correspond to negligible penalization, whereas large values lead to strong penalization. Using a particular prior for the regression coefficients, we obtain a Bayesian version of ridge regression. More specifically, we assume a priori independent regression coefficients β_j , $j = 1, \dots, k$, and set

$$\tilde{\boldsymbol{\beta}} \mid \tau^2 \sim \mathbf{N}(\mathbf{0}, \tau^2 \mathbf{I}). \quad (4.24)$$

Since the prior for the intercept is improper, the joint prior for $\boldsymbol{\beta} = (\beta_0, \tilde{\boldsymbol{\beta}})'$ is also improper. It can be identified with a singular multivariate normal distribution (see Appendix B.3.2) with mean $\mathbf{0}$ and precision matrix $\mathbf{K} = 1/\tau^2 \text{diag}(0, 1, \dots, 1)$.

We will show in the appendix of this chapter on p. 261 that maximizing the corresponding posterior with respect to $\boldsymbol{\beta}$ is equivalent to minimizing the penalized least squares criterion (4.23) for fixed $\lambda = \sigma^2/\tau^2$.

While in the frequentist approach to ridge regression, the penalty parameter λ is estimated outside of the optimization criterion, e.g., via cross validation, the Bayesian approach allows for simultaneous inference for the regression coefficients and the amount of penalization measured through τ^2 . This is obtained by defining an additional prior for τ^2 . A convenient and flexible choice is an inverse gamma distribution $\tau^2 \sim \text{IG}(a_{\tau^2}, b_{\tau^2})$, similar to the conjugate prior for σ^2 outlined above. The advantage of this specification is that the full conditional for τ^2 is again an inverse gamma distribution allowing for straightforward simulation-based MCMC inference; see p. 240 below.

Introducing an additional prior for τ^2 , however, changes the interpretation of the prior. This is illustrated in Fig. 4.21 (left panel) which displays the log-prior density, for a single parameter β_j , conditional on τ^2 and the marginal log-prior with the variance parameter τ^2 integrated out. The marginal log-prior is quite different to the Gaussian conditional log-prior. It shows a distinct peak at zero with sharp declines. To derive the marginal prior for $\tilde{\boldsymbol{\beta}}$ formally, we note that the joint prior for $\tilde{\boldsymbol{\beta}}, \tau^2$ is a NIG($\mathbf{0}, \mathbf{I}, a_{\tau^2}, b_{\tau^2}$) distribution. The marginal prior for $\tilde{\boldsymbol{\beta}}$ is then a multivariate t-distribution with $2a_{\tau^2}$ degrees of freedom, location parameter $\mathbf{0}$, and dispersion matrix $b_{\tau^2}/a_{\tau^2} \mathbf{I}$; see Property 5 in Appendix B.3.5. Note that the diagonal dispersion matrix implies that the regression coefficients are marginally uncorrelated. However, they are not stochastically independent as would be the case if the marginal distribution would be multivariate Gaussian rather than a multivariate t-distribution. In a multivariate t-distribution, a diagonal dispersion matrix implies uncorrelated but *not* stochastically independent components.

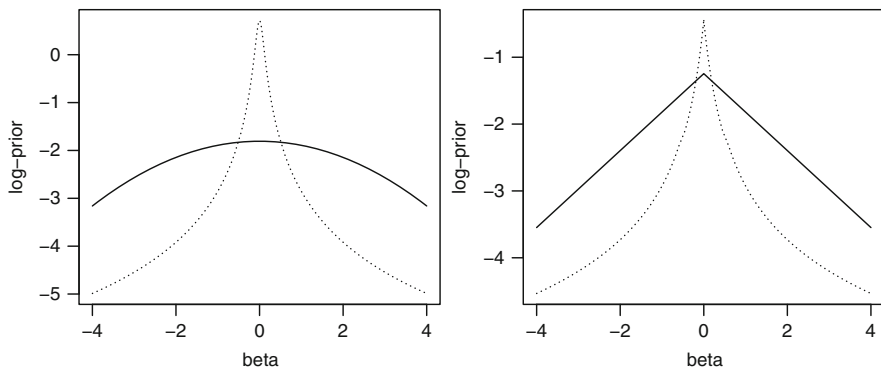


Fig. 4.21 Conditional (solid lines) and marginal (dashed lines) log-priors for the ridge (left panel) and the LASSO prior (right panel). In case of the ridge prior, the plots are based on $a = 0.28$ and $b = 0.005$ for the inverse gamma prior. In case of the LASSO, the plots correspond to $a = 0.08$ and $b = 0.001$. With these choices, roughly 90% of the probability mass are contained in the interval $[-4, 4]$. The hyperparameters are chosen such that the differences between conditional and marginal distributions and between ridge and LASSO are best visible

Bayesian LASSO

The LASSO replaces the quadratic penalty of ridge regression by the sum of absolute values leading to the penalized least squares criterion

$$\text{PLS}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^k |\beta_j|. \quad (4.25)$$

Similar to ridge regression, we define a prior for the regression coefficients such that the corresponding posterior mode is obtained by minimizing (4.25). We again assume (conditional) independence among the regression coefficients and arrive at the prior

$$\tilde{\boldsymbol{\beta}} \mid \tau_1^2, \dots, \tau_k^2 \sim \mathcal{N}(\mathbf{0}, \text{diag}(\tau_1^2, \dots, \tau_k^2)), \quad (4.26)$$

where now each regression coefficient β_j has its own variance τ_j^2 . The joint prior of $\boldsymbol{\beta} = (\beta_0, \tilde{\boldsymbol{\beta}}')$ can again be identified with a singular multivariate normal distribution with mean $\mathbf{0}$ and precision matrix $\mathbf{K} = \text{diag}(0, 1/\tau_1^2, \dots, 1/\tau_k^2)$.

The variance parameters are assumed mutually independent with priors

$$\tau_j^2 \mid \sim \text{Expo}(0.5\lambda^2);$$

see Definition B.10 of Appendix B.1 for the exponential distribution. The marginal distribution for β_j obtained by integrating over τ_j^2 is a Laplace distribution with

location parameter 0 and scale parameter $1/\lambda$, i.e., $p(\beta_j) \propto \exp(-\lambda|\beta_j|)$; see Definition B.8 in Appendix B.1 for the Laplace distribution. Based on this prior specification it can be shown that the posterior mode for β with fixed penalty parameter λ corresponds to minimizing the penalized least squares criterion (4.25). The proof is left to the reader as an exercise.

Similar to Bayesian ridge regression, we can assign a hyperprior for λ that allows for simultaneous estimation of the regression coefficients and the amount of penalization. Since the precision of the regression coefficients is given by $\text{Var}(\beta_j)^{-1} = 2\lambda^2$, we assign a gamma distribution to λ^2 , i.e., $\lambda^2 \sim G(a_\lambda, b_\lambda)$.

Summarizing, the joint prior for β , τ_j^2 , $j = 0, \dots, k$, and λ factors as

$$p(\beta, \tau_1^2, \dots, \tau_k^2, \lambda) = p(\beta_0) p(\tilde{\beta} | \tau_1^2, \dots, \tau_k^2) p(\tau_1^2 | \lambda) \cdots p(\tau_k^2 | \lambda) p(\lambda^2).$$

A summary of all prior assumptions for the Bayesian LASSO (and ridge regression) can be found in Box 4.6.

We finally compare the LASSO prior with the ridge prior. The right panel of Fig. 4.21 shows the log-prior $\log p(\beta_j | \lambda)$, for a single parameter β_j , conditional on the parameter λ together with the marginal log-prior with λ integrated out. As stated, the conditional prior is a Laplace distribution and therefore quite different from the Gaussian conditional log-prior in case of ridge regression (see the left panel of the figure). Somewhat surprisingly, the marginal log-priors appear to be similar although the LASSO prior still has heavier tails than the ridge prior. This is the reason why the Bayesian variants of ridge regression and the LASSO behave often very similar in empirical studies; see Example 4.14.

Posterior Inference

In case of regularization priors the posterior is analytically intractable because of the various variance hyperparameters involved. Therefore inference relies on MCMC simulation techniques. In both cases, a Gibbs sampler can be derived to subsequently draw from the full conditionals of the parameters. Box 4.7 summarizes the resulting algorithms. For the LASSO, the derivation is given in the appendix of this chapter on p. 262. For Bayesian ridge regression the derivation is in complete analogy and left to the reader as an exercise.

Example 4.14 Prices of Used Cars—Bayesian Ridge and LASSO



This example compares the Bayesian variants of ridge regression and the LASSO with their classical counterparts described in Sect. 4.2. Using the software package BayesX, we obtained the estimates displayed in Table 4.10 together with their 95 % credible intervals.

The Bayesian ridge estimates behave quite similar to classical ridge regression. Both variants provide similar results which are also very close to the unpenalized least squares estimate for these data. On the other hand, both LASSO variants show pronounced differences. Most striking is that the Bayesian LASSO does not allow removal of a covariate from the model, as is possible with the classical LASSO. The reason is that the Bayesian LASSO point estimator is the posterior mean or median (rather than the posterior mode) estimated via MCMC. Since the posterior for the regression coefficients

4.6 Regularization Priors

Observation Model

$$y \mid \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\boldsymbol{\beta}_0 \mathbf{1} + \tilde{\mathbf{X}} \tilde{\boldsymbol{\beta}}, \sigma^2 \mathbf{I})$$

Common Prior Assumptions

$$p(\boldsymbol{\beta}_0) \propto \text{const}$$

$$\sigma^2 \sim \text{IG}(a, b)$$

Ridge Prior

$$\tilde{\boldsymbol{\beta}} \mid \tau^2 \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$$

$$\tau^2 \sim \text{IG}(a_{\tau^2}, b_{\tau^2})$$

LASSO Prior

$$\tilde{\boldsymbol{\beta}} \mid \tau_1^2, \dots, \tau_k^2 \sim \mathcal{N}(\mathbf{0}, \text{diag}(\tau_1^2, \dots, \tau_k^2))$$

$$\tau_j^2 \mid \lambda \stackrel{iid}{\sim} \text{Expo}(0.5\lambda^2)$$

$$\lambda^2 \sim \mathcal{G}(a_\lambda, b_\lambda)$$

Software

Software package `BayesX`.

is typically skewed, the posterior mean and median will not coincide with the posterior mode, and, as a consequence, will always be different from zero. This is a distinct disadvantage of the Bayesian LASSO, as the main attraction of the classical LASSO is lost: the ability to perform variable selection. On the other hand, the sampling-based approach provides richer information regarding the posterior, such as posterior standard deviations and quantiles. A possible way to identify redundant covariates with the Bayesian LASSO (as well as Bayesian ridge regression) is to dismiss all covariates which have Bayesian credible intervals covering zero. Bayesian credible intervals are easily constructed from the corresponding posterior quantiles. For instance, the 95 % credible intervals displayed in Table 4.10 are composed of the 2.5 % quantile as the lower bound and the 97.5 % quantile as the upper bound. Based on the 95 % credible intervals only the variables *ageop1*,

4.7 Regularization Priors—Gibbs Sampler

1. *Initialization:*

- Define initial values $\boldsymbol{\beta}^{(0)}$, $(\sigma^2)^{(0)}$, and $(\tau^2)^{(0)}$ (ridge), $(\tau_1^2)^{(0)}, \dots, (\tau_k^2)^{(0)}$, $\lambda^{(0)}$ (LASSO).
- Set $t = 1$ and specify the number of iterations T .

2. *Sample $\boldsymbol{\beta}$:* Draw $\boldsymbol{\beta}^{(t)} \mid \cdot \sim N(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta)$ with $\boldsymbol{\mu}_\beta$ and $\boldsymbol{\Sigma}_\beta$ given by

$$\boldsymbol{\Sigma}_\beta = \left(\frac{1}{\sigma^2} \mathbf{X}'\mathbf{X} + \mathbf{K} \right)^{-1} \quad \boldsymbol{\mu}_\beta = \frac{1}{\sigma^2} \boldsymbol{\Sigma}_\beta \mathbf{X}'\mathbf{y}.$$

Here $\mathbf{K} = 1/\tau^2 \cdot \text{diag}(0, 1, \dots, 1)$ in case of ridge regression and $\mathbf{K} = (0, 1/\tau_1^2, \dots, 1/\tau_k^2)$ in case of LASSO.

3. *Sample σ^2 :* Sample $(\sigma^2)^{(t)}$ from the full conditional of σ^2 which is inverse gamma with parameters

$$a^{\text{new}} = a + \frac{n}{2}, \quad b^{\text{new}} = b + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

4. *Sample variance parameters:*

- For the ridge prior draw $(\tau^2)^{(t)} \mid \cdot$ from an inverse gamma distribution with parameters

$$a^{\text{new}} = a + \frac{k}{2}, \quad b^{\text{new}} = b + \frac{1}{2}\tilde{\boldsymbol{\beta}}'\tilde{\boldsymbol{\beta}}.$$

- For the LASSO prior, sample

$$(1/\tau_j^2)^{(t)} \mid \cdot \sim \text{InvGauss}\left(\frac{|\lambda|}{|\tilde{\beta}_j|}, \lambda^2\right),$$

$$(\lambda^2)^{(t)} \mid \cdot \sim G\left(a + k, b + \frac{1}{2} \sum_{j=1}^k \tau_j^2\right).$$

5. Stop if $t = T$, otherwise set $t = t + 1$ and proceed with step 2.

ageop2, *kilometerop1*, and *kilometerop2* are “relevant”, while the others are redundant (see Table 4.10).

Finally we note that the Bayesian LASSO appears to induce less shrinkage than the classical LASSO. In fact, Bayesian ridge regression and LASSO show quite similar shrinkage behavior. Indeed, Table 4.10 shows that both regularization variants, albeit conceptually different, produce almost identical posterior estimates. This is in agreement with our theoretical findings on p. 239; see in particular Fig. 4.21.

△

Table 4.10 Prices of used cars: posterior mean and 95 % credible intervals for Bayesian ridge regression and LASSO

Variable	Bayesian ridge		Bayesian LASSO		Ridge	LASSO	LS
	Coeff.	95 % CI	Coeff.	95 % CI	Coeff.	Coeff.	Coeff.
<i>ageop1</i>	−0.682	(−0.802, −0.559)	−0.694	(−0.813, −0.563)	−0.672	−0.682	−0.709
<i>ageop2</i>	0.165	(0.052, 0.280)	0.163	(0.052, 0.270)	0.164	0.150	0.172
<i>ageop3</i>	0.014	(−0.089, 0.128)	0.011	(−0.099, 0.113)	0.015	–	0.016
<i>kilometerop1</i>	−0.428	(−0.541, −0.308)	−0.424	(−0.545, −0.303)	−0.425	−0.412	−0.437
<i>kilometerop2</i>	0.140	(0.028, 0.252)	0.126	(0.012, 0.244)	0.138	0.110	0.142
<i>kilometerop3</i>	0.013	(−0.103, 0.125)	0.009	(−0.085, 0.108)	0.010	–	0.009
<i>TIA</i>	−0.005	(−0.021, 0.011)	−0.004	(−0.020, 0.012)	−0.005	–	−0.005
<i>extras1</i>	−0.093	(−0.332, 0.152)	−0.075	(−0.306, 0.124)	−0.104	−0.036	−0.114
<i>extras2</i>	−0.030	(−0.257, 0.211)	−0.022	(−0.240, 0.200)	−0.042	–	−0.031

For comparison the last three columns contain results for classical ridge and LASSO regression as well as the least squares estimator

4.4.3 Classical Bayesian Model Choice (and Beyond)

The classical approach to Bayesian model choice is to compare the models under consideration through their posterior model probabilities (PMP); see Appendix B.5.4 for a general introduction to Bayesian model choice. In this section, we apply this framework to linear models.

Suppose we are given a number of potential covariates and there is uncertainty as to which of the covariates should enter the model. For k possible regressors there are 2^k different models when the intercept β_0 is always included. Denote by M_r , $r = 1, \dots, 2^k$, the different models. More specifically, M_r is given by

$$y | \beta, \sigma^2, M_r = y | \beta_0, \tilde{\beta}_r, \sigma^2, M_r \sim N(\beta_0 \mathbf{1} + \tilde{X}_r \tilde{\beta}_r, \sigma^2 I),$$

where the $k_r \times n$ -design matrix $\tilde{X}_r$ consists of all k_r covariates included in M_r and $\tilde{\beta}_r$ is the corresponding vector of regression coefficients. As usual, the vector β is the full vector of regression coefficients (including the intercept). Those components of β not contained in $\tilde{\beta}_r$ are zero in model M_r .

We assign the normal-inverse gamma prior discussed in Sect. 4.4.1 to β_0 , $\tilde{\beta}_r$, and the error variance σ^2 , i.e., $\beta_0, \tilde{\beta}_r | \sigma^2, M_r \sim N(\mathbf{m}_r, \sigma^2 \mathbf{M}_r)$, and $\sigma^2 | M_r = \sigma^2 \sim \text{IG}(a, b)$. This results in the prior distribution

$$p(\beta_0, \tilde{\beta}_r, \sigma^2 | M_r) = p(\beta_0, \tilde{\beta}_r | \sigma^2, M_r) p(\sigma^2) = N(\mathbf{m}_r, \sigma^2 \mathbf{M}_r) \cdot \text{IG}(a, b).$$

For completeness, we combine the zero components of β in the $(k - k_r)$ -dimensional vector $\tilde{\beta}_{-r}$ with a Dirac prior at $(0, \dots, 0)'$

$$p(\tilde{\beta}_{-r} | \beta_0, \tilde{\beta}_r, \sigma^2, M_r) = p(\tilde{\beta}_{-r} | M_r) = \text{Dirac}(0, \dots, 0).$$

The Dirac prior concentrates all probability mass onto the point $(0, \dots, 0)'$, i.e., $P(\tilde{\beta}_{-r} = (0, \dots, 0)' | M_r) = 1$ and zero otherwise. Thus the prior for β and σ^2 under model M_r factors into

$$p(\beta, \sigma^2 | M_r) = p(\tilde{\beta}_{-r} | \beta_0, \tilde{\beta}_r, \sigma^2, M_r) p(\beta_0, \tilde{\beta}_r, \sigma^2 | M_r).$$

To compare models using their posterior probabilities, we additionally have to assign prior probabilities to each model M_r . A popular prior is

$$p(M_r) = \theta^{k_r} (1 - \theta)^{k - k_r}, \quad (4.27)$$

i.e., every possible covariate enters the model independently and with inclusion probability $\theta \in (0, 1)$. The “natural” choice $\theta = 1/2$ results in a uniform prior $p(M_r) = 1/2^k$, i.e., each model M_r has the same prior probability.

Prior (4.27) with inclusion probability θ implies a certain prior distribution on the size of the models, denoted by S . Let δ_j , $j = 1, \dots, k$, be inclusion indicators with $\delta_j = 1$ if covariate x_j is included in the model and 0 otherwise. Then δ_j has a Bernoulli distribution, i.e., $\delta_j \sim B(1, \theta)$, and the model size S has a binomial distribution with

$$S = \sum_{j=1}^k \delta_j \sim B(k, \theta),$$

resulting in a prior mean model size of $E(S) = \theta \cdot k$ and variance $\text{Var}(S) = \theta \cdot (1 - \theta) \cdot k$ (see Definition B.1 in Appendix B.1). In light of these results, the choice $\theta = 1/2$ seems less natural as suggested at first sight. In particular for a large number k of possible predictors, the prior expected model size appears to be much higher than one would typically expect in applications. For instance, for $k = 50$ potential covariates, the prior expected model size of $E(S) = 25$ seems to be far too high for most applications.

A convenient way to elicit the model prior (4.27) is to specify the prior mean model size $E(S)$ and then to set $\theta = E(S)/k$. For instance, if we have $k = 20$ potential regressors and assume a priori a model size of $E(S) = 5$, then we must set $\theta = 5/20 = 1/4$.

Based on our prior assumptions, the posterior probability for model M_r is given by

$$p(M_r | y) = \frac{p(y | M_r) p(M_r)}{\sum_{h=1}^{2^k} p(y | M_h) p(M_h)}, \quad (4.28)$$

where $p(y | M_r)$ is the marginal likelihood obtained as

$$p(\mathbf{y} | M_r) = \int p(\mathbf{y} | \beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2, M_r) p(\beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2 | M_r) d\beta_0 d\tilde{\boldsymbol{\beta}}_r d\sigma^2.$$

It can be shown that $p(\mathbf{y} | M_r)$ is multivariate t-distributed with $2a$ degrees of freedom, location parameter $\mathbf{X}_r \mathbf{m}_r$, and dispersion matrix $\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r'$, where $\mathbf{X}_r = (\mathbf{1} \tilde{\mathbf{X}}_r)$; see p. 264 in the appendix of this chapter for the derivation. The Bayes factor for two competing models M_r and M_s is obtained as

$$\text{BF}_{rs} = \frac{p(\mathbf{y} | M_r)}{p(\mathbf{y} | M_s)}.$$

With $\tilde{\mathbf{M}}_r = (\mathbf{X}_r' \mathbf{X}_r + \mathbf{M}_r^{-1})^{-1}$, it results in

$$\text{BF}_{rs} = \left(\frac{|\tilde{\mathbf{M}}_r| |\mathbf{M}_s|}{|\tilde{\mathbf{M}}_s| |\mathbf{M}_r|} \right)^{1/2} \left(\frac{2a + (\mathbf{y} - \mathbf{X}_s \mathbf{m}_s)' (\mathbf{I} - \mathbf{X}_s \tilde{\mathbf{M}}_s \mathbf{X}_s') (\mathbf{y} - \mathbf{X}_s \mathbf{m}_s)}{2a + (\mathbf{y} - \mathbf{X}_r \mathbf{m}_r)' (\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r') (\mathbf{y} - \mathbf{X}_r \mathbf{m}_r)} \right)^{a+n/2}; \quad (4.29)$$

see again p. 264 for a derivation.

Once the posterior probabilities $p(M_r | \mathbf{y})$ are computed for every model M_r under consideration, there are several ways to summarize the results. If the primary focus is on selecting one single (preferably sparse) model, often the model M_* with *highest posterior probability* is taken and inference for the regression coefficients is based on the posterior $p(\boldsymbol{\beta}_*, \sigma^2 | \mathbf{y}, M_*)$ conditional on model M_* . If model selection is done by minimizing the BIC, as in Sect. 3.4.2 of the previous chapter, we exactly follow this strategy. However, [Barbieri and Berger \(2004\)](#) point out that often the model M_* with highest posterior probability is not optimal in terms of prediction. They show that the optimal predictive model is often the *median probability model*. This model consists of those covariates with posterior probability of 1/2 and higher for being in the model.

Both approaches, however, ignore model uncertainty. In many applications, there are a number of models which are close in terms of posterior probabilities. If this is the case, inference for $\boldsymbol{\beta}$ (or any other quantity of interest) is better conducted by *model averaging* where the models are weighted by their posterior probability. More specifically, the posterior is given by

$$p(\boldsymbol{\beta}, \sigma^2 | \mathbf{y}) = \sum_{r=1}^{2^k} p(\boldsymbol{\beta}, \sigma^2 | \mathbf{y}, M_r) p(M_r | \mathbf{y}), \quad (4.30)$$

where $p(\boldsymbol{\beta}, \sigma^2 | \mathbf{y}, M_r)$ is the conditional posterior under model M_r and $p(M_r | \mathbf{y})$ is the corresponding posterior model probability given in Eq. (4.28). While models with high posterior probability make important contributions to the posterior, those with negligible posterior probability will contribute only very little information. If most models coincide in their posterior assessment of specific subvectors of $\boldsymbol{\beta}$, this assessment will also carry over to the model-averaged estimate.

While the computation of the posteriors $p(\boldsymbol{\beta}, \sigma^2 | \mathbf{y}, M_r)$ and $p(M_r | \mathbf{y})$ required to obtain Eq. (4.30) is straightforward for a particular model M_r , it may be prohibitive for *all models*. The problem is that the number of possible models grows exponentially with k . For $k \leq 25$ regressors, enumeration of all models under consideration is usually possible in the available software packages. If k exceeds 25, more sophisticated algorithms are necessary. The model space then can be explored via MCMC simulation techniques. In doing so, we usually do not visit all models but those models with relatively high posterior probabilities. One such Monte Carlo approach is the MC³ algorithm of Madigan and York (1995); see also Fernandez et al. (2001). The MC³ algorithm for exploring model space works as described in Box 4.8.

As already mentioned, eliciting the conjugate prior is difficult in practice as it contains four hyperparameters: the prior mean $\mathbf{m}_r$, the prior covariance $\mathbf{M}_r$ for the regression coefficients, as well as the inverse gamma parameters a and b of the variance prior. Therefore in applications and software packages a modified prior structure often is assumed that requires to specify less hyperparameters. Fully noninformative priors are not possible if variable selection should be an integral part of the analysis as then the Bayes factor is not determined; see Appendix B.5.4. A widely used choice is to assume a noninformative prior for the intercept, Zellner's g -prior for the regression coefficients in combination with a zero prior mean, and a noninformative prior for the variance. This leads to the prior structure

$$\begin{aligned} p(\beta_0) &\propto \text{const}, \\ p(\tilde{\boldsymbol{\beta}}_r | \sigma^2, M_r) &\sim \mathbf{N}(\mathbf{0}, \sigma^2 (g \tilde{\mathbf{X}}_r' \tilde{\mathbf{X}}_r)^{-1}), \\ p(\sigma^2) &\propto \frac{1}{\sigma^2}. \end{aligned} \quad (4.31)$$

This prior depends only on one hyperparameter which reduces initialization of the prior to the choice of the factor g . An alternative to Zellner's g -prior is to assume a priori independence among the regression coefficients leading to $p(\tilde{\boldsymbol{\beta}}_r | \sigma^2, M_r) \sim \mathbf{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. These priors take us outside the conjugate model framework because of the improper priors for the intercept and the overall variance. Although the prior is partially improper, posterior analysis including model choice is possible because the intercept and the variance are the same for all models, so that Bayes factors are uniquely determined. Moreover, the MC³ algorithm for exploring the model space can be easily adapted. Using Zellner's g -prior, the Bayes factor required to compute the acceptance rate (4.33) is given by

$$\text{BF}_{rs} = \left(\frac{g}{g+1} \right)^{\frac{k_r - k_s}{2}} \left(\frac{1+g-R_s^2}{1+g-R_r^2} \right)^{\frac{n-1}{2}}, \quad (4.32)$$

4.8 MC³ Algorithm

Algorithm

1. Choose a start model M_{r_0} with $r_0 \in \{1, \dots, 2^k\}$ and the number T of iterations. Set $t = 1$.
2. Propose a new model M_{r^*} randomly from the neighborhood models of the current model $M_{r_{t-1}}$. The neighborhood of $M_{r_{t-1}}$ consists of all models that are obtained by adding or deleting one variable. Accept the newly proposed model with acceptance probability

$$\begin{aligned} \alpha(M_{r^*} | M_{r_{t-1}}) &= \min \left\{ \frac{p(y | M_{r^*}) p(M_{r^*})}{p(y | M_{r_{t-1}}) p(M_{r_{t-1}})}, 1 \right\} \\ &= \min \left\{ \text{BF}_{r^*, r_{t-1}} \frac{p(M_{r^*})}{p(M_{r_{t-1}})}, 1 \right\} \end{aligned} \quad (4.33)$$

as the current model M_{r_t} . Otherwise set $M_{r_t} = M_{r_{t-1}}$.

3. Update the full parameter vector β by sampling from the full conditional of the model parameter vector β_{r_t} (see the derivation on p. 234) and setting all other parameters in β to zero.
4. Update the error variance σ^2 by sampling from the full conditional (see again the derivation on p. 234).
5. Stop if $t = T$, otherwise set $t = t + 1$ and go to 2.

Software

- R package BMS.
- Other R packages for Bayesian variable selection (partly based on other methodology than described here) are BAS and BMA.

where R_r^2 and R_s^2 are the coefficients of determination of models M_r and M_s . Sampling the regression parameters β and σ^2 is done by a Gibbs sampler that consecutively samples from the full conditionals of $\beta_r = (\beta_0, \tilde{\beta}_r)'$ and σ^2 . Since the prior for β_0 is improper, joint updating β_0 and $\tilde{\beta}_r$ is not straightforward. The joint prior for β_0 and $\tilde{\beta}_r$ can be written as

$$p(\beta_0, \tilde{\beta}_r | \sigma^2) \propto \exp \left(-\frac{1}{2\sigma^2} \beta_r' K \beta_r \right),$$

where the precision matrix K is given by

$$K = \begin{pmatrix} 0 & \mathbf{0} \\ \mathbf{0} & g \tilde{X}_r' \tilde{X}_r \end{pmatrix}.$$

Now the full conditional is expressed as

$$\begin{aligned} p(\boldsymbol{\beta}_r | \cdot) &\propto \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r)'(\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r)\right) \exp\left(-\frac{1}{2}\boldsymbol{\beta}_r' \mathbf{K} \boldsymbol{\beta}_r\right) \\ &\propto \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{\beta}_r - \hat{\boldsymbol{\beta}}_r)' \mathbf{X}_r' \mathbf{X}_r (\boldsymbol{\beta}_r - \hat{\boldsymbol{\beta}}_r)\right) \exp\left(-\frac{1}{2}\boldsymbol{\beta}_r' \mathbf{K} \boldsymbol{\beta}_r\right), \end{aligned}$$

where we have again used identity (4.18). Applying Theorem B.4 (p. 649) of Appendix B.3 with $\mathbf{A} = \frac{1}{\sigma^2} \mathbf{X}_r' \mathbf{X}_r$, $\mathbf{a} = \hat{\boldsymbol{\beta}}_r$, $\mathbf{B} = \mathbf{K}$, and $\mathbf{b} = \mathbf{0}$ shows that the full conditional is multivariate Gaussian with covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\beta}_r}$ and mean $\boldsymbol{\mu}_{\boldsymbol{\beta}_r}$ given by

$$\boldsymbol{\Sigma}_{\boldsymbol{\beta}_r} = \left(\frac{1}{\sigma^2} \mathbf{X}_r' \mathbf{X}_r + \mathbf{K} \right)^{-1} \quad \boldsymbol{\mu}_{\boldsymbol{\beta}_r} = \frac{1}{\sigma^2} \boldsymbol{\Sigma}_{\boldsymbol{\beta}_r} \mathbf{X}_r' \mathbf{y}.$$

The full conditional of σ^2 is given by

$$\begin{aligned} p(\sigma^2 | \cdot) &\propto \frac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r)'(\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r)\right) \\ &\quad \frac{1}{(\sigma^2)^{k_r/2}} \exp\left(-\frac{1}{2\sigma^2} \tilde{\boldsymbol{\beta}}_r' \mathbf{g} \tilde{\mathbf{X}}_r' \tilde{\mathbf{X}}_r \tilde{\boldsymbol{\beta}}_r\right) \frac{1}{\sigma^2}, \end{aligned}$$

which can be identified with an inverse gamma distribution with parameters

$$a = \frac{n + k_r}{2} \quad b = \frac{1}{2} \left((\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r)'(\mathbf{y} - \mathbf{X}_r\boldsymbol{\beta}_r) + \tilde{\boldsymbol{\beta}}_r' \mathbf{g} \tilde{\mathbf{X}}_r' \tilde{\mathbf{X}}_r \tilde{\boldsymbol{\beta}}_r \right).$$

Assuming $p(\tilde{\boldsymbol{\beta}}_r | \sigma^2, M_r) \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ instead of Zellner's g -prior, the Bayes factor and the Gibbs sampler are similar, details are left to the reader as an exercise.

Example 4.15 Prices of Used Cars—Bayesian Model Averaging (1)

We illustrate the Bayesian approach for model choice using the data on the price of used cars. As in Example 3.19 (p. 152), we assume possibly nonlinear effects for the variables *age* and *kilometer* modeled through the orthogonal cubic polynomials *ageop1*, *ageop2*, *ageop3* and *kilop1*, *kilop2*, *kilop3*; see Example 3.5 on p. 90 for orthogonal polynomials. Together with the regressors *TIA*, *extras1*, and *extras2* we have $k = 9$ potential covariates. We used the package BMS of R for the analysis; see Zeugner (2010) for a tutorial. We started with a uniform prior for the models, i.e., $\theta = 1/2$ in Eq. (4.27), and $g = 1/n$ for Zellner's g -prior. Table 4.11 provides a summary of the preliminary results. From left to right, the columns correspond to the variable names, the posterior inclusion probabilities (PIP), the posterior estimates for the regression coefficients together with their standard deviations, the probability of a positive sign for the respective coefficient, and (for comparison) the least squares estimates. The PIP is the ratio between the number of visited models that include a particular covariate and the total number of visited models. Similarly, the probability of a positive sign reflects the ratio between models with positive sign for a covariate and the total number of models.

For the covariates *ageop1*, *ageop2*, *kilop1*, and *kilop2* with high inclusion probabilities (> 0.5), the Bayesian estimator averaged over the models is quite close to the ordinary least squares estimator. For the remaining covariates, the inclusion probabilities are very



Table 4.11 Prices of used cars: posterior inclusion probabilities (PIP), model averaged estimated coefficients and standard deviations, probabilities of positive sign, and for comparison the ordinary least squares estimates

Variable	PIP	Mean	Std. dev.	Cond. pos sign	LS
<i>ageop1</i>	1.00	−0.7065	0.0621	0	−0.7085
<i>ageop2</i>	0.94	0.1823	0.0721	1	0.1716
<i>ageop3</i>	0.07	0.0009	0.0156	1	0.0162
<i>kilop1</i>	1.00	−0.4345	0.0616	0	−0.4366
<i>kilop2</i>	0.61	0.0872	0.0827	1	0.1417
<i>kilop3</i>	0.07	0.0011	0.0160	1	0.0090
<i>TIA</i>	0.08	−0.0003	0.0025	0	−0.0051
<i>extras1</i>	0.11	−0.0127	0.0565	0	−0.1135
<i>extras2</i>	0.07	−0.0029	0.0374	0	−0.0315

The results are based on a uniform prior for the models ($\theta = 0.5$) and $g = 1/n$ for Zellner's g

Table 4.12 Prices of used cars: top five models T1–T5 with highest posterior probabilities

Variable	T1	T2	T3	T4	T5
<i>ageop1</i>	+	+	+	+	+
<i>ageop2</i>	+	+	+	+	+
<i>ageop3</i>	−	−	−	−	−
<i>kilop1</i>	+	+	+	+	+
<i>kilop2</i>	+	−	+	+	+
<i>kilop3</i>	−	−	−	−	−
<i>TIA</i>	−	−	−	+	−
<i>extras1</i>	−	−	+	−	−
<i>extras2</i>	−	−	−	−	+
<i>PMP</i>	0.375	0.250	0.040	0.032	0.030

The results are based on a uniform prior for the models ($\theta = 0.5$) and $g = 1/n$ for Zellner's g . A plus (minus) sign indicates that the variable is included in (excluded from) the model. The last row displays the posterior model probabilities (PMP).

low and the model averaged estimates are shrunk to zero compared to least squares. As can be seen from Table 4.12, the covariates with high inclusion probability also build the two models with by far the highest posterior probabilities. The top model, with posterior model probability of 0.37, consists exactly of the four covariates with inclusion probabilities higher than 0.5. In this case, the median probability model, that consists of those covariates with posterior probability of 1/2 and higher for being in the model, coincides with the model with highest posterior probability. The second model, with posterior probability 0.25, additionally excludes the variable *kilop2*. All other models have comparably low posterior probabilities. Note that the model with highest posterior probability corresponds exactly to the best AIC model of Example 3.19. It is also close to the model obtained with the LASSO in Example 4.11, which additionally contains covariate *extras1*.

We conclude the example by an investigation of the sensitivity of results on prior assumptions, particularly on the prior expected model size $E(S)$. The upper panel of Fig. 4.22 compares the PIP for the three models with $\theta = 1/2$, $\theta = 2/9$, and $\theta = 8/9$ corresponding to expected model sizes $E(S) = 4.5$, $E(S) = 2$, and $E(S) = 8$. Clearly, the results are quite sensitive to the choice of θ and $E(S)$. Except for *ageop1*, *ageop2*,

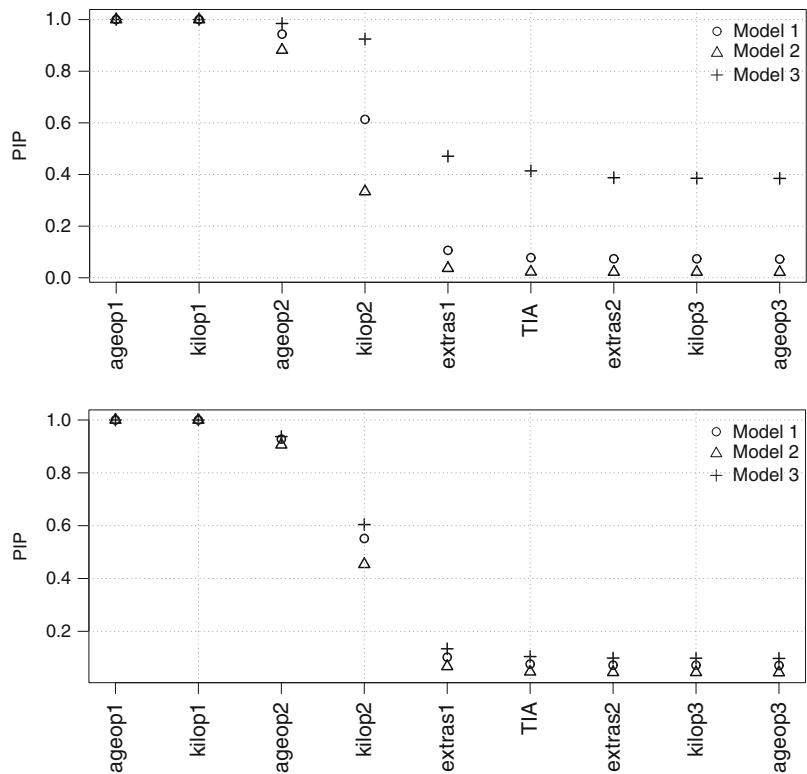


Fig. 4.22 Prices of used cars: comparison of posterior inclusion probabilities with $\theta = 1/2$ (corresponds to expected model size $E(S) = 4.5$), $\theta = 2/9$ ($E(S) = 2$), and $\theta = 8/9$ ($E(S) = 8$). The *upper panel* corresponds to fixed θ . The *lower panel* corresponds to a beta hyperprior for θ

and *kilop1*, the PIP differ considerably, e.g., for *kilop2* from 0.33 ($E(S) = 2$) over 0.61 ($E(S) = 4.5$) to 0.92 ($E(S) = 8$). We will explain the reasons of this undesirable behavior below.

△

The example shows that the results can be strongly affected by the prior choice for θ and the corresponding prior on the model size. Indeed, if θ is chosen as $\theta = E(S)/k$, then the induced prior for the model size S places relatively small probability mass on model sizes that are moderately far away from the mean $E(S)$. This is illustrated with Fig. 4.23 which shows, for $k = 10$ (left column) and $k = 40$ (right column) regressors, some priors for model size based on different choices for θ (dashed lines). In particular, with asymmetric prior expected model size $E(S)$ and for $k = 40$, a broad range of possible model sizes has virtually no prior probability mass. As a result, estimates are often quite sensitive to the choice of the prior expected model size.

A remedy to reduce the dependence of results on the prior choice is to introduce a hyperprior in a further stage of the hierarchy. In our case, a flexible and

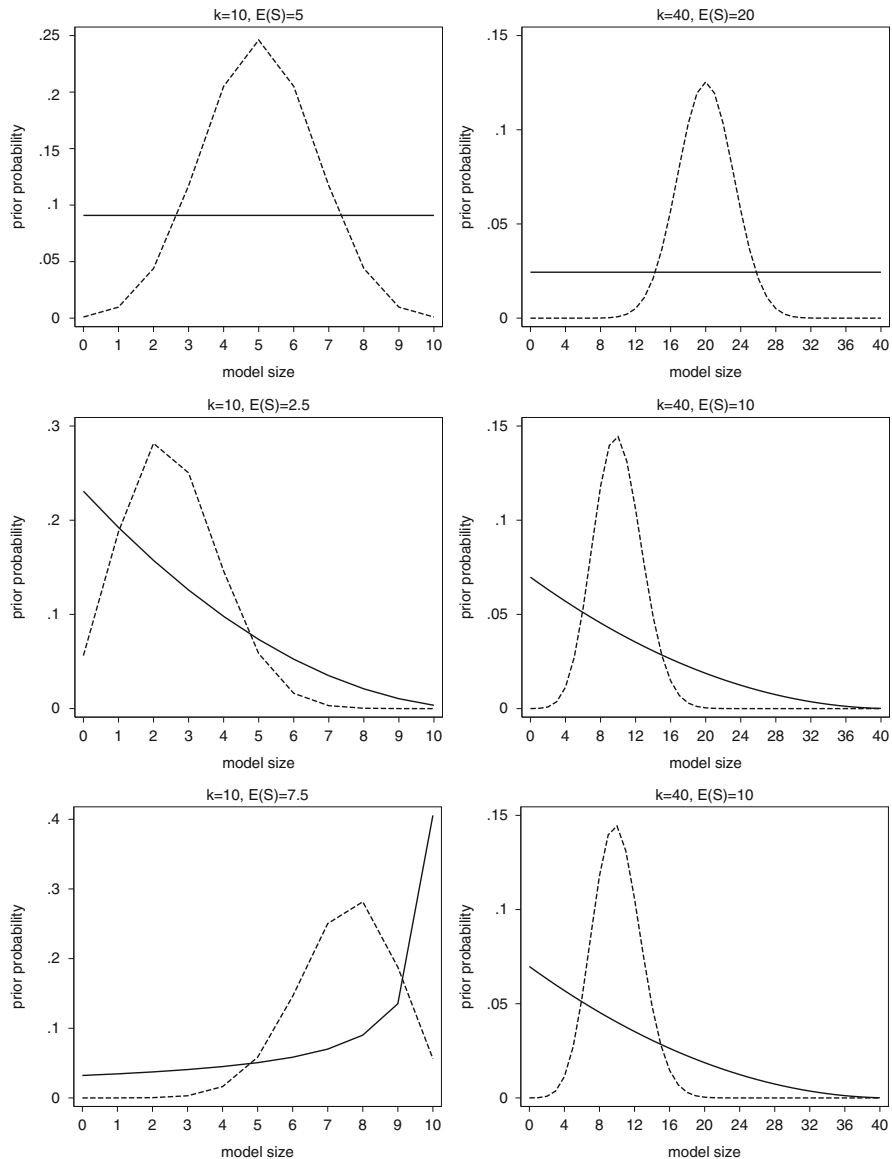


Fig. 4.23 Priors for model size. The *left column* shows various priors for model size with $k = 10$ potential regressors; the *right column* corresponds to $k = 40$. The *solid lines* correspond to a $\text{Beta}(a, b)$ hyperprior for θ . The *dashed lines* correspond to fixed θ

convenient choice for θ is to assume a beta distribution with hyperparameters a and b (Ley & Steel, 2009), i.e., $\theta \sim \text{Beta}(a, b)$. For the beta distribution see also Definition B.2 in Appendix B.1.

Since $S | \theta \sim \text{B}(k, \theta)$, the marginal distribution for S is beta-binomial (see Definition B.3 in Appendix B.1), i.e., $S \sim \text{BetaB}(k, a, b)$ with prior model size distribution

$$P(S = s) = \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)\Gamma(a + b + k)} \binom{k}{s} \Gamma(a + s)\Gamma(b + k - s),$$

and mean and variance given by

$$\begin{aligned} E(S) &= \frac{a}{a + b}k, \\ \text{Var}(S) &= \frac{ab(a + b + k)}{(a + b)^2(a + b + 1)}k. \end{aligned} \quad (4.34)$$

To ease initiation of the prior, we fix $a = 1$, which still allows for very flexible priors. To choose b , we rearrange $E(S)$ in Eq. (4.34) to obtain $b = (k - E(S))/E(S)$. Hence, similar to fixed θ , we can choose the prior expected model size to fully specify the prior. As can be seen from Fig. 4.23, the (marginal) prior for model size S shows much more variability compared to fixed θ so that all possible model sizes have positive probability mass.

Using a $\text{Beta}(a, b)$ hyperprior requires a slight modification of the MC^3 algorithm as the prior model odds in Eq. (4.33) has to be adapted. The marginal prior for model M_r is obtained by integrating over θ

$$p(M_r) = \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a + k_r)\Gamma(b + k - k_r)}{\Gamma(a + b + k)}; \quad (4.35)$$

see the appendix of this chapter, p. 266, for a derivation.

Using formula (4.35) the prior odds between two models M_r and M_s required for Eq. (4.33) in the MC^3 algorithm are easily computed as

$$\frac{p(M_r)}{p(M_s)} = \frac{\Gamma(a + k_r)\Gamma(b + k - k_r)}{\Gamma(a + k_s)\Gamma(b + k - k_s)}.$$

Example 4.16 Prices of Used Cars—Bayesian Model Averaging (2)

We rerun the three regressions of Example 4.15, again with prior expected model size $E(S) = 2, 4.5, 8$, but now with a beta hyperprior for θ . The bottom panel of Fig. 4.22 compares the PIP for the nine possible covariates. The probabilities are now quite close to each other. An exception are the PIP for *kilop2*, where there is some inclusion uncertainty. For two of the three model priors (with $E(S) = 4.5$ and $E(S) = 8$) the best model with highest posterior probability consists of the covariates *ageop1*, *ageop2*, *kilop1*, and *kilop2*. The model that excludes *kilop2* is the second best model. It is also the best model for prior expected model size $E(S) = 2$. All other models have very low posterior probabilities.

In summary, we are quite certain that only two covariates, the age of the car and the kilometer reading, are relevant predictors for the price. A second-order polynomial is



sufficient to model the nonlinear effects of the two covariates. There is some uncertainty whether a linear effect for kilometer reading is sufficient. Note also that the quadratic effect of kilometer reading is already close to linearity; see Fig. 3.22 (p. 155).

△

We finally note that there are also approaches to specify an additional hyperprior for the factor g . Details can be found in [Liang, Paulo, Molina, Clyde, and Berger \(2008\)](#). The approach is also implemented in the R package BMS.

4.4.4 Spike and Slab Priors

We again consider the Bayesian linear variable selection model of the previous section; see Box 4.9 for a summary. For simplicity, we restrict ourselves to a priori independent regression coefficients, i.e., $p(\boldsymbol{\beta}_r | \sigma^2, M_r) \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$.

We now present an equivalent model formulation that provides further insight and gives rise to interesting modifications. In the last section, we specified the prior $p(M_r | \theta) = \theta^{k_r} \cdot (1 - \theta)^{k - k_r}$ for the models M_r together with a Beta(a_θ, b_θ) prior for θ . Each model M_r can also be identified with a specific vector $\boldsymbol{\delta}_r$ of the inclusion indicators $\boldsymbol{\delta} = (\delta_1, \dots, \delta_k)'$. Recall that the j th component δ_j of $\boldsymbol{\delta}$ defines whether or not the j th covariate x_j is included in the model. The model prior can now be specified equivalently in terms of the inclusion indicator $\boldsymbol{\delta}$ by assuming

$$\delta_j | \theta \stackrel{i.i.d.}{\sim} B(1, \theta),$$

i.e., $P(\delta_j = 1) = \theta$ and $P(\delta_j = 0) = 1 - \theta$. Hence, we assume a priori that a particular covariate x_j enters the model with inclusion probability θ . Since posterior analysis is based on MCMC simulations, we can interpret the relative frequency of $\delta_j = 1$ in the MCMC samples as the posterior probability that x_j enters the model.

The priors for the regression coefficients β_j , $j = 1, \dots, k$, can now be defined in dependence of the inclusion vector $\boldsymbol{\delta}$ by assuming the mixture prior

$$p(\beta_j | \delta_j, \sigma^2) = (1 - \delta_j) \text{Dirac}(0) + \delta_j N(0, \sigma^2). \quad (4.36)$$

The mixture prior has the following interpretation: If $\delta_j = 0$ the first component in Eq. (4.36) is “activated” and the corresponding regression coefficient is zero with probability one. In other words, covariate x_j is removed from the model. We call this first part of the mixture prior the *spike component*. For $\delta_j = 1$, we have the usual normal prior for β_j . This part of the mixture prior is called the *slab component*.

We summarize the alternative formulation of the model in Box 4.10. This is fully equivalent to the model of the previous Sect. 4.4.3. We presented the alternative formulation in terms of spike and slab priors because it gives rise to various other approaches related to Bayesian variable selection, which may be preferable in some situations. The main disadvantage of the spike and slab mixture prior (4.36) (and the equivalent model of the previous section) is that the MCMC algorithms for Bayesian inference require computation of the marginal likelihood, which is only analytically



4.9 Bayesian Linear Variable Selection Model

Observation Model Conditional on Model M_r

$$\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2, M_r \sim N(\beta_0 \mathbf{1} + \tilde{\mathbf{X}}_r \tilde{\boldsymbol{\beta}}_r, \sigma^2 \mathbf{I}).$$

Prior Structure Conditional on Model M_r

$$\begin{aligned} p(\boldsymbol{\beta}, \sigma^2 \mid M_r) &= p(\tilde{\boldsymbol{\beta}}_{-r} \mid \beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2, M_r) p(\beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2 \mid M_r) \\ &= p(\tilde{\boldsymbol{\beta}}_{-r} \mid \beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2, M_r) p(\beta_0 \mid M_r) p(\tilde{\boldsymbol{\beta}}_r \mid \sigma^2, M_r) p(\sigma^2 \mid M_r), \end{aligned}$$

with

$$p(\tilde{\boldsymbol{\beta}}_{-r} \mid \beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2, M_r) = \text{Dirac}(0, \dots, 0),$$

$$p(\beta_0 \mid M_r) = p(\beta_0) \propto \text{const.},$$

$$p(\tilde{\boldsymbol{\beta}}_r \mid \sigma^2, M_r) \sim N(\mathbf{0}, \sigma^2 (g \tilde{\mathbf{X}}_r' \tilde{\mathbf{X}}_r)^{-1}),$$

$$p(\sigma^2 \mid M_r) = p(\sigma^2) \sim \frac{1}{\sigma^2}.$$

An alternative to Zellner's g -prior is to assume independent regression coefficients and assume $p(\tilde{\boldsymbol{\beta}}_r \mid \sigma^2, M_r) \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$.

Model Prior

Fixed inclusion parameter

$$p(M_r) = \theta^{k_r} (1 - \theta)^{k - k_r}.$$

Stochastic inclusion parameter

$$p(M_r \mid \theta) = \theta^{k_r} (1 - \theta)^{k - k_r}$$

$$\theta \sim \text{Beta}(a_\theta, b_\theta).$$

available in special cases, e.g., the linear model. In many other settings, for instance in the regression models with non-normal responses, discussed in Chaps. 5 and 6, the marginal likelihood is not analytically, and also not often numerically, tractable.

The recent literature has therefore proposed various alternatives. An approach due to [George and McCulloch \(1993\)](#) that is conceptually close replaces the mixture prior (4.36) by

$$p(\beta_j \mid \delta_j, \sigma^2) = (1 - \delta_j) N(0, \nu_0 \sigma^2) + \delta_j N(0, \sigma^2), \quad (4.37)$$

4.10 Spike and Slab Priors

Observation Model

$$y \mid \boldsymbol{\beta}, \sigma^2 \sim N(\boldsymbol{\beta}_0 \mathbf{1} + \tilde{\boldsymbol{\beta}} \tilde{\mathbf{X}}, \sigma^2 \mathbf{I})$$

Spike and Slab Prior for the Regression Coefficients

Dirac spike

$$p(\beta_j \mid \delta_j, \sigma^2) \stackrel{iid}{=} (1 - \delta_j) \text{Dirac}(0) + \delta_j N(0, \sigma^2) \text{ for } j = 1, \dots, k.$$

Continuous normal spike

$$p(\beta_j \mid \delta_j, \sigma^2) \stackrel{iid}{=} (1 - \delta_j) N(0, \nu_0 \sigma^2) + \delta_j N(0, \sigma^2), \text{ with } \nu_0 \text{ “small.”}$$

Spike and Slab Prior for the Variances

$$\beta_1, \dots, \beta_k \mid \tau_1^2, \dots, \tau_k^2 \sim N(\mathbf{0}, \text{diag}(\tau_1^2, \dots, \tau_k^2)).$$

Dirac spike

$$\tau_j^2 \mid \delta_j \stackrel{iid}{=} (1 - \delta_j) \text{Dirac}(0) + \delta_j \text{IG}(a_{\tau^2}, b_{\tau^2}).$$

Continuous inverse gamma spike

$$\tau_j^2 \mid \delta_j \stackrel{iid}{=} (1 - \delta_j) \text{IG}(a_{\tau^2}, \nu_0 b_{\tau^2}) + \delta_j \text{IG}(a_{\tau^2}, b_{\tau^2}) \text{ with } \nu_0 \text{ “small.”}$$

Common Prior Assumptions

$$p(\beta_0) \propto \text{const}$$

$$\sigma^2 \sim \text{IG}(a, b)$$

$$\delta_j \mid \theta \stackrel{iid}{\sim} \text{B}(1, \theta)$$

$$\theta \sim \text{Beta}(a_\theta, b_\theta)$$

Software

- Software package BayesX for continuous variance spikes.
- R package BMS for Dirac spikes for the regression coefficients.

where the Dirac spike at 0 is replaced by the $N(0, \nu_0 \sigma^2)$ distribution. The variance component ν_0 is a hyperparameter and is chosen very small (close to zero), such that the spike part has very small variance. Consequently if $\delta_j = 0$, then the small variance of the spike part, in combination with the zero mean, ensures that the corresponding regression coefficient is essentially zero. Hence, $\delta_j = 0$ still implies that covariate x_j is (essentially) removed from the model. The main advantage of the alternative mixture prior (4.37) is that MCMC inference is simplified considerably since the computation of marginal likelihoods can be avoided.

A different concept due to [Ishwaran and Rao \(2005\)](#) focuses on the prior variance of the regression coefficients. Recall the Bayesian LASSO, where we assigned independent normal distributions to the regression coefficients β_j conditional on the variance parameters τ_j^2 , i.e., $\beta_j \mid \tau_j^2 \sim N(0, \tau_j^2)$ or

$$\tilde{\beta} \mid \tau_1^2, \dots, \tau_k^2 \sim N(\mathbf{0}, \text{diag}(\tau_1^2, \dots, \tau_k^2)).$$

Spike and slab priors for the variances are based on the observation that the regression coefficient β_j is estimated essentially zero if the corresponding variance parameter τ_j^2 is zero or at least close to zero. The prior mean of zero, in combination with (near) zero variance, then guarantees that β_j cannot deviate far from zero, which in turn implies that the corresponding covariate x_j is essentially removed from the model.

The idea now is to specify a prior for the variances τ_j^2 that puts relatively large probability mass on near zero variance. This can again be obtained using a mixture distribution. One such approach is a mixture of a Dirac spike at 0 and an inverse gamma slab given by

$$\tau_j^2 \mid \delta_j = (1 - \delta_j) \cdot \text{Dirac}(0) + \delta_j \cdot \text{IG}(a_{\tau^2}, b_{\tau^2}),$$

where $\delta_j \in \{0, 1\}$ again is an indicator variable. Alternatively, we can use a mixture of two inverse gamma distributions. More specifically, we obtain

$$\tau_j^2 \mid \delta_j = (1 - \delta_j) \cdot \text{IG}(a_{\tau^2}, \nu_0 b_{\tau^2}) + \delta_j \cdot \text{IG}(a_{\tau^2}, b_{\tau^2}), \quad (4.38)$$

where the hyperparameter ν_0 is chosen to be small. This prior has the following interpretation: The mean, mode, and variance of an $\text{IG}(a, b)$ distribution are given by

$$\begin{aligned} E(X) &= \frac{b}{a-1} \quad a > 1, \\ \text{Mod}(X) &= \frac{b}{a+1}, \\ \text{Var}(X) &= \frac{b^2}{(a-1)^2(a-2)} \quad a > 2. \end{aligned}$$

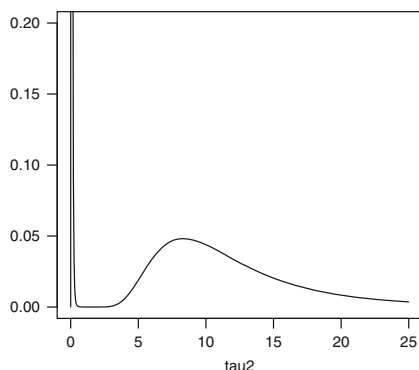


Fig. 4.24 Spike and slab priors: marginal density of the variance parameter τ_j^2 for some hyperparameter combinations

Provided that v_0 is very small, the first component in Eq. (4.38) has its mode (and mean if it exists) close to zero while the variability of the prior is limited. Thus, the first component makes sure that the marginal prior for τ_j^2 will have a *spike* near zero. The second component in Eq. (4.38), the *slab* part, can be seen as the “usual” variance prior if the corresponding covariate should be included in the model. To shed more light on the distribution of the variance component, Fig. 4.24 shows the marginal density (with δ_j integrated out) of the variance parameter τ_j^2 for some hyperparameter combinations. The characteristic feature is the “spike” at zero.

A summary of the various spike and slab approaches is given in Box 4.10. For illustration, in the appendix of this chapter (p. 266), we derive a Gibbs sampler for the continuous variance spike and slab prior (4.38). The derivation of Gibbs samplers for the other spike and slab variants is similar and left as an exercise to the reader. The great advantage of spike and slab priors for the variances, rather than the regression coefficients, is that updating of the variance and inclusion parameters is independent of the observation model. This means that the concept carries over to more complex models, e.g., the regression models for non-normal responses of Chaps. 5 and 6. If we are able to update the regressions coefficients, which is typically possible even in complex models, the remaining update steps remain unchanged.



4.5 Bibliographic Notes and Proofs

4.5.1 Bibliographic Notes

For modeling correlated errors in Sect. 4.1.4, we only examined autoregressive processes of first order. More complex error structures, especially ARMA models (Autoregressive Moving Average), are treated in greater detail in the literature

focusing on time series analysis. Standard books are [Brockwell and Davis \(2002\)](#) and [Hamilton \(1994\)](#).

A standard book for variable selection in linear models is [Miller \(2002\)](#). A good overview on regularization approaches, such as ridge regression and the LASSO, can be found in [Hastie, Tibshirani, and Friedman \(2009\)](#). Alternative regularization approaches based on different penalties are, for example, bridge regression, where general norms of coefficient vectors are used as penalties ([Fu, 1998](#)), or the elastic net that is a compromise between ridge regression and LASSO ([Zou & Hastie 2005](#)). The latter has the particular advantage to select complete blocks of correlated coefficients instead of only single representatives of such groups. Penalties with a similar aim are defined in the octogonal shrinkage and clustering algorithm for regression (OSCAR, [Bondell & Reich, 2008](#)) or with the smoothly clipped absolute deviation (SCAD, [Fan & Li, 2001](#)).

Boosting approaches in a regression context, including theoretical properties, are described in [Friedman \(2001\)](#), [Bühlmann and Yu \(2003\)](#), and [Bühlmann \(2006\)](#). [Bühlmann and Hothorn \(2007\)](#) give a general introduction into boosting based on a functional gradient interpretation. A different variant of boosting that relies on likelihood maximization techniques (likelihood-based boosting) is considered, for example, in [Tutz and Binder \(2006\)](#).

Bayesian linear models are covered in all introductory books on Bayesian analysis. Standard references are the textbooks of [Gelman, Carlin, Stern, and Rubin \(2003\)](#) and [O'Hagan \(1994\)](#). Still interesting is the classical book by [Box and Tiao \(1992\)](#). The Bayesian LASSO is due to [Park and Casella \(2008\)](#). An overview over other Bayesian regularization priors is given in [Fahrmeir, Kneib, and Konrath \(2010\)](#). Recent references to the Bayesian linear variable selection model of Box 4.9 are the papers by [Fernandez et al. \(2001\)](#), [Ley and Steel \(2009\)](#), and the many references therein. Spike and slab priors with continuous spike for the regression coefficients are introduced in [George and McCulloch \(1993\)](#) and [George and McCulloch \(1997\)](#). Spike and slab priors for the variances are proposed in [Ishwaran and Rao \(2003, 2005\)](#). A good overview of spike and slab priors is provided in [Malsiner-Walli and Wagner \(2011\)](#).

4.5.2 Proofs

Autocorrelated Errors of First Order (p. 192)

We first derive the so-called MA(∞) presentation for the errors ε_i . Based on this representation, we then derive means, variances, and covariances of errors.

MA(∞) Presentation

Our goal is to represent ε_i as a weighted infinite sum of the u_i . This is based on the idea that the process started far back in the past. Inserting first $\varepsilon_{i-1} = \rho \varepsilon_{i-2} + u_{i-1}$ into $\varepsilon_i = \rho \varepsilon_{i-1} + u_i$, we obtain

$$\varepsilon_i = \rho (\rho \varepsilon_{i-2} + u_{i-1}) + u_i = \rho^2 \varepsilon_{i-2} + \rho u_{i-1} + u_i.$$

Inserting $\varepsilon_{i-2} = \rho \varepsilon_{i-3} + u_{i-2}$ yields

$$\varepsilon_i = \rho^3 \varepsilon_{i-3} + \rho^2 u_{i-2} + \rho u_{i-1} + u_i.$$

Successive insertion finally results in the presentation

$$\varepsilon_i = \sum_{k=0}^{\infty} \rho^k u_{i-k} = u_i + \rho u_{i-1} + \rho^2 u_{i-2} + \dots \quad (4.39)$$

Mean of Errors

Using representation (4.39) we obtain

$$E(\varepsilon_i) = \sum_{k=0}^{\infty} \rho^k E(u_{i-k}) = 0.$$

Covariance of Errors

Using again representation (4.39) as well as $E(\varepsilon_i) = 0$ and $E(u_i u_j) = 0$ for $i \neq j$, we obtain

$$\begin{aligned} \text{Cov}(\varepsilon_i, \varepsilon_{i-j}) &= E(\varepsilon_i \varepsilon_{i-j}) \\ &= E((u_i + \rho u_{i-1} + \dots + \rho^j u_{i-j} + \dots)(u_{i-j} + \rho u_{i-j-1} + \dots)) \\ &= E(\rho^j u_{i-j}^2) + E(\rho^{j+2} u_{i-j-1}^2) + \dots \\ &= \rho^j \sum_{k=0}^{\infty} (\rho^2)^k E(u_{i-j-k}^2) \\ &= \rho^j \sum_{k=0}^{\infty} (\rho^2)^k \sigma_u^2 \\ &= \rho^j \frac{\sigma_u^2}{1 - \rho^2}. \end{aligned}$$

The special case $j = 0$ implies

$$\text{Var}(\varepsilon_i) = \frac{\sigma_u^2}{1 - \rho^2}.$$

□

Proof of the Least Squares Identity (p. 211)

We show

$$\begin{aligned} \text{LS}(\hat{\beta}) &= (\beta - \hat{\beta})' X' X (\beta - \hat{\beta}) + y' (I_n - X(X'X)^{-1}X') y \\ &= (\beta - \hat{\beta})' X' X (\beta - \hat{\beta}) + \hat{\varepsilon}' \hat{\varepsilon}. \end{aligned} \quad (4.40)$$

To verify Eq. (4.40) we first evaluate the products for the least squares criterion to obtain

$$(y - X\beta)'(y - X\beta) = y'y - 2\beta'X'y + \beta'X'X\beta. \quad (4.41)$$

On the other hand, we expand the quadratic form in Eq. (4.40) which gives

$$(\beta - \hat{\beta})'X'X(\beta - \hat{\beta}) = \beta'X'X\beta - 2\beta'X'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta}.$$

Inserting $\hat{\beta} = (X'X)^{-1}X'y$ yields

$$2\beta'X'X\hat{\beta} = 2\beta'X'X(X'X)^{-1}X'y = 2\beta'X'y$$

for the second addend and

$$\hat{\beta}'X'X\hat{\beta} = y'X(X'X)^{-1}X'X(X'X)^{-1}X'y = y'X(X'X)^{-1}X'y$$

for the third addend. We thus obtain

$$(\beta - \hat{\beta})'X'X(\beta - \hat{\beta}) = \beta'X'X\beta - 2\beta'X'y + y'X(X'X)^{-1}X'y.$$

Inserting this expression in the right-hand side of Eq. (4.40) gives exactly the right-hand side of Eq. (4.41) which proves the least squares identity. The second line of Eq. (4.40) states that

$$y'(I_n - X(X'X)^{-1}X')y = \hat{\varepsilon}'\hat{\varepsilon},$$

which is easily verified by expanding

$$\hat{\varepsilon}'\hat{\varepsilon} = (y - X\hat{\beta})'(y - X\hat{\beta})$$

and inserting $\hat{\beta} = (X'X)^{-1}X'y$.

Derivation of the Posterior in the Linear Model with NIG Prior (p. 227)

We first rewrite the expression

$$(y - X\beta)'(y - X\beta) + (\beta - m)'M^{-1}(\beta - m).$$

Defining

$$\begin{aligned} \tilde{M} &= (X'X + M^{-1})^{-1} \\ \tilde{m} &= \tilde{M}(M^{-1}m + X'y), \end{aligned}$$

we obtain

$$\begin{aligned}
 & (y - X\beta)'(y - X\beta) + (\beta - m)'M^{-1}(\beta - m) \\
 &= y'y - 2\beta'X'y + \beta'X'X\beta + \beta'M^{-1}\beta - 2\beta'M^{-1}m + m'M^{-1}m \\
 &= y'y + \beta'(X'X + M^{-1})\beta - 2\beta'(X'y + M^{-1}m) + m'M^{-1}m \\
 &= y'y + \beta'\tilde{M}^{-1}\beta - 2\beta'\tilde{M}^{-1}\tilde{M}(X'y + M^{-1}m) + m'M^{-1}m \\
 &= y'y + \beta'\tilde{M}^{-1}\beta - 2\beta'\tilde{M}^{-1}\tilde{m} + m'M^{-1}m \\
 &= y'y + (\beta - \tilde{m})'\tilde{M}^{-1}(\beta - \tilde{m}) - \tilde{m}'\tilde{M}^{-1}\tilde{m} + m'M^{-1}m.
 \end{aligned}$$

Defining

$$\tilde{a} = a + \frac{n}{2}$$

and

$$\tilde{b} = b + \frac{1}{2} \left(y'y + m'M^{-1}m - \tilde{m}'\tilde{M}^{-1}\tilde{m} \right),$$

insertion into the posterior distribution (4.17) gives

$$P(\beta, \sigma^2 | y) \propto \frac{1}{(\sigma^2)^{\frac{p}{2}}} \exp \left(-\frac{1}{2\sigma^2} (\beta - \tilde{m})'\tilde{M}^{-1}(\beta - \tilde{m}) \right) \frac{1}{(\sigma^2)^{\tilde{a}+1}} \exp \left(-\frac{\tilde{b}}{\sigma^2} \right).$$

Comparison with the density of a normal-inverse gamma distribution (compare the density (4.14) on p. 228) shows that the posterior is again a normal-inverse gamma distribution with parameters $\tilde{m}$, $\tilde{M}$, $\tilde{a}$, and $\tilde{b}$.

□

Derivation of the Posterior Mode for Bayesian Ridge Regression (p. 238)

As stated in the text, the derivation is based on fixed σ^2 and τ^2 . The posterior for β is then given by

$$\begin{aligned}
 p(\beta | y) &\propto p(y | \beta) p(\beta_0) p(\tilde{\beta}) \\
 &\propto \exp \left(-\frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta) \right) \exp \left(-\frac{1}{2\tau^2} \tilde{\beta}'\tilde{\beta} \right).
 \end{aligned}$$

Maximizing the posterior is equivalent to maximizing $\log p(\beta | y)$ or minimizing $-\log p(\beta | y)$. This computes to

$$\begin{aligned}
 -\log p(\beta | y) &= \frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta) + \frac{1}{2\tau^2} \tilde{\beta}'\tilde{\beta} \\
 &= \frac{1}{2\sigma^2} \left((y - X\beta)'(y - X\beta) + \lambda \tilde{\beta}'\tilde{\beta} \right),
 \end{aligned}$$

with $\lambda = \sigma^2/\tau^2$. Hence the posterior mode is for given λ obtained by minimizing the penalized least squares criterion

$$PLS(\lambda) = (y - X\beta)'(y - X\beta) + \lambda \tilde{\beta}'\tilde{\beta}.$$

Derivation of the Gibbs Sampler for the Bayesian LASSO (p. 239)

We develop a Gibbs sampler that subsequently draws from the full conditionals of $\beta = (\beta_0, \tilde{\beta})'$, σ^2 , $\tau_1^2, \dots, \tau_k^2$, and λ^2 . All full conditionals are proportional to the posterior which is given by

$$p(\beta, \sigma^2, \tau_1^2, \dots, \tau_k^2, \lambda | y) \propto p(y | \beta, \sigma^2) p(\beta_0) p(\tilde{\beta} | \tau_1^2, \dots, \tau_k^2) \prod_{j=1}^k p(\tau_j^2 | \lambda) p(\lambda^2).$$

To derive the specific distribution of the full conditionals we can always disregard those factors in the posterior that do not depend on the parameter of current interest.

Full Conditional of β

To derive the full conditional for β we first note that $p(\beta | \tau_1^2, \dots, \tau_k^2) = p(\beta_0) p(\tilde{\beta} | \tau_1^2, \dots, \tau_k^2)$ can be expressed as

$$p(\beta | \tau_1^2, \dots, \tau_k^2) \propto \exp\left(-\frac{1}{2}\beta'K\beta\right),$$

where $K = \text{diag}(0, 1/\tau_1^2, \dots, 1/\tau_k^2)$. Now the full conditional is given by

$$p(\beta | \cdot) \propto \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right) \exp\left(-\frac{1}{2}\beta'K\beta\right).$$

Since the least squares criterion can be written as

$$(y - X\beta)'(y - X\beta) = (\beta - \hat{\beta})'X'X(\beta - \hat{\beta}) + y'(I_n - X(X'X)^{-1}X')y$$

(see the derivation on p. 211) with the second summand independent of β , we obtain

$$p(\beta | \cdot) \propto \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})'X'X(\beta - \hat{\beta})\right) \exp\left(-\frac{1}{2}\beta'K\beta\right).$$

Invoking Theorem B.4 of Appendix B.3 with $A = \frac{1}{\sigma^2}X'X$, $a = \hat{\beta}$, $B = K$, and $b = \mathbf{0}$ shows that the full conditional is multivariate Gaussian with covariance matrix Σ_β and mean μ_β given by

$$\Sigma = \left(\frac{1}{\sigma^2}X'X + K\right)^{-1} \quad \mu_\beta = \frac{1}{\sigma^2}\Sigma_\beta X'y.$$

In summary we have $\beta | \cdot \sim N(\mu_\beta, \Sigma_\beta)$.

Full Conditional of σ^2

The full conditional for σ^2 is given by

$$\begin{aligned} p(\sigma^2 | \cdot) &\propto p(\mathbf{y} | \boldsymbol{\beta}, \sigma^2) p(\sigma^2) \\ &\propto \frac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right) \frac{1}{(\sigma^2)^{a+1}} \exp\left(-\frac{b}{\sigma^2}\right) \\ &= \frac{1}{(\sigma^2)^{a+n/2+1}} \exp\left(-\frac{1}{\sigma^2}\left(b + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right)\right), \end{aligned}$$

which can be identified as an inverse gamma distribution with parameters

$$a^{new} = a + \frac{n}{2} \quad b^{new} = b + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Full Conditional of τ_j^2

Define

$$\mu = \left(\frac{\lambda^2}{\beta_j^2}\right)^{1/2}.$$

The full conditional of τ_j^2 , $j = 1, \dots, k$, is then given by

$$\begin{aligned} p(\tau_j^2 | \cdot) &\propto p(\beta_j | \tau_j^2) p(\tau_j^2 | \lambda) \\ &\propto \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{1}{2\tau_j^2}\beta_j^2\right) \exp\left(-\frac{\lambda^2}{2}\tau_j^2\right) \\ &= \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{\beta_j^2}{2\tau_j^2} - \frac{\lambda^2\tau_j^2}{2}\right) \\ &\propto \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{\beta_j^2}{2\tau_j^2} + \frac{\lambda^2}{\mu} - \frac{\lambda^2\tau_j^2}{2}\right) \\ &\propto \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{\lambda^2\tau_j^2}{2\mu^2} \left(\frac{1}{(\tau_j^2)^2} - 2\frac{\mu}{\tau_j^2} + \mu^2\right)\right) \\ &= \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{\lambda^2\tau_j^2}{2\mu^2} \left(\frac{1}{\tau_j^2} - \mu\right)^2\right). \end{aligned}$$

This is not a standard distribution. However, it turns out that the distribution of $\omega_j = 1/\tau_j^2$ is inverse Gaussian with location parameter μ and scale parameter λ^2 . Updating of τ_j^2 is then obtained by first sampling $1/\tau_j^2$ from the inverse Gaussian distribution and then inverting the result to obtain τ_j^2 .

To derive the distribution of ω_j we apply the change of variables Theorem B.1 of Appendix B.1. With $\omega_j = g(\tau_j^2) = 1/\tau_j^2$, $g^{-1}(\omega_j) = 1/\omega_j$, and $g'(\tau_j^2) = -1/(\tau_j^2)^2$, we obtain

$$p(\omega_j | \cdot) \propto \left(\frac{1}{\omega_j^3} \right)^{1/2} \exp \left(-\frac{\lambda^2}{2\mu \omega_j} (\omega_j - \mu)^2 \right),$$

which has the form of the proposed inverse Gaussian distribution.

Full Conditional of λ^2

We update λ by drawing from the full conditional of λ^2 . It is given by

$$\begin{aligned} p(\lambda^2 | \cdot) &\propto \prod_{j=1}^k p(\tau_j^2 | \lambda) p(\lambda^2) \\ &\propto \left(\frac{\lambda^2}{2} \right)^k \exp \left(-\frac{1}{2} \lambda^2 \sum_{j=1}^k \tau_j^2 \right) (\lambda^2)^{a-1} \exp(-b\lambda^2) \\ &\propto (\lambda^2)^{a+k-1} \exp \left(-\lambda^2 \left(b + \frac{1}{2} \sum_{j=1}^k \tau_j^2 \right) \right), \end{aligned}$$

which can be identified with a gamma distribution. More specifically,

$$\lambda^2 | \cdot \sim G \left(a + k, b + \frac{1}{2} \sum_{j=1}^k \tau_j^2 \right).$$

Derivation of the Marginal Likelihood and the Bayes Factor in the Linear Model (p. 243)

We derive the marginal likelihood for model M_r given by

$$\mathbf{y} | \boldsymbol{\beta}, \sigma^2, M_r = \mathbf{y} | \beta_0, \tilde{\boldsymbol{\beta}}_r, \sigma^2, M_r \sim N(\beta_0 \mathbf{1} + \tilde{X}_r \tilde{\boldsymbol{\beta}}_r, \sigma^2 \mathbf{I}),$$

with priors $\beta_0, \tilde{\boldsymbol{\beta}}_r | \sigma^2, M_r \sim N(\mathbf{m}_r, \sigma^2 \mathbf{M}_r)$, and $\sigma^2 | M_r \sim \text{IG}(a, b)$; see Sect. 4.4.3.

To derive the marginal distribution we write model M_r as

$$\begin{aligned} \mathbf{y} &= \mathbf{X}_r \boldsymbol{\beta}_r + \boldsymbol{\varepsilon}_1, \quad \boldsymbol{\varepsilon}_1 \sim N(\mathbf{0}, \sigma^2 \mathbf{I}), \\ \boldsymbol{\beta}_r &= \mathbf{m}_r + \boldsymbol{\varepsilon}_2, \quad \boldsymbol{\varepsilon}_2 \sim N(\mathbf{0}, \sigma^2 \mathbf{M}_r), \end{aligned}$$

where $\mathbf{X}_r = (\mathbf{1} \tilde{X}_r)$, $\boldsymbol{\beta}_r = (\beta_0, \tilde{\boldsymbol{\beta}}_r)'$, and $\boldsymbol{\varepsilon}_1$ and $\boldsymbol{\varepsilon}_2$ are stochastically independent. Inserting $\boldsymbol{\beta}_r$ into the first equation gives

$$\mathbf{y} = \mathbf{X}_r(\mathbf{m}_r + \boldsymbol{\varepsilon}_2) + \boldsymbol{\varepsilon}_1 = \mathbf{X}_r \mathbf{m}_r + \mathbf{X}_r \boldsymbol{\varepsilon}_2 + \boldsymbol{\varepsilon}_1 \sim N(\mathbf{X}_r \mathbf{m}_r, \sigma^2(\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r')),$$

which is the conditional distribution of $\mathbf{y}$ given σ^2 . To derive the covariance matrix $\sigma^2(\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r')$, we used properties 5 and 6 of Theorem B.2 in Appendix B.2. Since $\mathbf{y} | \sigma^2 \sim \mathcal{N}(\mathbf{X}_r \mathbf{m}_r, \sigma^2(\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r'))$ and $\sigma^2 \sim \text{IG}(a, b)$, the joint distribution is $\mathbf{y}, \sigma^2 \sim \text{NIG}(\mathbf{X}_r \mathbf{m}_r, \mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r', a, b)$. Hence $\mathbf{y}$ is marginally multivariate t with $\nu = 2a$ degrees of freedom, location parameter $\boldsymbol{\mu} = \mathbf{X}_r \mathbf{m}_r$, and dispersion matrix $\boldsymbol{\Sigma} = \mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r'$; see property 5 in Definition B.3.5 of Appendix B.3 and the derivation on p. 228.

To obtain the Bayes factor we need to compute the inverse $\boldsymbol{\Sigma}^{-1}$ of the dispersion matrix. Using the matrix inversion lemma (Appendix A.3, Theorem A.14) with $\mathbf{A} = \mathbf{I}$, $\mathbf{B} = -\mathbf{X}_r$, $\mathbf{C} = \mathbf{X}_r$, and $\mathbf{D} = \mathbf{M}_r^{-1}$, we obtain

$$\boldsymbol{\Sigma}^{-1} = (\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r')^{-1} = \mathbf{I} - \mathbf{X}_r (\mathbf{X}_r' \mathbf{X}_r + \mathbf{M}_r^{-1})^{-1} \mathbf{X}_r' = \mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r', \quad (4.42)$$

with

$$\tilde{\mathbf{M}}_r = (\mathbf{X}_r' \mathbf{X}_r + \mathbf{M}_r^{-1})^{-1}.$$

The Bayes factor for two competing models M_r and M_s is obtained as

$$\text{BF}_{js} = \frac{p(\mathbf{y} | M_r)}{p(\mathbf{y} | M_s)}$$

and results in:

$$\begin{aligned} \text{BF}_{js} &= \frac{t_{2a}(\mathbf{X}_r \mathbf{m}_r, \mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r')}{t_{2a}(\mathbf{X}_s \mathbf{m}_s, \mathbf{I} + \mathbf{X}_s \mathbf{M}_s \mathbf{X}_s')} \\ &= \left(\frac{|\mathbf{I} + \mathbf{X}_s \mathbf{M}_s \mathbf{X}_s'|}{|\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r'|} \right)^{1/2} \left(\frac{2a + (\mathbf{y} - \mathbf{X}_s \mathbf{m}_s)'(\mathbf{I} - \mathbf{X}_s \tilde{\mathbf{M}}_s \mathbf{X}_s')(\mathbf{X}_s \mathbf{m}_s)}{2a + (\mathbf{y} - \mathbf{X}_r \mathbf{m}_r)'(\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r')(\mathbf{X}_r \mathbf{m}_r)} \right)^{a+n/2}. \end{aligned}$$

In the literature, the Bayes factor is usually given in a slightly different form. Using Appendix A.4 property 4 of Theorem A.17 and Eq. (4.42) we obtain

$$\left(\frac{|\mathbf{I} + \mathbf{X}_s \mathbf{M}_s \mathbf{X}_s'|}{|\mathbf{I} + \mathbf{X}_r \mathbf{M}_r \mathbf{X}_r'|} \right)^{1/2} = \left(\frac{|\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r'|}{|\mathbf{I} - \mathbf{X}_s \tilde{\mathbf{M}}_s \mathbf{X}_s'|} \right)^{1/2}.$$

Invoking Sylvester's theorem (Appendix A.4, Theorem A.17, property 6) with $\mathbf{A} = \tilde{\mathbf{M}}_r^{-1} = \mathbf{X}_r' \mathbf{X}_r + \mathbf{M}_r^{-1}$, $\mathbf{B} = \mathbf{X}_r'$, and $\mathbf{C} = -\mathbf{X}_r$ yields

$$\begin{aligned} |\tilde{\mathbf{M}}_r^{-1} - \mathbf{X}_r' \mathbf{X}_r| &= |\tilde{\mathbf{M}}_r^{-1}| \cdot |\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r'| \\ \Leftrightarrow |\mathbf{M}_r^{-1} + \mathbf{X}_r' \mathbf{X}_r - \mathbf{X}_r' \mathbf{X}_r| &= \frac{1}{|\tilde{\mathbf{M}}_r|} |\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r'| \\ \Leftrightarrow \frac{|\tilde{\mathbf{M}}_r|}{|\mathbf{M}_r|} &= |\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r'|. \end{aligned}$$

We finally arrive at the Bayes factor

$$\text{BF}_{rs} = \left(\frac{|\tilde{\mathbf{M}}_r| |\mathbf{M}_s|}{|\tilde{\mathbf{M}}_s| |\mathbf{M}_r|} \right)^{1/2} \left(\frac{2a + (\mathbf{y} - \mathbf{X}_s \mathbf{m}_s)' (\mathbf{I} - \mathbf{X}_s \tilde{\mathbf{M}}_s \mathbf{X}_s') (\mathbf{y} - \mathbf{X}_s \mathbf{m}_s)}{2a + (\mathbf{y} - \mathbf{X}_r \mathbf{m}_r)' (\mathbf{I} - \mathbf{X}_r \tilde{\mathbf{M}}_r \mathbf{X}_r') (\mathbf{y} - \mathbf{X}_r \mathbf{m}_r)} \right)^{a+n/2}.$$

Derivation of the Marginal Model Prior If a Beta Prior Is Assumed for θ (p. 252)

We obtain

$$\begin{aligned} p(M_r) &= \int_0^1 p(M_r | \theta) p(\theta) d\theta \\ &= \int_0^1 \theta^{k_r} (1 - \theta)^{k-k_r} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1} d\theta \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \theta^{a+k_r-1} (1 - \theta)^{b+k-k_r-1} d\theta \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a+k_r)\Gamma(b+k-k_r)}{\Gamma(a+b+k)} \int_0^1 \text{Beta}(a+k_r, b+k-k_r) d\theta \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a+k_r)\Gamma(b+k-k_r)}{\Gamma(a+b+k)}. \end{aligned}$$

The integration has been done by expanding the term within the integral sign such that the density of a $\text{Beta}(a+k_r, b+k-k_r)$ distribution results. Then the corresponding integral over the unit interval is one.

Derivation of the Gibbs Sampler for the Continuous Spike and Slab Variance Prior (p. 253)

We develop a Gibbs sampler that consecutively samples from the full conditionals of $\boldsymbol{\beta}$, σ^2 , $\tau_1^2, \dots, \tau_k^2$, $\delta_1, \dots, \delta_k$, and θ .

The full conditionals of $\boldsymbol{\beta}$ and σ^2 are identical to those of the Bayesian LASSO. It remains to derive the full conditionals of the variances τ_j^2 , the indicators δ_j , and the prior inclusion parameter θ .

Full Conditional of τ_j^2

Suppose first that $\delta_j = 1$. We then have

$$\begin{aligned} p(\tau_j^2 | \cdot) &\propto \frac{1}{(\tau_j^2)^{1/2}} \exp\left(-\frac{1}{2\tau_j^2} \beta_j^2\right) \frac{1}{(\tau_j^2)^{a_{\tau^2}+1}} \exp\left(-\frac{b_{\tau^2}}{\tau_j^2}\right) \\ &= \frac{1}{(\tau_j^2)^{a_{\tau^2}+1/2+1}} \exp\left(-\frac{1}{\tau_j^2} \left(b_{\tau^2} + \frac{1}{2}\beta_j^2\right)\right), \end{aligned}$$

which is the $IG(a_{\tau^2} + 1/2, b_{\tau^2} + 1/2\beta_j^2)$ distribution.

For $\delta_j = 0$, we have to exchange b_{τ^2} by $v_0 b$ and arrive at the $IG(a_{\tau^2} + 1/2, v_0 b_{\tau^2} + 1/2\beta_j^2)$ distribution. Summarizing we have

$$\tau_j^2 \mid \cdot = (1 - \delta_j)IG(a_{\tau^2} + 1/2, v_0 b_{\tau^2} + 1/2\beta_j^2) + \delta_j IG(a_{\tau^2} + 1/2, b_{\tau^2} + 1/2\beta_j^2).$$

Full Conditional of δ_j

For $\delta_j = 1$, we have

$$P(\delta_j = 1 \mid \cdot) \propto \theta \cdot IG(\tau_j^2, a, b),$$

where $IG(\tau_j^2, a, b)$ denotes the density of the respective inverse gamma distribution evaluated at the current value τ_j^2 . For $\delta_j = 0$ the corresponding probability is given as

$$P(\delta_j = 0 \mid \cdot) \propto (1 - \theta) \cdot IG(\tau_j^2, a, v_0 b).$$

Thus $\delta_j \mid \cdot \sim B(1, \theta^{new})$ with

$$\theta^{new} = \frac{\theta \cdot IG(\tau_j^2, a, b)}{\theta \cdot IG(\tau_j^2, a, b) + (1 - \theta) \cdot IG(\tau_j^2, a, v_0 b)}.$$

Full Conditional of θ

Let

$$s = \sum_{j=1}^k \delta_j$$

be the number indicators with $\delta_j = 1$. Then the full conditional is given by

$$\begin{aligned} p(\theta \mid \cdot) &\propto \prod_{j=1}^k p(\delta_j \mid \theta) p(\theta) \\ &\propto \theta^s (1 - \theta)^{k-s} \theta^{a-1} (1 - \theta)^{b-1} \\ &= \theta^{a+s-1} (1 - \theta)^{b+k-s-1}, \end{aligned}$$

which is a beta distribution. More specifically,

$$\theta \mid \cdot \sim \text{Beta}(a + s, b + k - s).$$